

Commutative Algebra

Joe Orii

Contents

1	Rings and Modules	1
1.1	Commutative Rings and Ideals	1
1.2	Prime and Maximal Ideals	8
1.3	Geometric P.O.V.	12
1.4	Krull's Lemma	17
1.5	Local Rings	21
1.6	Modules	23
1.7	More on Localization	26
1.8	Exercise from [2]	28
2	Noetherian and Artinian Rings	31
2.1	Noetherian Rings	31
2.2	Hilbert Basis Theorem	33
2.3	Cayley-Hamilton and Nakayama	36
2.4	Artin Rings	39
2.5	Dimension of a Ring	42
2.6	Primary Decomposition	47
2.7	Exercises from [2]	54
3	The Nullstellensatz	57
3.1	Algebras	57
3.2	Normal Rings and Integral Closure	60
3.3	Noether Normalization Lemma	63
3.4	Nullstellensatz Theorem	66
3.5	Ax Theorem	71
4	Dimension of Local Rings	75
4.1	Krull's Principal Ideal Theorem	78
4.2	Krull's Height Theorem	81
5	Discrete Valuation Rings	87
6	Completions	93
6.1	Hensel's Lemma	98
	Bibliography	102

1 Rings and Modules

§1.1 Commutative Rings and Ideals

§1.1.i Introduction

Recall the basic definitions of a commutative ring. We shall let the ring R to have multiplicative identity.

Remark 1.1.1 — We will often drop the word ‘commutative’ in the setting for rings because this course is about commutative rings.

Example 1.1.2

Let R be a ring; then, $0 = 1$ if and only if $R = \{0\}$.

Definition 1.1.3. Let R be a ring. Then, $a \in R$ is a **unit** if it is nonzero and there exists $ab = 1$. b is called the **inverse** of a .

Definition 1.1.4. A ring R is a **field** if every element is a unit.

Definition 1.1.5. An element $a \in R$ is a **zero divisor** if there exists a nonzero $b \in R$ such that $ab = 0$.

Definition 1.1.6. A ring R is an **integral domain** if there exists no nonzero zero divisors in R .

Definition 1.1.7. Given two rings R_1 and R_2 , a **morphism of rings** is a map $f: R_1 \rightarrow R_2$ such that $f(1) = 1$, $f(a + b) = f(a) + f(b)$, and $f(a \cdot b) = f(a)f(b)$.

Example 1.1.8

Let $m > 1$ be an integer. We denote by $\mathbb{Z}/m\mathbb{Z}$ to be the set of integers modulo m . Then, $\mathbb{Z}/m\mathbb{Z}$ is a commutative ring and the following are equivalent:

1. $\mathbb{Z}/m\mathbb{Z}$ is a field.
2. $\mathbb{Z}/m\mathbb{Z}$ is an integral domain.
3. m is prime.

Generally speaking, the units of $\mathbb{Z}/m\mathbb{Z}$ are the elements $[a] \in \mathbb{Z}/m\mathbb{Z}$ such that $1 \leq a \leq m - 1$ and $\gcd(a, m) = 1$. The zero divisors are, taking the same representative, those such that $\gcd(a, m) > 1$.

Example 1.1.9

Given a field k , $k[X]$ is an integral domain. The units of $k[X]$ the nonzero elements of k , so $k[X]$ is never a field.

Definition 1.1.10. A field k is **algebraically closed** if for any nonconstant polynomial $f \in k[X]$, there exists $a \in k$ such that $f(a) = 0$.

Example 1.1.11

$\mathbb{C}$ is algebraically closed.

Example 1.1.12

We can define, for any ring S , the ring $R = S[X]$. Inductively, we can define $S[X_1, \dots, X_n] = S[X_1, \dots, X_{n-1}][X_n]$.

To conclude the introduction, note that we will always be interested in the geometric analogue of the algebra we’re studying. Often, in commutative algebra, we will be interested in polynomial rings; on the other hand, we will be interested in ‘shapes’—more specifically Affine spaces.

§1.1.ii Localizations and Fraction Fields

Definition 1.1.13. Let R be a commutative ring and $S \subset R$ be a subset. Then, S is called a **multiplicative** if $1 \in S$ and for any $a, b \in S$, we have that $a \cdot b \in S$.

Proposition 1.1.14

Let R be a commutative ring and let $S \subset R$ be a multiplicative set. Define the equivalence $\sim$ on $R \times S$ so that $(a, s) \sim (b, t)$ if and only if there exists $r \in S$ such that $r(at - bs) = 0$. We denote by $\frac{a}{s}$ the equivalence class of (a, s) . Then, the set

$$S^{-1}R := \left\{ \frac{a}{s} : a \in R, s \in S \right\}$$

admits two natural operations so that it is a commutative ring. If R is an integral domain, then there is a natural inclusion $R \hookrightarrow S^{-1}R$.

Proof. We can equip addition and multiplication as

$$\begin{aligned} \frac{a}{s} + \frac{b}{t} &= \frac{at + bs}{st} \\ \frac{a}{s} \cdot \frac{b}{t} &= \frac{ab}{st} \end{aligned}$$

The inclusion $R \hookrightarrow S^{-1}R$ maps $r \mapsto \frac{r}{1}$. □

Definition 1.1.15. In the above, $S^{-1}R$ is called the **localization** of R with respect to S .

Remark 1.1.16 — Note that R being an integral domain for the injection $R \hookrightarrow S^{-1}R$ is crucial.

Proposition 1.1.17

Let R be an integral domain. Choosing $S = R \setminus \{0\}$ gives

$$\text{Frac}(R) := \left\{ \frac{a}{b} : a, b \in R, b \neq 0 \right\}.$$

This admits two natural operations so that it is a field which contains R as a subring.

Proof. Follows from the previous. $\square$

Definition 1.1.18. In the above, we refer to $\text{Frac}(R)$ as the **field of fractions** of R .

Remark 1.1.19 — Another way to say $\text{Frac}(R)$ is that it is the smallest field that contains R .

Example 1.1.20

If $R = \mathbb{Z}$, then $\text{Frac}(R) = \mathbb{Q}$.

Example 1.1.21

If $R = k[X]$, where k is a field, then $\text{Frac}(R) = k(x)$.

Exercise 1.1.22.

1. Assume that S is such that $a \cdot s = 0$ for $a \in R$, $s \in S$ implies $a = 0$. Show that in this case the natural homomorphism

$$R \rightarrow S^{-1}R$$

is injective.

2. Construct an example of R and S such that the above homomorphism is not injective.

Solution: 1. Let $r \in R$. Suppose that $\frac{r}{1} = 0$. Then, there exists some $s \in S$ such that $s(r - 0) = 0$. Then, $sr = 0$, so $r = 0$. Thus, injective.

2. Let $R = \mathbb{Z}/6\mathbb{Z}$. Then, consider the multiplicative subset $S = (\mathbb{Z}/6\mathbb{Z}) \setminus (2\mathbb{Z}/6\mathbb{Z})$. Then, $S = \{[1], [3], [5]\}$. Then, suppose $\frac{[2]}{[1]} = 0$. Then, there exists some element $s \in S$ such that $s[2] = 0$. Then, $s = [3]$ annihilates $[2]$, so the natural map is not injective. $\square$

Exercise 1.1.23. Let S be a multiplicative subset of a ring R . Show that the fraction field of $S^{-1}R$ coincides with $\text{Frac}(R)$.

Solution: This is almost immediate. Let

$$S' = \left\{ \frac{r}{s} \in S^{-1}R : \frac{r}{s} \neq 0 \right\} = \left\{ \frac{r}{s} \in S^{-1}R : r \neq 0 \right\}.$$

Now, let $\frac{r}{u} \in \text{Frac}(R)$. It is clear then that $\frac{r}{u} \in (S')^{-1}(S^{-1}R)$, as this is equivalent to the quotient $\frac{\frac{r}{u}}{\frac{1}{1}}$. We are done. $\square$

§1.1.iii Nilpotent Elements

Definition 1.1.24. Let R be a commutative ring. Then, $a \in R$ is called **nilpotent** if there exists a positive integer n such that $a^n = 0$.

Example 1.1.25

$[2] \in \mathbb{Z}/8\mathbb{Z}$ is nilpotent. We see that $[2]^3 = 0$.

Exercise 1.1.26. Let R be a commutative ring and let $a \in R$ be a nonzero element. Define

$$S := \{a^n : n \geq 0\}.$$

Then, show that S is multiplicative and, denoting $R_a := S^{-1}R$, show that $R_a = \{0\}$ if and only if a is nilpotent.

Solution: S is clearly multiplicative since $1 \in S$ and for any $a^n, a^m \in S$, $a^{n+m} \in S$. Suppose that $R_a = 0$. Then, $\frac{1}{1} = \frac{0}{1}$ in R_a . In other words, there exists some $t \in S$ such that $t(1 - 0) = 0$. That is, $a^n = 0$, for some n . Conversely, suppose a is nilpotent. Then, $a^n = 0$ for some n , so $0 \in S$. Then, $\frac{1}{1} = \frac{0}{1}$, since $0(1 - 0) = 0$ works. Thus, the R_a is the zero ring. $\square$

Proposition 1.1.27

Let R be a commutative ring and let $a \in R$ be nilpotent. Then, $1 + a$ is a unit.

Proof. Let n be a positive integer such that $a^n = 0$. Then, we can check that

$$\sum_{i=0}^{n-1} (-a)^i = 1 - a + a^2 + \cdots + (-a)^{n-1}$$

is the inverse of $1 + a$. $\square$

Exercise 1.1.28. Show that the sum of a unit and a nilpotent element is a unit.

Solution: If u is a unit (with inverse v) and a is nilpotent, then we can write $u + a = u(1 + u^{-1}a)$. $1 + u^{-1}a$ is a unit. and u is a unit, so we are done. $\square$

Exercise 1.1.29. Let A be a ring and let $A[[x]]$ be the ring of formal power series

$$f = \sum_{n=0}^{\infty} a_n x^n$$

with coefficients in A . Show that:

1. f is a unit in $A[[x]]$ if and only if a_0 is a unit in A .
2. If f is nilpotent, then a_n is nilpotent for all $n \geq 0$. Is the converse true?
3. f belongs to the Jacobson radical of $A[[x]]$ if and only if a_0 belongs to the Jacobson radical of A .

Solution: 1. If f is a unit, then there is some formal power series g such that $fg = 1$, so if g is indexed using b_i , then $a_0b_0 = 1$. Conversely, if a_0 is a unit, there exists some b_0 such that $a_0b_0 = 1$. We now iteratively build a formal power series. So suppose this is done up to $k - 1$ -th degree. Then the k -th degree

$$a_0b_k + \cdots + a_kb_0 = 0 \implies b_k = -b_0^{-1}(a_0b_k + \cdots + a_{k-1}b_1),$$

so the induction is complete, and we can build on. Note that b_0 is a unit, which is what makes this possible.

2. We claim that if A is an integral domain, then $A[[X]]$ is an integral domain. Let $f, g \neq 0$, where $f = \sum_{n \geq n_0} a_n x^n$ and $g = \sum_{m \geq m_0} b_m x^m$. Then, in particular, $a_{n_0}b_{m_0} \neq 0$, because each of the a_{n_0}, b_{m_0} are not zero and A is an integral domain. Now, let $\mathfrak{p}$ be an arbitrary prime of A . Then, $A/\mathfrak{p}$ is an integral domain, so so is $(A/\mathfrak{p})[[X]]$. Consider the image of f in $(A/\mathfrak{p})[[X]]$, $\bar{f}$. Since f is nilpotent, it follows that $\bar{f}^n = 0$ in $(R/\mathfrak{p})[[X]]$, so $\bar{f} \in \mathfrak{p}[[X]]$ in the first place. In particular, this implies that all the coefficients of f are in $\mathfrak{p}$. Since $\mathfrak{p}$ is an arbitrary prime ideal of A , it follows that all $a_n \in \sqrt{(0)}$.

3. f is in the Jacobson radical of $A[[X]]$ if and only if $1 - fg$ is a unit in $A[[X]]$. This is true if and only if $1 - a_0c$ is a unit for all $c \in A$. However, this is true if and only if $a_0 \in \mathcal{J}(A)$. $\square$

§1.1.iv Ideals

Definition 1.1.30. Let R be a commutative ring. An **ideal** of R is a subset $I \subset R$ such that

1. I is an additive subgroup of R .
2. for any $r \in R$ and $a \in I$, we have $ra \in I$.

An ideal $I \subset R$ is **proper** if $I \neq R$.

Proposition 1.1.31

If R is a commutative ring and $I \subset R$ is an ideal, then R/I is also a commutative ring with a natural morphism of rings $R \rightarrow R/I$.

Proof. Clear. $\square$

Remark 1.1.32 — Note that this provides a inclusion reversing bijection between the ideals of R/I and ideals of R which contain I .

Example 1.1.33

If $R = \mathbb{Z}$ and $I = m\mathbb{Z}$, then I is an ideal of R and $R/I = \mathbb{Z}/m\mathbb{Z}$.

Definition 1.1.34. Let R be a commutative ring and let $S \subset R$ be a subset. The ideal **generated** by S is

$$(S) := \{r_1s_1 + \cdots r_ns_n : r_1, \dots, r_n \in R, s_1, \dots, s_n \in S\}.$$

If $S = (s_1, \dots, s_n)$ is finite, then we denote $(s_1, \dots, s_n) := (S)$.

Remark 1.1.35 — As is clear, (S) is the smallest ideal containing S .

Exercise 1.1.36. Let R be a commutative ring. Then, show that:

1. If $S \subset R$ is a subset, then (S) is an ideal of R .
2. $(1) = R$
3. Given $x \in R$, x is a unit if and only if $(x) = R$.

Solution: We first show that for a subset $S \subset R$, (S) is an ideal. Recall that

$$(S) = \{r_1s_1 + \cdots + r_ns_n : r_i \in R, s_j \in S\}.$$

This is clear, since this is an additive subgroup and is closed under multiplication by an element in R .

We need to show that $(1) = R$. This is clear, since $r \cdot 1 \in (1)$, so $R \subseteq (1)$. The converse is clear.

We need to show that $x \in R$ is a unit if and only if $(x) = R$. The forward is clear, because if x is a unit, there exists nonzero $y \in R$ such that $xy = 1 \in R$, and so we’ve recovered $(1) = R$. For the converse, if $(x) = R$, then there exists some $y \in R$ such that $xy = 1$, so x is a unit. $\square$

Exercise 1.1.37. Let $I, J \subseteq R$ be ideals. Show that $I \cap J = IJ$ if $I + J = R$. Further, if $I + J \neq R$, find counter examples to the statement $I \cap J = IJ$. Is the converse of the original statement true? That is, if $I \cap J = IJ$, can we get that $I + J = R$?

Solution: We show the first. Note that we always have $IJ \subseteq I \cap J$. Let $\sum a_ib_i \in IJ$. Then, since each term is in I and J , of course the entire sum is in I and J . Thus, $IJ \subseteq I \cap J$. Given this, we’re actually trying to prove that $I + J = R$ implies $I \cap J \subseteq IJ$. Assuming $I + J = R$, we get

$$I \cap J \subseteq (I \cap J)R = (I \cap J)(I + J) = I \cdot (I \cap J) + J \cdot (I \cap J) \subseteq IJ + JI = IJ.$$

Now for a counter example. We can search for one by noting that in PIDs, $(a) \cap (b) = (\text{lcm}(a, b))$. Take $R = \mathbb{Z}$, and consider the ideals (p) and (p^2) . Then, $IJ = (p^3)$, and $I \cap J = (p^2)$.

Finally, no. We can take $I = J = (0)$, but this is certainly not true for general rings R . $\square$

§1.1.v Characterizing Domains

Definition 1.1.38. Given a commutative ring R , an ideal I is **principal** if there exists $x \in R$ such that $I = (x)$. A ring is a **principal ideal domain** (PID) if all its ideals are principal.

Definition 1.1.39. An integral domain R is a **Euclidean domain** if there exists a norm $|\cdot| : R \rightarrow \mathbb{N}$ such that, for any $a, b \in R$, with $b \neq 0$, there exist $q, r \in R$ such that

$$a = bq + r \quad \text{and} \quad |r| < |b|.$$

Exercise 1.1.40. Show that Euclidean domain is a PID.

Proof. Let R be the Euclidean with the ‘Euclidean function’ N . Take a nonzero ideal $I \subset R$, and take $q \in I$ with $N(q)$ being minimal (which exists, since $\mathbb{Z}_{>0}$ is well ordered) and any other element $a \in I$. Then, we can write

$$a = bq + r,$$

with $b, r \in R$ and $N(r) < N(q)$ or $r = 0$. However, we chose $N(q)$ to be minimal, so $r = 0$. Thus, $a = bq$, implying that $a \in (q)$. In particular, there was no choice on $a \in I$, so $I = (q)$. $\square$

Example 1.1.41

$\mathbb{Z}$ is a PID.

Example 1.1.42

If k is a field, then $R = k[X]$ is a PID.

Definition 1.1.43. An integral domain R is a **unique factorization domain** (UFD) if for any $a \in R$, there exist irreducible elements $p_1, \dots, p_m \in R$ such that $a = p_1 \dots p_m$. Further, such a decomposition is unique up to the choice of the order $p_1, \dots, p_m$ and multiplication by units.

Remark 1.1.44 — Any PID is a UFD.

Exercise 1.1.45. Show that $\mathbb{Z}[\sqrt{2}]$ is a PID.

Solution: We need to show that $\mathbb{Z}[\sqrt{2}]$ is a PID. For this, we show that $\mathbb{Z}[\sqrt{2}]$ is a Euclidean domain equipped with the function $N: \mathbb{Z}[\sqrt{2}] \rightarrow \mathbb{Z}_{>0}$ defined by

$$N(a + b\sqrt{2}) = |a^2 - 2b^2|.$$

We just need to show that this is indeed a Euclidean function now. N is indeed nonzero, so we just need to check multiplicativity and that the Euclidean algorithm holds. Multiplicativity is clear from complex multiplication, so we just need to check that the Euclidean algorithm holds. Let $\alpha, \beta \in \mathbb{Z}[\sqrt{2}]$, and write $\frac{\alpha}{\beta} = p + q\sqrt{2}$, where $p, q \in \mathbb{Q}$. Then, we can find $r, s \in \mathbb{Z}$ such that

$$|p - r|, |q - s| < \frac{1}{2}$$

by rounding the rationals p and q to the nearest integer. Then,

$$N\left(\frac{\alpha}{\beta} - (r + s\sqrt{2})\right) = |(p - r)^2 + 2(q - s)^2| \leq \frac{3}{4},$$

where the last inequality is done by the triangle inequality. Therefore, $N(\alpha - \beta(r + s\sqrt{2})) \leq \frac{3}{4}N(\beta)$, so we’ve found our remainder term, which is $\alpha - \beta(r + s\sqrt{2})$. Thus, we have the Euclidean algorithm:

$$\alpha = \beta(r + s\sqrt{2}) + \underbrace{(\alpha - \beta(r + s\sqrt{2}))}_{\text{remainder}}$$

$\square$

§1.1.vi Constructing Ideals from Ideals

Definition 1.1.46. Let R be a commutative ring and $I, J \subset R$ be ideals. Then, we can define:

- Sum ideal: $I + J := \{a + b : a \in I, b \in J\} \subset R$
- Product ideal: $I \cdot J := \{a_1b_1 + \cdots + a_nb_n : a_i \in I, b_i \in J\} \subset R$
- Quotient ideal: $(I : J) := \{x \in R : xJ \subset I\}$.

Remark 1.1.47 — For products of ideals, if $I = (x)$ for some $x \in R$, then we denote $xJ := (x) \cdot J \subset R$.

Definition 1.1.48. Let I be an ideal of a commutative ring R . Given $k \geq 1$, the k -th power of I is the ideal I^k generated by the set

$$S = \{r_1 \cdots r_k : r_1, \dots, r_k \in I\}.$$

Further, the ideal $I \subset R$ is **nilpotent** if there exists a positive integer k such that $I^k = (0)$.

Exercise 1.1.49. Let R be a commutative ring and $I, J \subset R$ be ideals. Show that:

1. $I + J, I \cdot J, (I : J)$ are actually ideals
2. if $J \subset I$, then $(I : J) = R$.
3. $(I : R) = I$.
4. $(R : I) = R$.

Solution: 1. The sum and product are clear, so I will only verify the quotient ideal $(I : J)$. We first check that it's a subgroup of $(R, +)$. Let $x, y \in (I : J)$. Then, $xJ \subset I$ and $yJ \subset I$, so indeed $(x + y)J = xJ + yJ \subset I$, so it is closed under addition; this clearly implies associativity of addition of elements as well. Next, it's clear that $0 \in (I : J)$, since $0J = 0 \subset I$. Next, $xJ \subset I$ implies that $(-x)J \subset I$, since $xJ \subset I$ means that there is some element $j \in J$ such that $xj \in I$. Then, of course the negative of that, $(-x)j$, is in I , since I is an additive subgroup of R . We finally show that for any $r \in R$, then $rx \in (I : J)$. This is clear, because $rxJ \subset rI = I$.

2. If $J \subset I$, then every element in $r \in R$ satisfies $rJ \subset I$.

3. Let $x \in (I : R) = \{r \in R : rR \subset I\}$. Then, $x \in I$, because if otherwise, then $xR \not\subset I$. The converse is clear since I is an ideal.

4. Let $x \in (R : I)$. Then, since I is an ideal, $x \in R$. The converse is also clear by I being an ideal. $\square$

§1.2 Prime and Maximal Ideals

Definition 1.2.1. Let R be a commutative ring and $\mathfrak{m} \subset R$ be a proper ideal. Then, $\mathfrak{m}$ is a **maximal ideal** if for any ideal $I \subset R$ such that $\mathfrak{m} \subset I$, we have $\mathfrak{m} = I$ or $I = R$.

Exercise 1.2.2. Given a commutative ring R and $\mathfrak{m} \subset R$ a proper ideal, show that $\mathfrak{m}$ is a maximal ideal if and only if $R/\mathfrak{m}$ is a field.

Solution: This follows by the reverse-inclusion property of ideals and the quotient of rings by ideals. Then, if we focus on the ideals of $R/\mathfrak{m}$, it follows that the only ideals here are (0) and $R/\mathfrak{m}$ itself. In other words, every element is a unit, so $R/\mathfrak{m}$ is a field. Conversely, if we have $R/\mathfrak{m}$ is a field, then the only ideals of $R/\mathfrak{m}$ are (0) and itself. Thus, the only ideals which contain $\mathfrak{m}$ are $\mathfrak{m}$ and R , making $\mathfrak{m}$ maximal. $\square$

Example 1.2.3

Let $R = \mathbb{Z}$, and $m \in R$ be a non-zero number and let $I = (m) = m\mathbb{Z}$. Then, we have that $\mathbb{Z}/I = \mathbb{Z}/m\mathbb{Z}$ is a field if and only if m is prime. Therefore, I is maximal if and only if m is prime.

Example 1.2.4

Let k be a field and $R = k[X]$. Let $a \in k$ and $\mathfrak{m} = (X - a) \subset R$; we claim that this is maximal. Indeed, the ‘evaluation at a map’ $R/\mathfrak{m} \rightarrow k$ defined by $[f(X)] \mapsto f(a)$ is a well-defined isomorphism. Therefore, $R/\mathfrak{m}$ is a field and it follows that $\mathfrak{m}$ is maximal.

Example 1.2.5

We now generalize the previous example. Let k be a field, and let $R = k[X_1, \dots, X_m]$ and $a_1, \dots, a_m \in k$, then

$$\mathfrak{m} := (X - a_1, \dots, X - a_m) \subset R$$

is a maximal ideal. We prove this by using the evaluation map once more to show that $R/\mathfrak{m} \cong k$.

Remark 1.2.6 — Later, we will show that if k is an algebraically closed field, then these are the only maximal ideals. On the other hand, if $k = \mathbb{R}$, then $\mathfrak{m} := (x^2 + 1) \subset \mathbb{R}[x]$ is maximal, despite not being of the form $(x - a_1, \dots, x - a_m)$. Note that if $I := (x^2 - 1) \subset \mathbb{R}[x]$, then I is not maximal, since $I \subset (x - 1)$.

Zorn’s Lemma and Existence of Maximal Ideals

Definition 1.2.7. Let P be a set. Then, P is **partially ordered** if it admits a relation $\leq$ such that

- reflexive: $a \leq a$ for any $a \in P$.
- anti-symmetric: if $a, b \in P$ such that $a \leq b$ and $b \leq a$, then $a = b$.
- transitive: if $a, b, c \in P$ such that $a \leq b$ and $b \leq c$, then $a \leq c$.

Furthermore, P is **totally ordered** if it is partially ordered and for any $a, b \in P$, $a \leq b$ or $b \leq a$.

Definition 1.2.8. Given a partially ordered set P and $T \subset P$, an **upper bound** of T is an element $x \in P$ such that $a \leq x$ for all $a \in T$.

Definition 1.2.9. Let P be a partially ordered set and $m \in P$. Then, m is **maximal** if for any $a \in P$ such that $m \leq a$, we have $a = m$.

Theorem 1.2.10 (Zorn's Lemma)

Let P be a partially ordered set such that every totally ordered subset $T \subset P$ has an upper bound. Then, P admits a maximal element $m \in P$.

Remark 1.2.11 — This is equivalent to the axiom of choice.

Proposition 1.2.12

Let R be a commutative ring and let $I \subset R$ be a proper ideal. Then, there exists a maximal ideal $\mathfrak{m} \subset R$ containing I .

Proof. Let P be the set of proper ideals in R containing I . Then, we can order the elements of P via inclusion, and so we see that P is a partially ordered set. Let $T \subset P$ be a totally ordered subset. Then, consider

$$\tilde{I} := \cup_{J \in T} J \subset R,$$

We claim this is a proper ideal of R . If $x, y \in \tilde{I}$, then since T is totally ordered, then we have that x, y are in the same $J \in T$, so indeed $x + y \in \tilde{I}$. $rx \in \tilde{I}$ is clear; thus, $\tilde{I}$ is an ideal. Moreover, it is proper because if it weren't, then $1 \in \tilde{I}$ and so $1 \in J$ for some $J \in T$, implying that J is not proper—a contradiction. Now, $I \subset \tilde{I}$, so $\tilde{I}$ is an element in P which is an upper bound of T . Thus, by Zorn's Lemma, P admits a maximal element, which is maximal ideal of R . $\square$

Remark 1.2.13 — We can check this in the case $R = k[X_1, \dots, X_n]$. Let $V(I) \subset \mathbb{A}_k^n$ is a variety defined by an ideal $I \subset R$ such that $V(I) \neq \emptyset$. Then, there exists a point $z \in V(I)$. Then, z defines a maximal ideal $\mathfrak{m}$ such that $I \subset \mathfrak{m}$. The inclusion reversing property indeed proves this.

Later, we shall see that if $k = \bar{k}$, the algebraic closure of k , then $V(I) \neq \emptyset$ if and only if $I \subsetneq R$.

Exercise 1.2.14. Show that every non-unit of R is contained in a maximal ideal.

Solution: Let x not be a unit. Then, $(x) \subsetneq R$, so (x) is contained in a maximal ideal. $\square$

Prime Ideals

Definition 1.2.15. Let R be a commutative ring and let $\mathfrak{p} \subset R$ be a proper ideal. Then, $\mathfrak{p}$ is a **prime ideal** if for any $x, y \in R$ such that $xy \in \mathfrak{p}$, we have that $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$.

Exercise 1.2.16. Let R be a commutative ring and $I \subset R$ be a proper ideal. Show that:

1. I is prime if and only if R/I is an integral domain.
2. every maximal ideal is prime.

Solution: 1. Let I be prime, and consider the quotient R/I ; this contains elements of the form $r + I$, where $r \in R$. Let $(x + I)(y + I) = I$, for $x, y \in R$. Then, it must be that $xy \in I$. Since I is prime, then it cannot be that $x, y \notin I$, so there are no zerodivisors. Conversely, if R/I is an integral domain, then consider $(x + I)(y + I) = I$ again. Then, $xy \in I$, but since there are no zerodivisors, it follows that $x \in I$ or $y \in I$. In other words, I is prime.

2. Let $\mathfrak{m}$ be a maximal ideal in R . Then, $R/\mathfrak{m}$ is a field, and, in particular, an integral domain. Therefore, $\mathfrak{m}$ is a prime ideal. $\square$

Exercise 1.2.17. Let $f: R_1 \rightarrow R_2$ be a morphism of rings and let $I \subset R_2$ be an ideal. Show that:

1. if I is prime then $f^{-1}(I)$ is also prime.
2. if I is maximal, then $f^{-1}(I)$ is not necessarily maximal.
3. Let $f: R_1 \rightarrow R_2$ be surjective now, and let $I \subset R_2$ be maximal. Show that $f^{-1}(I)$ is maximal.

Proof. 1. Let $xy \in f^{-1}(I)$. Then, $f(xy) \in I$, so $f(x)f(y) \in I$. Thus, $f(x) \in I$ or $f(y) \in I$, implying that x or y is in $f^{-1}(I)$.

2. Consider $R_1 = \mathbb{Z}$ and $R_2 = \mathbb{Q}$, with f being the inclusion. (0) is maximal in $\mathbb{Q}$ because $\mathbb{Q}$ is a field, but $f^{-1}((0)) = (0)$, which is not maximal because $\mathbb{Z}$ is not a field. More generally, we can take a ring R that is not a field, and let $f: R \hookrightarrow \text{Frac}(R)$ works, and consider the preimage of the zero ideal.

3. We know that $R_1/\ker(f) \cong R_2$ and R_2/I is a field. Therefore, this is equivalent to the following: take $g: S_1 \rightarrow S_2$ such that S_2 is a field, and $I = (0) \subset S_2$. Then, we need to show that $g^{-1}(I)$ is maximal. $S_2 \cong S_1/\ker g$, so $S_1/\ker g$ is a field, so $g^{-1}(I)$ is maximal. $\square$

Example 1.2.18

Let $R = \mathbb{Z}$ and $m \in R$ be a nonzero element. Then, the following are equivalent:

- $(m) \subset R$ is prime.
- $(m) \subset R$ is maximal.
- m is prime.

Example 1.2.19

$(0) \subset \mathbb{Z}$ is a prime ideal but not maximal.

Example 1.2.20

Let k be a field, and take $R = k[X, Y]$, and consider $I = (X) \subset R$. Then, $R/I \cong k[Y]$ is an integral domain but not a field, so it is prime but not maximal.

Example 1.2.21

Let k be a field and let $R = k[X, Y]$. Let $I = (X) \subset R$. Then, the function $R/I \rightarrow k[Y]$ defined by $[f(X, Y)] \mapsto f(0, Y)$ is an isomorphism. We have that $k[Y]$ is an integral domain, but not a field; thus, I is a prime ideal in R but not maximal. On the other hand, the ideal $(X^2) \subset R$ and $(XY) \subset R$ are not prime.

The Radical Ideal

Definition 1.2.22. Let R be a ring and $I \subset R$ be an ideal. Then, we define the **radical** of I as

$$\sqrt{I} := \{r \in R : r^m \in I \text{ for some } m > 0\}.$$

If $I = \sqrt{I}$, then we say that I is a **radical ideal**.

Proposition 1.2.23

Let R be a ring and $I \subset R$ be an ideal. Then, $\sqrt{I} \subset R$ is an ideal.

Proof. The hardest part is to show that $\sqrt{I}$ is closed under addition. Let $x, y \in \sqrt{I}$ so that $x^n, y^m \in I$ for some n, m . Then, we can use the binomial formula:

$$(x + y)^{n+m-1} = \sum_{i=0}^{n+m-1} r_i x^i y^{n+m-1-i},$$

for some $r_i \in R$. For all $i = 0, \dots, n+m-1$, $i \geq n$ or $n+m-1-i \geq m$. In both cases, $x^i y^{n+m-1-i} \in I$, so $(x+y)^{n+m-1} \in I$, so $x+y \in \sqrt{I}$. $\square$

§1.3 Geometric P.O.V.

Definition 1.3.1. Let k be a field and $n > 0$ be an integer. Then, we denote

$$\mathbb{A}_k^n := \{(c_1, \dots, c_n) : c_i \in k \text{ for } i = 1, \dots, n\}$$

to be the n -th dimensional affine space over k^n .

Definition 1.3.2. Given a field k , let $S \subset k[X_1, \dots, X_n]$ be a set of polynomials. Then, the zero locus

$$V(S) := \{x \in \mathbb{A}_k^n : f(x) = 0, \forall f \in S\} \subset \mathbb{A}_k^n$$

is called the **affine variety** (or simply variety). If $S = \{f_1, \dots, f_k\}$ is finite, then we just denote $V(S) = V(f_1, \dots, f_k)$.

Remark 1.3.3 — Note that if $S_1 \subset S_2 \subset k[X_1, \dots, X_n]$ then $V(S_2) \subset V(S_1)$. If $I := (S) \subset k[X_1, \dots, X_n]$ is the ideal generated by a subset $S \subset k[X_1, \dots, X_n]$ then $V(S) = V(I)$. Thus, we can always assume that a variety is defined by an ideal I of $k[X_1, \dots, X_n]$.

Definition 1.3.4. If $V, W \subset \mathbb{A}_k^n$ are varieties such that $W \subset V$, then W is called a **subvariety** of V .

Example 1.3.5 (Trivial Examples)

Let $I = \mathbb{R} = (1)$. Then, $V(I) = \emptyset$. Conversely, if $I = (0)$, then $V(I) = \mathbb{A}_k^n$. Thus, $\mathbb{A}_k^n$ and $\emptyset$ are affine varieties.

Lemma 1.3.6

Let k be a field and let $I, J \subset k[X_1, \dots, X_n]$ be ideals and let $V = V(I) \subset \mathbb{A}_k^n$ and $W = V(J) \subset \mathbb{A}_k^n$ be associated varieties. Then,

$$\begin{aligned} V \cap W &= V(I + J) \\ V \cup W &= V(I \cdot J) = V(I \cap J) \end{aligned}$$

Particularly, the intersection and union of two varieties is also a variety.

Proof. Clear. □

Remark 1.3.7 — It follows that the set of varieties form a topological space on $\mathbb{A}_k^n$ whose closed sets are varieties inside $\mathbb{A}_k^n$. This is called the **Zariski topology**.

Example 1.3.8

Take $k = \mathbb{R}$ and $n = 2$, and let $f(X_1, X_2) = X_1^2 + X_2^2 - 1$ and $g(X_1, X_2) = X_1 - X_2$. Then,

$$V(f) \cap V(g) = \{(X_1, X_2) \in \mathbb{A}_k^2 : X_1^2 + X_2^2 - 1 = X_1 - X_2 = 0\},$$

and

$$V(f) \cup V(g) = V(f \cdot g) = \{(X_1, X_2) \in \mathbb{A}_k^2 : f \cdot g = 0\}$$

Example 1.3.9

Let $I_\alpha \subset R = k[X_1, \dots, X_n]$ be an ideal for all $\alpha \in A$, some indexing set (possibly infinite). We consider

$$\bigcap_{\alpha \in A} V(I_\alpha) = V(+_{\alpha \in A} I_\alpha),$$

where

$$+_{\alpha \in A} I_\alpha = \left\{ \sum_{i=1}^N f_i : f_i \in I_{\alpha_i} \text{ for some } \alpha_i \in A \right\}.$$

We can check this by tracing the definitions.

Example 1.3.10

Is it true that the union of infinitely many varieties is a variety? No. Consider $R = k[X]$. Then, the infinite union

$$\bigcup_{a \in k} V(x - a)$$

is not a variety. For instance, $\mathbb{Z} \subset \mathbb{R} = \mathbb{A}_{\mathbb{R}}^1$ is not a variety. The crux behind this is that the maximum number of solutions to the polynomial is the degree.

Definition 1.3.11. Let k be a field. We now assume that $X \subset \mathbb{A}_k^n$ is a subset, and we define

$$I(X) := \{f \in k[X_1, \dots, X_n] : f(x) = 0, \forall x \in X\}.$$

We can easily check that $I(X) \subset k[X_1, \dots, X_n]$ is an ideal, which we will call the **ideal** of X .

Remark 1.3.12 (Reverse-Inclusion Property of Ideals and Varieties) — Note that if $X_1 \subset X_2 \subset \mathbb{A}_k^n$, then $I(X_2) \subset I(X_1)$. Conversely, if $I \subset J$ as ideals of $k[X_1, \dots, X_n]$, then $V(J) \subset V(I)$. We note that these maps V and I do *not* provide a bijection.

Lemma 1.3.13

Let k be a field and let $X \subset \mathbb{A}_k^n$ be a subset. Then $V(I(X))$ is the smallest variety in $\mathbb{A}_k^n$ which contains X . In particular, if X is a variety, then $V(I(X)) = X$. The second statement is clear.

Proof. Let $x \in X$. Then $f(x) = 0$ for all $f \in I(X)$ and, therefore, $x \in V(I(X))$. Thus,

$$X \subset V(I(X)).$$

Assume now that $W \subset \mathbb{A}_k^n$ is a variety such that $X \subset W$. We want to show that $V(I(X)) \subset W$. Suppose otherwise. Since W is a variety, let $W = V(J)$ for some ideal $J \subset k[X_1, \dots, X_n]$. Then J is not contained in $I(X)$ and there exists $f \in J$ which is not contained in $I(X)$. Therefore, there exists $x \in X$ such that $f(x) \neq 0$, implying that $x \notin W$, which is a contradiction, since f vanishes on W . $\square$

Remark 1.3.14 — We can rephrase the previous lemma in the following manner: $V(I(X))$ is the closure of X in $\mathbb{A}_k^n$ with respect to the Zariski topology.

Lemma 1.3.15

Let k be a field and let $I \subset k[X_1, \dots, X_n]$. Then, we have:

- (i) $V(I) = V(\sqrt{I})$
- (ii) $\sqrt{I} \subset I(V(I))$

Proof. (i): Note that $I \subset \sqrt{I}$, so $V(\sqrt{I}) \subset V(I)$. Now, we want to show the reverse inclusion. Let $z \in V(I)$ and suppose $f \in \sqrt{I}$. Then, $f^n \in I$, so $f^n(z) = 0$. k is a field, so there are no nonzero zerodivisors, so $f(z) = 0$, so $z \in V(\sqrt{I})$. Thus, we achieve equality.

(ii). By the previous lemma, it is clear that $\sqrt{I} \subset I(V(\sqrt{I}))$. Then, by the first part, we have that $\sqrt{I} \subset I(V(I))$. $\square$

Remark 1.3.16 — Later, we show that if k is algebraically closed and $I \subset k[X_1, \dots, X_n]$ is an ideal then $I(V(I)) = \sqrt{I}$.

Definition 1.3.17. Given $X \subset \mathbb{A}_k^n$, we define

$$A(X) := k[X_1, \dots, X_n]/I(X)$$

to be the **coordinate ring** of X .

Remark 1.3.18 — Note that any $[f] \in A(X)$ defines a function $\tilde{f}: X \rightarrow k$ defined by $x \mapsto f(x)$. It is easy to check that $\tilde{f}$ is well defined.

Definition 1.3.19. Let $X \subset \mathbb{A}_k^n$ and $Y \subset \mathbb{A}_k^m$, and let $F: \mathbb{A}_k^n \rightarrow \mathbb{A}_k^m$ be defined by $F = (f_1, \dots, f_m)$, where $f_i \in k[X_1, \dots, X_n]$ and such that $F(X) \subset Y$. The restricted function

$$\varphi := F|_X: X \rightarrow Y$$

is called a **regular mapping** or a **morphism between varieties**.

Remark 1.3.20 — We have an induced morphism of rings

$$A(Y) \rightarrow A(X), \quad [g] \mapsto [g \circ F] = [g(f_1, \dots, f_m)].$$

Example 1.3.21

Let k be a field and $a := (a_1, \dots, a_n) \in \mathbb{A}_k^n$ be a point. Then, if $X = \{a\}$, the ideal of X is

$$I(X) = ((x_1 - a_1), \dots, (x_n - a_n)).$$

Note that we showed that this is a maximal ideal, so we have that the ideal of a point is a maximal ideal in $k[X_1, \dots, X_n]$

Remark 1.3.22 — We note that the converse of the example above is not true. That is, if we have a maximal ideal in $k[X_1, \dots, X_n]$, the variety of that ideal is not necessarily a point. The following example highlights this.

Example 1.3.23

Take $k = \mathbb{R}$, and take $I = (X^2 + 1) \subset R = \mathbb{R}[X]$. We get that I is maximal, since there is no polynomial $g \in \mathbb{R}[X]$ such that $(X^2 + 1) \subsetneq (g) \subsetneq R$. However, $V(I) = \emptyset$, since there are no points which vanish on $X^2 + 1$. If we take $k = \mathbb{C}$ (or any algebraically closed field k), this is a point; in fact, in this case, $(x^2 + 1)$ is not maximal in $\mathbb{C}[X]$ since $(x^2 + 1) \subset (x + i)$. This last point will be elaborated on later.

Definition 1.3.24. Let k be a field and $I \subset k[X_1, \dots, X_n]$ be an ideal and let $V := V(I)$. Then, V is said to be **irreducible** if V cannot be written as a union $V = V_1 \cup V_2$, where $V_i \subset V$ is a proper affine subvariety for $i = 1, 2$. That is, if $V = V_1 \cup V_2$, then $V = V_1$ or $V = V_2$.

Example 1.3.25

Take $I = (X(X^2 + Y^2 - 1)) \subset k[X, Y]$. This is a product, so $V(I)$ is union of a line $V(x)$ and a circle $V(X^2 + Y^2 - 1)$. However, both the line nor the circle are equal to $V(I)$.

Lemma 1.3.26

Let k be a field and $V \subset \mathbb{A}_k^n$ be an affine variety. Then, V is irreducible if and only if $I(V)$ is a prime ideal.

Proof. Assume first that V is irreducible. Let $f, g \in k[X_1, \dots, X_n]$ be such that $f \cdot g \in I(V)$. Let $V_1 := V(f) \cap V$ and let $V_2 := V(g) \cap V$. By lemma 1.3.6, we have that V_1 and V_2 are affine varieties. We first show that

$$V_1 \cup V_2 = V.$$

Clearly, $V_1 \cup V_2 \subset V$. Now assume that $x \in V$. Since $f \cdot g \in I(V)$, we have that $(f \cdot g)(x) = 0$. Therefore, we have that $f(x) = 0$ or $g(x) = 0$, which implies that $x \in V_1$ or $x \in V_2$, so the claim follows. Since V is irreducible, it follows that $V_1 = V$ or $V_2 = V$. If $V_1 = V$, then $V \subset V(f)$, implying that $f \in I(V)$. Similarly, if $V_2 = V$, then $g \in I(V)$. Thus, $I(V)$ is a prime ideal.

It's time for the converse. Assume now that $I(V)$ is a prime ideal and suppose by contradiction that $V = V_1 \cup V_2$, where $V_i \subset V$ is a proper affine variety. Since V_1 is proper, Lemma 1.3.13 implies that $V(I(V))$ is not contained in $V(I(V_1))$. Therefore, there exist $f \in I(V_1)$ such that $f \notin I(V)$. Similarly, there exists $g \in I(V_2)$ such that $g \notin I(V)$. On the other hand, since $V = V_1 \cup V_2$, it follows that $V_1 \cup V_2 = V(I(V_1) \cdot I(V_2))$. Therefore, we have that $f \cdot g \in I(V)$, which contradicts the fact that $I(V)$ is a prime ideal. $\square$

Exercise 1.3.27. Let $\mathfrak{p} \subset \mathbb{Z}[X]$ be a prime ideal such that $\mathfrak{p} \neq (0)$.

1. Show that $\mathfrak{p} \cap \mathbb{Z}$ is a prime ideal in $\mathbb{Z}$. Thus, $\mathfrak{p} \cap \mathbb{Z} = (p)$ where $p = 0$ or p is a prime number.
2. Show that if $\mathfrak{p} \cap \mathbb{Z} = (0)$ then $\mathfrak{p} = (f)$ where $f \in \mathbb{Z}[X]$ is an irreducible polynomial.
3. Show that if $\mathfrak{p} \cap \mathbb{Z} = (p)$ where p is a prime number, then $\mathfrak{p} = (p, f)$ where $f = 0$ or $f \in \mathbb{Z}[X]$ is a monic polynomial and its reduction modulo p , $[f] \in \mathbb{Z}/p\mathbb{Z}[X]$, is irreducible.

Solution: 1. Let $a, b \in \mathbb{Z}$ such that $a \cdot b \in \mathfrak{p} \cap \mathbb{Z}$. Then, since $\mathfrak{p}$ is prime, $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. Thus, $a \in \mathfrak{p} \cap \mathbb{Z}$ or $b \in \mathfrak{p} \cap \mathbb{Z}$.

2. Let $\mathfrak{p} \cap \mathbb{Z} = (0)$. This implies that there are no integers in the prime ideal $\mathfrak{p}$, so $\mathfrak{p}$ consists of only polynomials. Now note that $\mathbb{Z}[X]$ is a UFD, since $\mathbb{Z}$ is a UFD (apply Gauss lemma). Suppose f is in $\mathfrak{p}$. Then, by unique factorization, if f were reducible,

then $f = gh$, and so $g \in \mathfrak{p}$ or $h \in \mathfrak{p}$; this is unique up to units. Continuing in this fashion, given any polynomial, we can *trace back* to any irreducible polynomial p , where there exists $q \in \mathbb{Z}[X]$ such that $pq = f$. This tells us that irreducible polynomials exist in $\mathfrak{p}$. Now suppose that there were two irreducible polynomials $f, g \in \mathfrak{p}$ and suppose $\deg(f) \geq \deg(g)$. Then, considering these to be polynomials in $\mathbb{Q}[X]$, since $\mathbb{Q}[X]$ is a Euclidean domain, we know that there exists a polynomial such that $a \in \mathbb{Q}[X]$ such that

$$f = ag + r,$$

where $\deg(r) < \deg(g)$. We can repeat this algorithm until $\deg(r)$. If $r = 0$, then we get a polynomial $a \in \mathbb{Q}[X]$ such that $f = ag$. If $r \neq 0$ (and is a constant of course), then we can find polynomials $a, b \in \mathbb{Q}[X]$ such that $af + bg = r$. In either case, we can multiply through by the denominator to get a relation with coefficients purely in $\mathbb{Z}$. In the first case, we get that f is actually not irreducible, which is a contradiction. In the second, it follows that $\mathfrak{p} \cap \mathbb{Z} \neq (0)$, since r multiplied the common denominator is in $\mathbb{Z}$ and $\mathfrak{p}$. Therefore, it must be that there is only one irreducible polynomial in $\mathfrak{p}$. Therefore, given any polynomial in $\mathfrak{p}$, it can be written with f being in the product, i.e. $\mathfrak{p} = (f)$.

3. We have $\mathfrak{p} \cap \mathbb{Z} = (p)$. If $\mathfrak{p} = (p)$, we're done. Thus, assume

$$(p) \subsetneq \mathfrak{p} \subsetneq \mathbb{Z}[X].$$

It is clear that

$$\mathbb{Z}[X]/(p) \xrightarrow{\sim} \mathbb{Z}/p\mathbb{Z}[X].$$

Under this map $\mathfrak{p} \mapsto \bar{\mathfrak{p}} = \{x \bmod (p) : x \in \mathfrak{p}\}$. We note that this is a nonzero ideal of $\mathbb{Z}/p\mathbb{Z}[X]$. $\mathbb{Z}/p\mathbb{Z}[X]$ is a Euclidean domain, so in particular a PID; thus, $\bar{\mathfrak{p}} = (\bar{f})$, and we can assume this to be monic since we can just multiply by the inverse of the leading coefficient if not. It is also irreducible because if not, we can just replace by an irreducible factor. Now let f be a lift of $\bar{f}$ in $\mathbb{Z}[X]$. We note that f is irreducible, because if it weren't then $\bar{f}$ is reducible. We now want to show that $\mathfrak{p} = (p, f)$. Consider the map

$$\varphi: \mathbb{Z}[X]/(p, f) \rightarrow (\mathbb{Z}/p\mathbb{Z}[X])/(\bar{f}).$$

We claim that this is an isomorphism. Let $g \in \mathbb{Z}[X]$ such that $\varphi(g) \in (\bar{f})$. Then, $\varphi(g) = \bar{f} \cdot \bar{h}$, so $g - fh = p \cdot k$, for some integer k . It is clear that $g \in (p, f)$ then, so we have injectivity. For surjectivity, we can just take arbitrary lifts, so this is clear. Thus, this is an isomorphism. $\bar{f}$ is irreducible, so the ideal $(\bar{f})$ is maximal (since $\mathbb{Z}/p\mathbb{Z}[X]$ is a PID), so the RHS is a field. Therefore, $\mathbb{Z}[X]/(p, f)$ is also a field, so (p, f) is maximal. Since $(p, f) \subset \mathfrak{p}$, we get equality then. $\square$

§1.4 Krull's Lemma

Proposition 1.4.1 (Krull's Lemma)

Let R be a commutative ring and let $S \subset R$ be a multiplicative set. Let $I \subset R$ be an ideal which is disjoint from S . Then, there exists a prime ideal $\mathfrak{p} \subset R$ containing I and such that $\mathfrak{p}$ is disjoint from S .

Proof. Let P be the set of ideals in R containing I and disjoint from S . Then, we can order the elements of P by inclusion. As in the proof of Proposition 1.2.12, we have that any totally ordered subset T of P admits an upper bound. Therefore, by Zorn's Lemma

(Theorem 1.2.10), there exists a maximal element $\mathfrak{p}$ in P . Particularly, $\mathfrak{p}$ is disjoint from S . We need to show that $\mathfrak{p}$ is prime in R , so let $x, y \in R$ such that $xy \in \mathfrak{p}$, and assume for the sake of contradiction that $x, y \notin \mathfrak{p}$. Then, $\mathfrak{p}$ is strictly included in $\mathfrak{p} + (x)$ and since $\mathfrak{p}$ is maximal in P , it follows that $\mathfrak{p} + (x)$ intersects S . Similarly, we have that $\mathfrak{p} + (y)$ intersects S , and since S is multiplicative, we have that $(\mathfrak{p} + (x))(\mathfrak{p} + (y))$ also intersects S . On the other hand, we have that

$$(\mathfrak{p} + (x))(\mathfrak{p} + (y)) \subset \mathfrak{p} + (xy) = \mathfrak{p}.$$

Therefore, $\mathfrak{p}$ intersects S , which is a contradiction. $\square$

§1.4.i The Nilradical and Jacobson's Radical

Proposition 1.4.2

Let R be a commutative ring and $I \subset R$ be an ideal. Then,

$$\sqrt{I} = \bigcap_{I \subset \mathfrak{p}} \mathfrak{p},$$

where the intersection runs over all the prime ideals $\mathfrak{p}$ containing I .

Proof. If $\mathfrak{p}$ is a prime ideal containing I and $x \in R$ such that $x^n \in I$ for some positive integer n , then $x^n \in \mathfrak{p}$. Since $\mathfrak{p}$ is prime, by induction on n , it follows that $x \in \mathfrak{p}$. Therefore, $\sqrt{I} \subset \mathfrak{p}$. For the converse, assume that $x \notin \sqrt{I}$. Then, we define $S := \{1, x, x^2, \dots\}$, and it is easy to see that S is a multiplicative set which is disjoint from I . By Proposition 1.4.1, there exists a prime ideal $\mathfrak{p}$ of R containing I and which is disjoint from S . Therefore, $x \notin \mathfrak{p}$. $\square$

Corollary 1.4.3

Taking a commutative ring R , then $\sqrt{(0)}$ is equal to the intersection of all prime ideals of R .

Definition 1.4.4. Let R be a commutative ring. Then, we define:

- the **nilradical** $\mathcal{N}(R)$ of R is the radical of the ideal (0) , i.e. $\mathcal{N}(R)$ is the set of all the nilpotent elements in R .
- the **Jacobson radical** $\mathcal{J}(R)$ of R is the intersection of all the maximal ideals of R .

Remark 1.4.5 — Note that it follows the Corollary of Proposition 1.4.2 that the nilradical of a commutative ring R is the intersection of all the prime ideals of R .

Exercise 1.4.6. Let R be a commutative ring. Show that the following are equivalent:

1. R has exactly one prime ideal.
2. every element of R is either a unit or nilpotent.
3. $\sqrt{(0)}$ is a maximal ideal.

Solution: (1 $\implies$ 2): Let $\mathfrak{p} \subset R$ be the unique prime ideal. Then, by the previous proposition, $\sqrt{(0)} = \mathfrak{p}$. Now, suppose that x is not a unit. Then, we know that the principal ideal $(x) \subsetneq R$. By Zorn's lemma, (x) is contained in a maximal ideal $\mathfrak{m}$, but $\mathfrak{m}$ is prime so (x) is actually contained in $\mathfrak{m} = \mathfrak{p} = \sqrt{(0)}$. Then, it follows that $x \in \sqrt{(0)}$, so it's nilpotent.

(2 $\implies$ 3): Let $\sqrt{(0)} \subset \mathfrak{m}$ be a maximal ideal, and suppose that there exists an element $x \in \mathfrak{m} \setminus \sqrt{(0)}$. $x \notin \sqrt{(0)}$, so it's not nilpotent; thus, it's a unit. However, $\mathfrak{m}$ is maximal so a proper ideal, which is a contradiction.

(3 $\implies$ 1): Let $\mathfrak{p}$ be any prime ideal; since (0) is contained in $\mathfrak{p}$, it follows that $\sqrt{(0)} \subset \mathfrak{p}$. However, $\sqrt{(0)}$ is maximal, so in fact $\sqrt{(0)} = \mathfrak{p}$. It follows that $\sqrt{(0)}$ is the only prime ideal in R . $\square$

Exercise 1.4.7. Let R be a commutative ring. Show that $x \in \mathcal{J}(R)$ if and only if $1 - xy$ is a unit in R for all $y \in R$.

Solution: Let $x \in \mathcal{J}(R)$, so that x is in every single maximal ideal of R . Suppose $1 - xy$ is not a unit. Then, $1 - xy$ is contained in some maximal ideal $\mathfrak{m}$. Then, since $x \in \mathfrak{m}$, it follows that $1 \in \mathfrak{m}$, but this contradicts the maximality of $\mathfrak{m}$.

For the converse direction now. Suppose $x \notin \mathcal{J}(R)$. Then, there exists a maximal ideal $\mathfrak{m}$ such that $x \notin \mathfrak{m}$. Then, $\mathfrak{m} + (x) = R$ by maximality of $\mathfrak{m}$. Thus, there exists $y \in R, m \in \mathfrak{m}$ such that $m + yx = 1$. But then, m is a unit, is a contradiction. $\square$

Exercise 1.4.8. Let A be a ring and let $A[x]$ be the ring of polynomials in an indeterminate x , with coefficients in A . Let

$$f = a_0 + a_1x + \cdots + a_nx^n \in A[x].$$

Prove that

1. f is a unit in $A[x] \iff a_0$ is a unit in A and $a_1, \dots, a_n$ are nilpotent. (If $b_0 + b_1x + \cdots + b_mx^m$ is the inverse of f , prove by induction on r that $a_n^{r+1}b_{m-r} = 0$. Hence show that a_n is nilpotent)
2. f is nilpotent $\iff a_0, a_1, \dots, a_n$ are nilpotent.
3. f is a zero-divisor $\iff$ there exists $a \neq 0$ in A such that $af = 0$. (Choose a polynomial $g = b_0 + b_1x + \cdots + b_mx^m$ of least degree m such that $fg = 0$. Then $a_nb_m = 0$, hence $a_ng = 0$ (because a_ng annihilates f and has degree $< m$). Now show by induction that $a_{n-r}g = 0$ ($0 \leq r \leq n$).)
4. f is said to be primitive if $(a_0, a_1, \dots, a_n) = (1)$. Prove that if $f, g \in A[x]$, then fg is primitive $\iff f$ and g are primitive.

Solution: 1. Suppose there exists some $g \in A[x]$ such that $fg = 1$. Write $g = b_0 + \cdots + b_mx^m$. Then, it is clear that $a_0b_0 = 1$, so a_0 is a unit. To show that $a_1, \dots, a_n$ are nilpotent, we shall show that $a_n^{r+1}b_{m-r} = 0$. First, for $r = 0$, this is clear because this is the highest coefficient of fg . Now, suppose this is true up to some $r = k - 1$. Focus on the coefficient of x^{n+m-k} , which is

$$\sum_{i+j=n+m-k} a_ib_j.$$

This must equal 0, since $fg = 1$. Now, consider

$$a_n^k \cdot \left(\sum_{i+j=n+m-k} a_i b_j \right) = a_n^k (a_n b_{m-k} + a_{n-1} b_{m-k+1} + \cdots + a_{n-k} b_m).$$

Then, all terms except the first in the bracket disappear due to the inductive hypothesis. Therefore,

$$a_n^{k+1} b_{m-k} = 0.$$

Now, b_0 is a unit, so let $r = m$. This tells us that $a_n^{m+1} b_0 = 0$, but this implies that $a_n^{m+1} = 0$.

Now for the converse. First, note that if the $a_1, \dots, a_n$ are nilpotent, then $a_i x^i$ are nilpotent for $1 \leq i \leq n$. Then,

$$N(x) = \sum_{i=1}^n a_i x^i$$

is nilpotent. Then, $f(x) = a_0 + N(x)$, which is a sum of a unit and a nilpotent element. Therefore, $f(x)$ is a unit, by Exercise 1.1.28.

2. Suppose f is nilpotent. Then, xf is nilpotent. Then, $1 + xf$ is a unit, so $a_0, \dots, a_n$ are nilpotent; alternatively, induct on the degree of f , and note that a_n is nilpotent by focus on the highest term. Then, $f(x) - a_n x^n$ is also nilpotent. The converse is clear. Each $a_i x^i$ is nilpotent, and a sum of nilpotents is nilpotent.

3. If there exists some nonzero $a \in A$ such that $af = 0$, then f is clearly a zero divisor. Thus, we prove the converse. Suppose that f is a zero divisor, and let $g \in A[x]$ be of minimal degree such that $fg = 0$. Write $f(x) = \sum_{i=0}^n a_i x^i$, $g(x) = \sum_{j=0}^m b_j x^j$. We shall show by induction that $a_{n-r}g = 0$, for all $0 \leq r \leq n$. For the base case, we find that $a_n b_m = 0$, so $a_n g$ is of degree less than g and $a_n g f = 0$, so $a_n g = 0$ by minimality of the degree of g . Now assume that the result is true for up to a_{k-1} . We again want to show that $a_k g$ is of degree less than g , so that $a_k g f = 0$, giving $a_k g = 0$. For this, we want to show that $a_k b_m = 0$. For this, focus on the $k+m$ -th index of fg :

$$0 = \sum_{i+j=k+m} a_i b_j = a_k b_m + a_{k+1} b_{m-1} + \cdots$$

Note that the sum above is finite, since we are summing over i, j positive. The inductive hypothesis then says that $a_\ell g = 0$ for $\ell \leq k-1$, so in particular, $a_\ell b_p = 0$ for all p . Therefore, the sum above becomes $0 = a_k b_m$. $a_k g = 0$ by the argument stated at the start of the induction. In particular, we find from this that $a_{n-r}g = 0$, for all $0 \leq r \leq n$. Now, choose one b_ℓ . Since $a_{n-r}g = 0$ for all r , it follows that $b_\ell f = 0$. Thus, we've found an element in A such that $af = 0$.

4. Let $\mathbf{a} = (a_0, \dots, a_n)$ be the coefficients of f and $\mathbf{b} = (b_0, \dots, b_m)$ for g . Let $\mathbf{c} = (\sum_{i+j=k} a_i b_j)_{0 \leq k \leq m+n}$ be the ideal generated by the coefficients of the products. We want to show that $\mathbf{a} = \mathbf{b} = (1) \iff \mathbf{c} = (1)$. We note that $\mathbf{c} \subseteq \mathbf{a} \cap \mathbf{b}$, so if $\mathbf{c} = (1)$, then it is immediate $\mathbf{a}, \mathbf{b} = (1)$ because these both contain $\mathbf{a} \cap \mathbf{b}$. Conversely, suppose that $\mathbf{a}, \mathbf{b} = (1)$ and $\mathbf{c} \neq (1)$. Then, there is a maximal ideal $\mathbf{m} \subseteq R$ containing $\mathbf{c}$. Then, fg maps to zero under the quotient $(R/\mathbf{m})[x]$. This is because $\mathbf{m}$ contains all coefficients of the product. However, $R/\mathbf{m}$ is a field, so an integral domain, so either f or g must be contained in $\mathbf{m}[x]$. However, if this is the case, then the coefficients of f and g are in $\mathbf{m}$, so there is no way that $\mathbf{a}$ and $\mathbf{b}$ are (1) , a contradiction. $\square$

Remark 1.4.9 — Having seen 4, there is an abstract method for 1, 2, 3, which relies on the fact that if R is an integral domain, then $R[X]$ is an integral domain and the units of $R[X]$ are those in R . The units of $R[X]$ are precisely those in R for when R is an integral domain, so the first becomes easy. As for the second, we find that $\bar{f}^n = 0$ in $(R/\mathfrak{p})[X]$, so since $R/\mathfrak{p}$ is an integral domain, $\bar{f} = 0$ in the quotient. Thus, all coefficients are in $\mathfrak{p}$.

Exercise 1.4.10. In the ring $R[x]$, show that the Jacobson radical is equal to the nilradical.

Solution: First, note that the nilradical is always a subset of the Jacobson radical, so we want to show the inclusion $\mathcal{J}(R[x]) \subseteq \mathcal{N}(R[x])$. Now, $f \in \mathcal{J}(R[x])$ implies that $1 - fg$ is a unit for all $g \in R[x]$. Thus, $1 + xf$ is a unit, so $a_0, a_1, \dots, a_n$ is nilpotent. Then, it is clear that $f(x)$ is nilpotent, since we can just take an arbitrarily large power of f which kills off everything. Therefore, $f \in \mathcal{N}(R[x])$. $\square$

§1.5 Local Rings

Definition 1.5.1. Let R be a commutative ring. Then, R is a **local ring** if has a unique maximal ideal $\mathfrak{m}$. The field $k := R/\mathfrak{m}$ is called the **residue field** of R .

Notation 1.5.2. A local ring R with maximal ideal $\mathfrak{m}$ is often denoted as $(R, \mathfrak{m})$.

Remark 1.5.3 — Particularly, if $(R, \mathfrak{m})$, then $\mathfrak{m} = \mathcal{J}(R)$.

Proposition 1.5.4

Let R be a local ring. Then, the units in R are exactly the elements in $R \setminus \mathfrak{m}$.

Proof. This follows from Proposition 1.2.12. $\square$

Proposition 1.5.5

Let R be a ring and $\mathfrak{m} \neq (1)$ an ideal of R such that every $x \in R \setminus \mathfrak{m}$ is a unit in R . Then, R is a local ring and $\mathfrak{m}$ is its maximal ideal.

Proof. First, $\mathfrak{m}$ is maximal. Considering an ideal $I \supset \mathfrak{m}$, if $x \in I \setminus \mathfrak{m}$, then it is a unit, so $I = (1)$. To show that $\mathfrak{m}$ is the unique maximal ideal, suppose we have another maximal ideal, say $\mathfrak{m}'$. Then, since $\mathfrak{m}' \neq \mathfrak{m}$, there exists some $x \in \mathfrak{m}' \setminus \mathfrak{m}$, but then x is a unit. Thus, $\mathfrak{m}'$ is not a maximal ideal. Thus, $\mathfrak{m}$ is unique, so R is a local ring. $\square$

Exercise 1.5.6. Let R be a ring and $\mathfrak{m}$ a maximal ideal of R , such that every element of $1 + \mathfrak{m}$ is a unit in R . Then, R is a local ring.

Solution: Let $x \in R \setminus \mathfrak{m}$. By the maximality of $\mathfrak{m}$, the ideal $(x, \mathfrak{m}) = (1)$. Thus, there exists $y \in R$ and $t \in \mathfrak{m}$ such that $xy + t = 1$. Thus, $xy = 1 - t$ belongs to $1 + \mathfrak{m}$, so is a unit by assumption. Then, we can apply the previous exercise. $\square$

Example 1.5.7

If R is a field, then R is a local ring with maximal ideal $\mathfrak{m} = (0)$.

Proposition 1.5.8

Let R be an integral domain, and $K = \text{Frac}(R)$, and $\mathfrak{p} \subset R$ be a prime ideal. Then,

$$R_{\mathfrak{p}} := \left\{ \frac{f}{g} \in K : f, g \in R, g \notin \mathfrak{p} \right\}$$

is a local ring with maximal ideal

$$\mathfrak{m}_{\mathfrak{p}} := \left\{ \frac{f}{g} \in K : f, g \in R, f \in \mathfrak{p}, g \notin \mathfrak{p} \right\}.$$

$(R_{\mathfrak{p}}, \mathfrak{m}_{\mathfrak{p}})$ is called the **localization** of R at $\mathfrak{p}$.

Proof. This follows from the Proposition 1.5.5. Every element of $R_{\mathfrak{p}} \setminus \mathfrak{m}_{\mathfrak{p}}$ is a unit. $\square$

Remark 1.5.9 — Note that if $S = R \setminus \mathfrak{p}$, then S is multiplicative and $R_{\mathfrak{p}} = S^{-1}R$.

Example 1.5.10

Let $p \in \mathbb{Z}$ be prime. Then,

$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \in \mathbb{Q} : a, b \in \mathbb{Z}, p \nmid b \right\}.$$

Note that this is different from the ring

$$\mathbb{Z} \left[\frac{1}{p} \right] = \left\{ \frac{a}{p^k} : a \in \mathbb{Z}, k \geq 0 \right\}.$$

Definition 1.5.11. Let k be a field. The **formal power series ring** is the ring

$$k[[X]] = \left\{ \sum_{i=0}^{\infty} a_i x^i : a_i \in k \right\}.$$

The elements of $k[[X]]$ are called formal power series with coefficients in k .

Remark 1.5.12 — Note that we don't impose any condition on the convergence of such series.

Example 1.5.13 (Formal Power Series Ring)

Let k be a field, and consider $R = k[[X]]$. We claim that R is a local ring with maximal ideal $\mathfrak{m} = (x)$. By Proposition 1.5.5, it's sufficient to show that any element in $R \setminus \mathfrak{m}$ is invertible. Let

$$f = \sum_{i=0}^{\infty} a_i x^i \in R \setminus \mathfrak{m}, \quad a_0, a_1, \dots \in k.$$

Since $f \notin \mathfrak{m}$, it follows that $a_0 \neq 0$. Thus, we can write

$$f = a_0(1 + xg),$$

where

$$g = \sum_{i=0}^{\infty} a_{i+1} a_0^{-1} x^i \in R.$$

Then, if we define

$$f^{-1} := a_0^{-1}(1 - xg + x^2g^2 - \dots),$$

it is easy to check that $f^{-1} \in R$ is the inverse of f in R . Therefore, our claim follows. Note that if f is a polynomial, then f^{-1} is the Taylor series of the function $\frac{1}{f}$ at 0.

§1.6 Modules

Definition 1.6.1. Let R be a commutative ring. A **R -module** is an abelian group M with a multiplication map $\cdot : R \times M \rightarrow M$ such that for any $r, s \in R$ and $m, n \in M$, we have:

1. $r \cdot (m \pm n) = r \cdot m \pm r \cdot n$
2. $(r + s) \cdot m = r \cdot m + s \cdot m$
3. $(rs) \cdot m = r \cdot (s \cdot m)$
4. $1 \cdot m = m$

Remark 1.6.2 — Note that every abelian group is a $\mathbb{Z}$ -module and if R is a field, then an R -module is just a vector space over R .

Example 1.6.3

Let R be a ring and $M = R[X]$ be the ring of polynomials in one variable. Then, M is a R -module.

Definition 1.6.4. Let R be a commutative ring and M be a R -module. Then, M is **finitely generated** if there exist $m_1, \dots, m_n \in M$ such that for all $m \in M$, there exist $r_1, \dots, r_n \in R$ such that

$$m = \sum_{i=1}^n r_i m_i.$$

The elements $m_1, \dots, m_n$ are called the **generators** of M over R .

Example 1.6.5

If R is a commutative ring and $I \subset R$ is an ideal, then I is a R -module. Particularly, I is finitely generated if and only if there exist $m_1, \dots, m_n \in I$ such that $I = (m_1, \dots, m_n)$.

Definition 1.6.6. Let R be a commutative ring and M be a R -module. A subset $N \subset M$ is called a **submodule** if $r \cdot m + s \cdot n \in N$ for all $m, n \in N$ and $r, s \in R$.

Exercise 1.6.7. Show that if M is a module over a commutative ring R and $N \subset M$ is a submodule, then the quotient M/N is also a module over R .

Solution: We just need to define an action. Let $m + N \in M/N$. Then, we define

$$r \cdot (m + N) := r \cdot m + N.$$

This satisfies all the desired properties. □

Example 1.6.8

Let R be a commutative ring and M be a R -module. If $I \subset R$ is an ideal, then

$$I \cdot M = \left\{ \sum_{i=1}^k r_i \cdot m_i : r_1, \dots, r_k \in I, m_1, \dots, m_k \in M \right\}$$

is a submodule of M .

Remark 1.6.9 — A submodule of finitely generated module does not need to be finitely generated; this is different from vector spaces.

Example 1.6.10

Let k be a field and let $R = k[X_1, X_2, \dots]$ be the ring of polynomials in infinitely many variables over k . Then, $M := R$ is finitely generated as a module over itself by the element $1 \in R$. On the other hand, the ideal $I = (X_1, X_2, \dots) \subset R$ defines a submodule of M which is not finitely generated over R . Indeed, assume that $f_1, \dots, f_k \in I$ generates I . Then, each f_i involves only finitely many variables and therefore exists $q \geq 1$ such that $X_q \in I$ is not generated by $f_1, \dots, f_k$.

Definition 1.6.11. Let R be a commutative ring and M, N be R -modules. Then, $f: M \rightarrow N$ is **R -linear** (or a homomorphism of R -modules) if

$$f(r \cdot m + s \cdot n) = r \cdot f(m) + s \cdot f(n),$$

for all $r, s \in R$ and $m, n \in M$. The morphism f is called an **isomorphism** if it is a bijection.

Definition 1.6.12. Let R be a commutative ring, and M be an R -module. We define the **annihilator** of M to be

$$\text{ann}(M) := \{r \in R : rm = 0, \forall m \in M\} \subset R.$$

Exercise 1.6.13. Let R be a commutative ring and M be an R -module.

1. Show that $\text{ann}(M) \subset R$ is an ideal.
2. Let $I \subset R$ be an ideal. Show that $I = \text{ann}(R/I)$.
3. Assume that M is generated by a single element $x \in M$ over R . Show that M is isomorphic to $R/\text{ann}(M)$.

Solution: 1. This is pretty straightforward.

2. This is pretty straightforward too, by considering containment from both ends.

3. Let $M = R \cdot x$. Then consider the map $\varphi: R \rightarrow M$ given by $r \mapsto rx$. This is surjective, since $M = R \cdot x$. The kernel is given by $\text{ann}(M)$, since the kernel is the set elements of R such that $rx = 0$. Note that $\text{ann}(M)$ precisely looks like $\{s \in R: s(rx) = 0 \forall s, r \in R\}$, but $sr \in R$, so we actually have $\text{ann}(M) = \{r \in R: rx = 0\}$. $\square$

Definition 1.6.14. Let R be a commutative ring and M be a module over R . Then, M is **simple** if the only submodules of M are $\{0\}$ and M .

Proposition 1.6.15

Let R be a commutative ring and M be a module over R . Then, M is simple if and only if M is isomorphic to $R/\mathfrak{m}$ for some maximal ideal $\mathfrak{m} \subset R$.

Proof. If $\mathfrak{m} \subset R$ is a maximal ideal, then $R/\mathfrak{m}$ is a field, so it can't have non-trivial submodules; thus, $R/\mathfrak{m}$ is a simple module.

Assume now that M is simple; let $v \in M$ be nonzero. Then,

$$R \cdot v = \{r \cdot v: r \in R\} \subset M$$

is a nonzero submodule of M . Therefore, $R \cdot v = M$ since M is simple; thus, v is a generator of M . Then, the previous exercise tells us that if $I = \text{ann}(M)$, then $M \cong R/I$. Particularly, R/I is a simple module over R . Therefore, by Proposition 1.2.12, there exists a maximal ideal $\mathfrak{m} \subset R$ which contains I . Then, $\mathfrak{m}/I$ is a submodule of R/I and since R/I is simple, we must have that $I = \mathfrak{m}$. Therefore, $M \cong R/\mathfrak{m}$. $\square$

Exercise 1.6.16. Let R be a commutative ring, and M be an R -module. Then, the following are equivalent.

1. $M = 0$
2. $M_{\mathfrak{m}} = 0$, for all maximal ideals $\mathfrak{m}$ of R .
3. $M_{\mathfrak{p}} = 0$, for all prime ideals $\mathfrak{p}$ of R .

Solution: The implication $1 \implies 2$ and 3 is clear, and the implication $3 \implies 2$ is clear, since all maximal ideals are prime. Thus, if we can show that $2 \implies 1$, then we are done. Let $M_{\mathfrak{m}} = 0$, for all maximal ideals $\mathfrak{m}$ of R . Let $m \in M$. Then, $\text{ann}(m) \subsetneq R$. Note that this is a proper ideal because $1 \cdot m \neq 0$. Thus, $\text{ann}(m) \subseteq \mathfrak{m}$, some maximal ideal $\mathfrak{m}$ of R . Therefore, if we consider the localization at $\mathfrak{m}$, we get that $M_{\mathfrak{m}} = 0$. Thus, in the localization, $\frac{m}{1} = \frac{0}{1}$, so there exists some nonzero $u \in R \setminus \mathfrak{m}$ such that $um = 0$. Thus, $u \in \text{ann}(m)$. However, this contradicts that $\text{ann}(m) \subseteq \mathfrak{m}$. Therefore, $M = 0$. $\square$

§1.7 More on Localization

Here, we discuss a bit more about localization; this is mostly taken from [1]. This part is not a part of the official notes, and I have independently decided to include this. Let R be a ring and S be a multiplicative subset. First, we study the ideals in R and those in $S^{-1}R$. We call the unit ideal $I = R$ 'trivial'. For an ideal $I \subseteq R$ we define the ideal

$$\text{res}(I) \subset S^{-1}R$$

also denoted $S^{-1}I$, to consist of elements of the form

$$\frac{a}{s}, \quad a \in I.$$

It is easy to check that this is indeed an ideal.

Exercise 1.7.1. Show that $S^{-1}I$ is trivial if and only if $I \cap S \neq \emptyset$.

Solution: If $S^{-1}I$ is trivial, there exists some $a \in I$ such that $\frac{a}{s} = \frac{1}{1}$. Thus, there exists $t \in S$ such that $ta = ts$. This implies that $ta \in S$ by multiplicativity. Further, $ta \in I$ because $a \in I$ and I is an ideal. Thus, true. For the converse, suppose $I \cap S \neq \emptyset$. Then, let $a \in I \cap S$. Then, $\frac{a}{a} = \frac{1}{1} \in S^{-1}I$ because there exists $t \in S$ such that $ta = ta$; namely, $t = 1$ works. $\square$

Exercise 1.7.2. Show that the quotient ring $S^{-1}R/S^{-1}I$ is the localization of R/I with respect to $S' \subset R/I$, where S' is the image of S under $R \rightarrow R/I$. Therefore, $S^{-1}R/S^{-1}I \cong (S')^{-1}(R/I)$. By abuse of notation, we may just write $S^{-1}R/S^{-1}I \cong S^{-1}(R/I)$.

Solution: We want to show that $S^{-1}R/S^{-1}I \cong (S')^{-1}(R/I)$. We consider the map $\phi: S^{-1}R/S^{-1}I \rightarrow (S')^{-1}(R/I)$ which maps

$$\frac{r}{s} + S^{-1}I \mapsto \frac{r + I}{s + I}.$$

We verify that this is a morphism of rings. This clearly respects multiplication, so we just check addition:

$$\phi\left(\frac{r}{s} + \frac{r'}{s'} + S^{-1}I\right) = \frac{rs' + r's + I}{ss' + I} = \frac{r + I}{s + I} + \frac{r' + I}{s' + I} = \phi\left(\frac{r}{s} + S^{-1}I\right) + \phi\left(\frac{r'}{s'} + S^{-1}I\right)$$

There's the inverse to it $\psi: (S')^{-1}(R/I) \rightarrow S^{-1}R/S^{-1}I$ defined by

$$\frac{r + I}{s + I} \mapsto \frac{r}{s} + S^{-1}I.$$

This is also an map of rings. It's evident that these are inverses, so we are done. $\square$

Exercise 1.7.3.

1. Show that if $\mathfrak{p} \subset R$ is prime, and $\mathfrak{p} \cap S = \emptyset$, then $S^{-1}\mathfrak{p}$ is also prime.
2. Show that if $\mathfrak{m} \subset R$ is maximal then $S^{-1}\mathfrak{m}$ is also maximal, and in this case the map $R/\mathfrak{m} \rightarrow S^{-1}R/S^{-1}\mathfrak{m}$ is an isomorphism.

Solution: 1. Let $\frac{x}{s} \cdot \frac{y}{t} \in S^{-1}\mathfrak{p}$. Then, $xy \notin S$, so $xy \in \mathfrak{p}$ and so x or $y \in \mathfrak{p}$. As s, t were assumed to be in S in the first place, so $\frac{x}{s}$ or $\frac{y}{t}$ is in $S^{-1}\mathfrak{p}$.

2. Note that we want to show that $S^{-1}R/S^{-1}\mathfrak{m}$ is a field. Then, $(S')^{-1}(R/\mathfrak{m}) \cong S^{-1}R/S^{-1}\mathfrak{m}$, where S' is the image of S through $R \rightarrow R/I$. However, $R/\mathfrak{m}$ is a field, and localizing by S' doesn't change $R/\mathfrak{m}$, so $(S')^{-1}(R/\mathfrak{m}) = R/\mathfrak{m}$. Thus, the quotient $S^{-1}R/S^{-1}\mathfrak{m}$ is a field. $\square$

For an ideal $J \subset S^{-1}R$ we define the ideal

$$\text{cl}(J) \subset A$$

as the preimage of J under the homomorphism $A \rightarrow S^{-1}A$. Clearly, if J is non-trivial, then

$$\text{cl}(J) \cap S = \emptyset,$$

for otherwise J would contain an invertible element. Therefore, this map is well-defined.

Exercise 1.7.4.

1. Show that for an ideal $J \subset S^{-1}R$, $\text{res}(\text{cl}(J)) = J$.
2. Show that for an ideal $I \subset R$, $I \subset \text{cl}(\text{res}(I)) = \{r \in R : \exists s \in S \text{ such that } rs \in I\}$.
3. Give an example for $I \neq \text{cl}(\text{res}(I))$.

Solution: 1. This is immediate from the definition of res and cl .

2. Let $x \in I$. Then, $\text{res}(x) = \frac{x}{1}$. Then, $x \in \text{cl}(\text{res}(I))$. To show the equality on the right, we first have that if $r \in \text{cl}(\text{res}(I))$, then x is the preimage of some $\frac{r}{s} \in S^{-1}I$ under the map $R \rightarrow S^{-1}R$. Therefore, $\frac{x}{1} = \frac{r}{s}$, so there exists some $t \in S$ such that $t(sx - r) = 0$. Then, $xst = tr \in I$, so we have that x is in the right. Conversely, let r be in the set on the right. Then, $rs \in I$. Therefore, $rs = x$, for some $x \in I$. Then, $rs - x = 0$, so $\frac{r}{1} = \frac{x}{s}$ in $S^{-1}I$. Therefore, r is the preimage of some element in $S^{-1}I$.

3. As an example of $I \neq \text{cl}(\text{res}(I))$, take $R = \mathbb{Z}$, and $S = \{1, 2, 4, \dots\}$, $I = 2\mathbb{Z}$. Then, $S^{-1}R = \mathbb{Z}[\frac{1}{2}]$. Since 2 is now a unit, $S^{-1}I = S^{-1}R$. Thus, $I \neq \text{cl}(\text{res}(I)) = R$. $\square$

Remark 1.7.5 — We note that if we have a morphism of rings, then the preimage of a prime ideal is always prime. However, preimages of maximal ideals need not be maximal. Therefore, cl maps prime ideals in $S^{-1}R$ to prime ideals in R , but maximal ideals in $S^{-1}R$ can go to a non-maximal prime ideal in R .

Proposition 1.7.6

If $\mathfrak{p} \subset A$ is a prime ideal such that $S \cap \mathfrak{p} = \emptyset$, then $\mathfrak{p} = \text{cl}(\text{res}(\mathfrak{p}))$.

Proof. If $x \in \text{cl}(\text{res}(\mathfrak{p}))$, then $xa \in \mathfrak{p}$, for some $a \in S$. $\mathfrak{p}$ is prime and $S \cap \mathfrak{p} = \emptyset$, so $x \in \mathfrak{p}$. Thus, we achieve equality. $\square$

Exercise 1.7.7. Let $f \in A$ and $S := \{1, f, f^2, \dots\}$. Then, $A_f := S^{-1}A$ is isomorphic to $A[X]/(Xf - 1)$.

Solution: Consider the map $\phi: A[X] \rightarrow A_f$ by $a \mapsto \frac{a}{1}$ and $X \mapsto \frac{1}{f}$. This is a morphism of rings, and is also well defined, as one can check. This is also clearly surjective with kernel $(Xf - 1)$, so we are done. $\square$

Exercise 1.7.8. Let R be a ring. Prove that $R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}} \cong \text{Frac}(R/\mathfrak{p})$, where $\mathfrak{p} \subseteq R$ is a prime.

Solution: This comes from Exercise 1.7.2. Let $S = R \setminus \mathfrak{p}$. Then, we know that $R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}} \cong S^{-1}R/S^{-1}\mathfrak{p} \cong S^{-1}(R/\mathfrak{p})$. However, in the quotient, S corresponds to $(R/\mathfrak{p}) \setminus \{0\}$, so the RHS is the fraction field of $R/\mathfrak{p}$. $\square$

§1.8 Exercise from [2]

Exercise 1.8.1. A ring A is such that every ideal not contained in the nilradical contains a non-zero idempotent, that is, an element e such that $e^2 = e \neq 0$. Prove that the nilradical and Jacobson radical of A are equal.

Solution: Suppose for the sake of contradiction, that $\mathcal{N}(A) \subsetneq \mathcal{J}(A)$. Then, by hypothesis, there is a nonzero idempotent $e \in \mathcal{J}(A)$. Then, $1 - e$ is a unit, and we have that $(1 - e)^2 = 1 - 2e + e^2 = 1 - e$. Now, we can find that $e(1 - e) = e - e^2 = 0$, which implies that e is a unit, since we can multiply the LHS with the inverse of $1 - e$. This is a contradiction. Therefore, $\mathcal{J}(A) = \mathcal{N}(A)$. $\square$

Exercise 1.8.2. Let R be a ring in which every element x satisfies $x^n = x$ for some $n > 1$ depending on x . Show that every prime ideal in R is maximal.

Solution: Let $\mathfrak{p}$ be a prime. Then, we want to show that $R/\mathfrak{p}$ is a field. Let $x \in R \setminus \mathfrak{p}$. Then, under its quotient, we see that $\bar{x}^n = \bar{x} \in R/\mathfrak{p}$ is not zero. Now $R/\mathfrak{p}$ is an integral domain. Therefore,

$$\bar{x}(\bar{x}^{n-1} - 1) = 0 \implies \bar{x} = 0 \text{ or } \bar{x}^{n-1} - 1 = 0.$$

Since $\bar{x} \neq 0$, we find that $\bar{x}^{n-1} = 1$. Therefore, $\bar{x}^{n-1} = 1$, so $\bar{x}$ is a unit. $\square$

Exercise 1.8.3. Let R be a nonzero ring. Show that the set of prime ideals of R has minimal elements with respect to inclusion.

Solution: Consider the set $\text{Spec}(R)$, and order $\mathfrak{q} \leq \mathfrak{p}$ if $\mathfrak{p} \subseteq \mathfrak{q}$. This is certainly nonzero because R is nonzero, and there exists primes in R . If R is a field, then we only have 0, but if R is not a field, then every nonunit is contained in a maximal ideal, which is prime. Now, take a totally ordered chain given by $\mathfrak{p}_i$. Then, $\cap \mathfrak{p}_i$ is an upper bound (with respect to the order $\leq$). Indeed, if $xy \in \cap \mathfrak{p}_i$, then $xy \in \mathfrak{p}_i$, for all i . Thus, $x \in \mathfrak{p}_i$ or $y \in \mathfrak{p}_i$, so this is prime. Now, Zorn's lemma implies that $\text{Spec}(R)$ has a minimal element with respect to inclusion then, so we are done. $\square$

Exercise 1.8.4. A ring A is *Boolean* if $x^2 = x$ for all $x \in A$. In a Boolean ring A , show that

1. $2x = 0$ for all $x \in A$;
2. every prime ideal $\mathfrak{p}$ is maximal, and $A/\mathfrak{p}$ is a field with two elements;
3. every finitely generated ideal in A is principal.

Solution: 1. $x + y = (x + y)^2 = x^2 + 2xy + y^2 = x + 2xy + y$. Therefore, $2xy = 0$, for any $x, y \in A$. Thus, $2x = 0$ for any $x \in A$.

2. This is the same as Exercise 1.8.2, but with a more restricting condition. Now, let $x \in A \setminus \mathfrak{p}$, so that $\bar{x} \neq 0$ in the quotient. Since $\bar{x}^2 = \bar{x}$, it follows that $\bar{x}(\bar{x} - 1) = 0$, and

since it's a field (so an ID), $\bar{x} = 0$ or 1 in the field.

3. Now, let $I = (a_1, \dots, a_m) \subseteq A$ be a finitely generated ideal of A . We prove that there is some element in the ideal generating the whole thing. This is done by induction. If $m = 2$, then $I = (a, b)$. Then, $a + ab + b$ generates the whole ideal. Indeed, $a(a + ab + b) = a^2 + a^2b + ab = a + ab + ab = a + 2ab = a$. Similarly, we have the same relation for b . Therefore, $I = (a + ab + b)$. If $I = (a_1, \dots, a_m)$, then we note that $I' = (a_1, \dots, a_m)$ can be generated by some element a . Then, $I = (a, a_m)$, so $a + aa_m + a_m$ works. Thus, we're done. $\square$

Exercise 1.8.5. A local ring contains no idempotent $\neq 0, 1$.

Solution: Let $x \in R$ be idempotent, so $x^2 = x$. If x is a unit, then $x = 1$. If not then x is contained inside the unique maximal ideal $\mathfrak{m}$ of R . Then, $1 - x$ is a unit, and $x(1 - x) = 0$. Therefore, $x = 0$. $\square$

Exercise 1.8.6. In a ring A , let Σ be the set of all ideals in which every element is a zero-divisor. Show that the set Σ has maximal elements and that every maximal element of Σ is a prime ideal. Hence the set of zero-divisors in A is a union of prime ideals.

Solution: We use Zorn's lemma. Σ is nonempty because $(0) \in \Sigma$. Now, take a totally ordered chain in Σ . Then, the union of such ideals is an upper bound. This is indeed an ideal, and if x is in this union, then it is in some smaller ideal, so it satisfies the criterion. Therefore, Zorn's lemma says that Σ admits a maximal ideal $\mathfrak{p}$ with respect to inclusion such that every element is a zero divisor. We claim this is prime. Let $xy \in \mathfrak{p}$ and suppose that $x, y \notin \mathfrak{p}$. This means that xy is a zero divisor but x and y are not zero divisors themselves. This is certainly not possible, because if $(xy)a = 0$, then $x(ya) = 0$ and $y(xa) = 0$. Since $\mathfrak{p}$ is a prime ideal, it is certainly a union of primes. $\square$

Exercise 1.8.7. Let $f: A \rightarrow B$ be a homomorphism of rings and let S be a multiplicatively closed subset of A . Let $T = f(S)$. Show that

$$S^{-1}B \quad \text{and} \quad T^{-1}B$$

are isomorphic as $S^{-1}A$ -modules.

Solution: First, we note that since f is a morphism of rings, T is multiplicative subset of B ; in particular, $0 \notin T$, $1 \in T$, and if $f(a), f(b) \in T$, then $f(ab) = f(a)f(b) \in T$. Then, we consider the map $\varphi: S^{-1}B \rightarrow T^{-1}B$ defined by

$$\frac{b}{s} \mapsto \frac{b}{f(s)},$$

with inverse $\psi: T^{-1}B \rightarrow S^{-1}B$ defined by

$$\frac{b}{t} \mapsto \frac{b}{f^{-1}(t)}.$$

This are clearly $S^{-1}A$ -module maps; that is, they respect the $S^{-1}A$ -action and is also a map of abelian groups. To deduce that these are inverses to each other, we make a remark about what occurs if a_1, a_2 have the same image in T . There is no issue then, because $a_1b = a_2b$ and $f(a_1)b = f(a_2)b$. Thus, the conclusion is clear. $\square$

Exercise 1.8.8. Let A be a ring. Suppose that, for each prime ideal $\mathfrak{p}$, the local ring $A_{\mathfrak{p}}$ has no nonzero nilpotent elements. Show that A has no nilpotent elements. If each $A_{\mathfrak{p}}$ is an integral domain, is A necessarily an integral domain?

Solution: We apply Exercise 1.6.16 to $M = \mathcal{N}(A)$. This tells us that $\mathcal{N}(A) = 0$ if and only if $\mathcal{N}(A_{\mathfrak{p}}) = 0$ for all primes $\mathfrak{p} \subseteq R$.

As for the second statement, this is not true. Consider $k \times k$. This is not an integral domain. However, its prime ideals are $(0) \times k$ and $k \times (0)$, so its localizations are isomorphic to k , which is an integral domain. $\square$

Exercise 1.8.9. Let A be a ring $\neq 0$ and let Σ be the set of all multiplicatively closed subsets S of A such that $0 \notin S$. Show that Σ has maximal elements, and that $S \in \Sigma$ is maximal if and only if $A \setminus S$ is a minimal prime ideal of A .

Exercise 1.8.10. If $A[x]$ is Noetherian, is A necessarily Noetherian?

Solution: Yes; $A[x]/(X) \cong A$, and quotients of Noetherian rings are Noetherian. $\square$

Exercise 1.8.11 (Nagata). Let A be a ring such that

1. for each maximal ideal $\mathfrak{m}$ of A , the local ring $A_{\mathfrak{m}}$ is Noetherian;
2. for each $x \neq 0$ in A , the set of maximal ideals of A which contain x is finite.

Show that A is Noetherian. *Hint:* Let $\mathfrak{a} \neq 0$ be an ideal in A . Let $\mathfrak{m}_1, \dots, \mathfrak{m}_r$ be the maximal ideals which contain $\mathfrak{a}$. Choose $x_0 \neq 0$ in $\mathfrak{a}$ and let $\mathfrak{m}_1, \dots, \mathfrak{m}_{r+s}$ be the maximal ideals which contain x_0 . Since $\mathfrak{m}_{r+1}, \dots, \mathfrak{m}_{r+s}$ do not contain $\mathfrak{a}$, there exist $x_j \in \mathfrak{a}$ such that

$$x_j \notin \mathfrak{m}_{r+j} \quad (1 \leq j \leq s).$$

Since each $A_{\mathfrak{m}_i}$, $1 \leq i \leq r$, is Noetherian, the extension of $\mathfrak{a}$ in $A_{\mathfrak{m}_i}$ is finitely generated. Hence there exist

$$x_{s+1}, \dots, x_t \in \mathfrak{a}$$

whose images in $A_{\mathfrak{m}_i}$ generate $A_{\mathfrak{m}_i}\mathfrak{a}$ for $i = 1, \dots, r$. Let

$$\mathfrak{a}_0 = (x_0, \dots, x_t).$$

Show that $\mathfrak{a}_0$ and $\mathfrak{a}$ have the same extension in $A_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$, and deduce that $\mathfrak{a}_0 = \mathfrak{a}$.

Solution: Let I be an ideal of A . Suppose I is contained in $\mathfrak{m}_1, \dots, \mathfrak{m}_r$, all maximal. Now take some $x_0 \in I$. Then, let $\mathfrak{m}_1, \dots, \mathfrak{m}_{r+s}$ be the maximal ideals containing x_0 . Since $\mathfrak{m}_{r+1}, \dots, \mathfrak{m}_{r+s}$ don't contain I , there exist elements $x_j \in I \setminus \mathfrak{m}_{r+j}$. Then, the extension of I in $A_{\mathfrak{m}_i}$ is finitely generated, for every $1 \leq i \leq r$. Therefore, there exist $x_{s+1}, \dots, x_t \in I$ whose images in $A_{\mathfrak{m}_i}$ generate the extension of I in $A_{\mathfrak{m}_i}$, for $i = 1, \dots, r$. Now, define $J := (x_0, \dots, x_t)$. Then, J and I have the same extension in $A_{\mathfrak{m}}$, for all $\mathfrak{m}$. If $\mathfrak{m} = \mathfrak{m}_i$, for some $1 \leq i \leq r$, then this is immediate. If not, then if $i = r+1, \dots, s$, then I and J both extend to the trivial ideal because they contain elements that are not contained in $\mathfrak{m}_i$. For any other maximal ideal, both I and J have an element contained in $R \setminus \mathfrak{m}$, so they extend to the trivial ideal. Thus, since *surjectivity is a local property*, we are done. $\square$

2 Noetherian and Artinian Rings

§2.1 Noetherian Rings

Definition 2.1.1. A commutative ring R is **Noetherian** if all its ideals are finitely generated.

Definition 2.1.2. Let R be a commutative ring and let

$$I_1 \subset I_2 \subset I_3 \subset \dots$$

be an increasing chain of ideals of R . Then, the chain satisfies the **ascending chain condition** (**ACC** for short) if there exists a positive integer k such that $I_l = I_k$ for all $l \geq k$.

Proposition 2.1.3

Let R be a commutative ring; then, the following are equivalent.

- (i) R is Noetherian.
- (ii) Every increasing chain of ideals satisfies the ACC.
- (iii) Every nonempty set of ideals has a maximal element (with respect to inclusion; not necessarily a maximal ideal).

Proof. (i) $\implies$ (ii): Let $I_1 \subset I_2 \subset I_3 \subset \dots$ be an increasing chain of ideals of R . We can then check that

$$I = \bigcup_{i \geq 1} I_i$$

is an ideal of R . Since R is Noetherian, I is finitely generated and there exist $m_1, \dots, m_k \in I$ such that $I = (m_1, \dots, m_k)$. Let $l \geq 1$ such that $m_1, \dots, m_k \in I_l$. Then, $I_i = I_l$ for any $i \geq l$.

(ii) $\implies$ (iii): Let S be a nonempty set of ideals. Assume for the sake of contradiction that S doesn't admit a maximal element. Then, there exists an increasing chain of ideals in S , $I_1 \subset I_2 \subset I_3 \subset \dots$, such that $I_i \neq I_{i+1}$ for all i , contradicting the ACC.

(iii) $\implies$ (i): Let $I \subset R$ be an ideal and assume for the sake of contradiction that I is not finitely generated. Let $m_1 \in I$. Then, $I \neq (m_1)$ and we may pick $m_2 \in I \setminus (m_1)$. Proceeding this way, we find an increasing chain of ideals

$$I_1 \subset I_2 \subset I_3 \subset \dots,$$

which does not admit a maximal element, contradicting the assumption. $\square$

Example 2.1.4

Let k be a field. Then, k is a Noetherian ring.

Example 2.1.5

Let k be a field. Then consider $R = k[X_1, X_2, \dots]$ be the ring of polynomials in infinitely many variables. Then, R is not Noetherian. Indeed, if $I_i = (X_1, \dots, X_i)$, then $I_1 \subset I_2 \subset I_3 \subset \dots$ is an increasing chain which doesn’t satisfy the ACC.

Exercise 2.1.6 (Cohen’s Theorem). Let R be a commutative ring such that all its prime ideals are finitely generated. Show that R is Noetherian.

Solution: For this, we use Zorn’s lemma. Let P be the set of ideals of R not finitely generated. Assume P is nonempty. Then, if we order this w.r.t. inclusion, P admits a maximal element, say I . I is not finitely generated, so in particular, it cannot be prime. Therefore, there exists $x, y \in R$ such that $xy \in I$ and $x, y \notin I$. Then $I + (x)$ and $(I : y)$ are not contained in I , so they are finitely generated. Say we write $I + (x) = (x, y_1, \dots, y_r)$ and $(I : y) = (z_1, \dots, z_s)$. Then, the elements $(y_i z_j)$ clearly generated I . $\square$

Remark 2.1.7 — The contrapositive of the above exercise is in [2] Exercise 7.1: R be a non-Noetherian ring and let Σ be the set of ideals in A which are not finitely generated. Show that Σ has maximal elements and that the maximal elements of Σ are prime ideals.

Remark 2.1.8 — We note that it is incorrect to proceed by taking some ideal I , saying it’s contained in a maximal ideal, so I is finitely generated. The point of this is that a sub-ideal (or more generally a submodule) of a finitely generated ideal (module) need not be finitely generated in its own right. For instance, consider the polynomial ring of infinitely variables: $k[x_1, x_2, \dots]$. Then, (x_1) is a finitely generated ideal, and $(x_1 x_2, x_1 x_3, \dots)$ is a sub-ideal of (x_1) , but is not a finitely generated ideal. However, Noetherian rings are where the nice statement ‘sub-ideals of a finitely generated ideal is finitely generated’ hold.

Lemma 2.1.9

Let R be a Noetherian ring and let $I \subset R$ be an ideal. Then, there exists a positive integer m such that I contains the m -th power of $\sqrt{I}$, i.e. $(\sqrt{I})^m \subset I$.

Proof. Since R is Noetherian, $\sqrt{I}$ is finitely generated. Let $x_1, \dots, x_k$ be generators. Then, $x_i^{n_i} \in I$ for some positive integer n_i , for all $i = 1, \dots, k$. Let

$$m := \sum_{i=1}^k (n_i - 1) + 1.$$

Then, $(\sqrt{I})^m$ is generated by the products $x_1^{r_1} \dots x_k^{r_k}$, where $r_1, \dots, r_k$ are nonnegative integers such that $\sum_{i=1}^k r_i = m$. Then, by the definition of m , we have that $r_i \geq n_i$ for some $i = 1, \dots, k$. Thus, $x_1^{r_1} \dots x_k^{r_k} \in I$, so our claim follows. $\square$

Proposition 2.1.10

Let R be a Noetherian ring. Then, the nilradical $\mathcal{N}(R)$ of R is nilpotent; that is, there exists a positive integer m such that $\mathcal{N}(R)^m = (0)$.

Proof. Let $I = (0)$. Recall that $\mathcal{N}(R) = \sqrt{I}$. Thus, Lemma 2.1.9 implies the claim. $\square$

Remark 2.1.11 — This is not true in general.

Remark 2.1.12 — If we have an inclusion of rings $A \subset R$ with R being Noetherian, A need not be Noetherian. Exercise 2.2.7 gives an example with $R = k[X, Y]$ and the subring A to be

$$A = \left\{ \sum_{i,j} a_{i,j} X^i Y^j : a_{i,j} \in k \text{ such that } a_{0,i} = 0 \text{ if } i > 0 \right\}.$$

Exercise 2.1.13. Let S be a multiplicative subset of R . If R is Noetherian, then $S^{-1}R$ is Noetherian. In particular, if $\mathfrak{p}$ is a prime ideal of R , then $R_{\mathfrak{p}}$ is Noetherian.

Solution: Ideals of $S^{-1}R$ are in bijection with restricted (or contracted) ideals of R . Therefore, if we take some proper ideals of $S^{-1}R$, it will correspond to some ideal of R which has empty intersection with S . This is finitely generated, so its extension is finitely generated. Thus, Noetherian. $\square$

Remark 2.1.14 — The converse of this is not true.

Exercise 2.1.15. This exercise provides a partial converse to Exercise 1.1.29. Let R be a Noetherian ring and let

$$f = \sum_{n=0}^{\infty} a_n x^n \in R[[x]].$$

Prove that f is nilpotent if and only if each a_n is nilpotent.

Solution: We note that if f is nilpotent, then we know that each a_n is nilpotent from Exercise 1.1.29. Recall that this was shown by using that if we have an arbitrary prime ideal $\mathfrak{p}$ of R , then $(R/\mathfrak{p})[[x]]$ is an integral domain, so if $f^m = 0$, then f^m is zero in $(R/\mathfrak{p})[[x]]$, implying that $f = 0$ in the quotient $(R/\mathfrak{p})[[x]]$, so $f \in \mathfrak{p}[[x]]$. In other words, f had coefficients in $\mathfrak{p}$ to begin with, and so $f \in \mathcal{N}(R)$. Conversely, suppose every a_n is nilpotent. Then, R is Noetherian, so $\mathcal{N}(R)$ is a nilpotent ideal. Therefore, suppose $\mathcal{N}(R)^q = 0$. Then, $f^q = 0$ because every coefficient will have q -terms from $\mathcal{N}(R)$ in it. $\square$

§2.2 Hilbert Basis Theorem

Definition 2.2.1. Let R be a commutative ring. For any polynomial $f(x) = \sum_{i=0}^n a_i x^i \in R[X]$ of degree n ($a_n \neq 0$), we define the **leading term** of f as

$$LT(f) := a_n.$$

Theorem 2.2.2 (Hilbert Basis Theorem)

Let R be a Noetherian ring. Then, $R[X]$ is also Noetherian.

Proof. Let $J \subset R[X]$ be a nonzero ideal. For any $n \geq 0$, we define

$$I_n := \{r \in R : \exists f \in J \text{ such that } \deg f = n \text{ and } LT(f) = r\}.$$

Then, $I_0 \subset I_1 \subset \dots$ is an increasing chain of ideals in R . Indeed, we can verify this because if $f \in J$, then $xf \in J$ and $\deg(xf) = \deg f + 1$ and $LT(xf) = LT(f)$. R is Noetherian, so there exists $n \geq 0$ such that $I_i = I_n$ for any $i \geq n$. Moreover, since R is Noetherian, each I_i is finitely generated. Particularly, we can write

$$I_i = (a_{i,1}, \dots, a_{i,\ell_i}),$$

where $a_{i,1}, \dots, a_{i,\ell_i} \in I_i$ for any $i \leq n$. Therefore, for each $i = 1, \dots, n$ and $j = 1, \dots, \ell_i$, there exists $f_{i,j} \in J$ such that $\deg f_{i,j} = i$ and $LT(f_{i,j}) = a_{i,j}$.

We want to show that J is generated by the elements $f_{i,j}$, for $i = 1, \dots, n$ and $j = 1, \dots, \ell_i$. Let $f \in J$ and let $m = \deg f$. Then $a := LT(f) \in I_m$. We proceed by induction on m . If $m = 0$ then $f \in I_0$ and we are done.

We now assume that $m \geq 1$. There are two cases to consider; when $m \geq n$ or $m < n$. If $m \geq n$ then $I_m = I_n$ and therefore

$$a = \sum_{i=1}^{\ell_n} r_i a_{n,i} \quad \text{for some } r_1, \dots, r_{\ell_n} \in R.$$

We define

$$g := f - \sum_{i=1}^{\ell_n} r_i x^{m-n} f_{n,i}.$$

then, $g \in J$ and $\deg g < \deg f$. Next is when $m < n$. We can set

$$a = \sum_{i=1}^{\ell_m} r_i a_{m,i} \quad \text{for some } r_1, \dots, r_{\ell_m} \in R.$$

We define

$$g := f - \sum_{i=1}^{\ell_m} r_i f_{m,i}.$$

Also in this case, we have $g \in J$ and $\deg g < \deg f$. In both cases, we may assume by induction that g is generated by the elements $f_{i,j}$, for $i = 1, \dots, n$ and $j = 1, \dots, \ell_i$. Thus, the same holds for f and we are done. $\square$

Corollary 2.2.3

Let R be Noetherian. Then, $R[X_1, \dots, X_n]$ is also Noetherian.

Example 2.2.4

Let k be a field. Then Theorem 2.2.2 easily implies by induction that $k[X_1, \dots, X_n]$ is a Noetherian ring for all $n \geq 1$.

Exercise 2.2.5. Suppose that $R[X]$ is Noetherian. Is R Noetherian?

Solution: Yes. This is because $R \cong R[X]/(X)$ as rings, and quotients of Noetherian rings are Noetherian. $\square$

Remark 2.2.6 — In general, a subring of a Noetherian ring is not always Noetherian, as the following exercise shows.

Exercise 2.2.7. Let k be a field. Then, by Hilbert's Theorem, $k[X, Y]$ is a Noetherian ring. Show that if

$$R := \left\{ \sum_{i,j \geq 0} a_{i,j} X^i Y^j : a_{i,j} \in k \text{ such that } a_{0,i} = 0 \text{ if } i > 0 \right\} \subset k[X, Y]$$

then R is not Noetherian. *Hint: show that if*

$$I_n := (X, XY, \dots, XY^n) \subset R$$

then

$$I_0 \subset I_1 \subset I_2 \subset \dots$$

is an increasing chain of ideals of R which does not satisfy ACC

Solution: To give a simple description of R , it is the ring of polynomials with no monomials purely in the variable Y . We note that the following is actually an ideal:

$$I_n = (X, XY, \dots, XY^n)$$

The chain of inclusions

$$I_0 \subset I_1 \subset I_2 \subset \dots$$

is also clear, since each iterative ideal has an extra generator compared to the previous. Suppose that this chain stabilizes. Then, there must be some N such that $I_m = I_N$, for all $m \geq N$. Let $m > N$ now. Then, this condition says that

$$(1, XY, \dots, XY^m) = (X, XY, \dots, XY^N).$$

If $XY^m \in (X, XY, \dots, XY^N)$, then there exists polynomials $f_0, \dots, f_N \in R$ such that

$$\begin{aligned} XY^m &= f_0(X, Y)X + f_1(X, Y)XY + \dots + f_N(X, Y)XY^N \\ &= X(f_0(X, Y) + f_1(X, Y)Y + \dots + f_N(X, Y)Y^N). \end{aligned}$$

Since $k[X, Y]$ is an integral domain, it follows that

$$Y^m = f_0(X, Y) + f_1(X, Y)Y + \dots + f_N(X, Y)Y^N.$$

However, each $f_i(X, Y)$ is a polynomial that must be in both X and Y , which is not possible. Thus, the chain of ideals doesn't stabilize. Actually, note that this chain is not a strictly increasing chain, since $I_n = (X)$, so this doesn't contradict that $k[X, Y]$ is Noetherian. $\square$

Exercise 2.2.8. Let k be an infinite field and let $f \in k[X_1, \dots, X_n]$. Show that $f = 0$ if and only if $f(a_1, \dots, a_n) = 0$, for all $a_1, \dots, a_n \in k$. Give a counter example to this statement when k is a finite field.

Solution: The forward direction is clear, so all we need to show is the converse. Let $f \in k[X_1, \dots, X_n]$. We induct on n . If $n = 1$, this is clear, since a polynomial over an infinite field has finitely many roots. Assume true up to $n - 1$. Then, write f using $\sum_{i=0}^d x_1^i f_i(x_2, \dots, x_n)$. Then, there is some i such that f_i is nonzero, so there exists some $a_2, \dots, a_n$ such that $f_i(a_2, \dots, a_n) \neq 0$. Then, $f(x_1, a_2, \dots, a_n)$ is a polynomial in one variable over an infinite field, and this has finitely many roots, so we can just choose a nonroot.

In $\mathbb{F}_p[X]$, the polynomial $X(X - 1) \dots (X - (p - 1))$ vanishes at all points in $\mathbb{A}_{\mathbb{F}_p}^1$. $\square$

Exercise 2.2.9. Suppose k is a field. Show that the ring $k[[X]]$ of formal power series over k is Noetherian.

Solution: First note that if $f(X) = \sum_{i=0}^{\infty} a_i X^i$, and $a_0 \neq 0$, then $f(X)$ is a unit. Let $J \subset k[[X]]$. Then, we want to show that J is finitely generated. Define

$$I_n := \{a_n \in k : \exists f(X) \in J \text{ such that } f(X) = a_n X^n + a_{n+1} X^{n+1} + \dots\}.$$

Then, $I_n \subset I_{n+1}$, because if $f(X) \in J$, then $Xf(X) \in J$. Since k is a field and $I_n \subset k$ is an ideal, there exists m such that $I_n = 0$ if $n < m$ and $I_n = k$ if $n \geq m$. From the inclusion $I_n \subset I_{n+1}$, this then follows. This implies that $J = (X^m)$. Then, the remaining part would be a unit by the remark at the beginning. $\square$

Remark 2.2.10 — We can generalize this; if R is a Noetherian ring, then $R[[X]]$ is Noetherian. The way we finish off the proof would be slightly different. We would use the fact that the chain defined by I_n s stabilize. Then, with some work, we get that J is finitely generated, with generators being those lowest terms up to the stabilizing index. This gives the Noetherian property.

§2.2.i Noetherian Topological Spaces

Definition 2.2.11. A topological space is called **Noetherian** if any decreasing sequence of closed subsets $V_1 \supset V_2 \supset V_3 \supset \dots$ satisfies the **descending chain condition (DCC)**, i.e. $V_k = V_l$ for all $l \geq k$.

Corollary 2.2.12

$\mathbb{A}_k^n$ is Noetherian with respect to the Zariski topology.

Proof. Indeed, if we have a decreasing chain of closed subsets $V_1 \supset V_2 \supset \dots$, then taking the ideal of them forms an ascending chain, which stabilizes since $k[X_1, \dots, X_n]$ is Noetherian. We then just apply the result that $V(I(V_i)) = V_i$. $\square$

§2.3 Cayley-Hamilton and Nakayama

We note that an F -vector space is the same as a F -module (when F is a field), so it isn't hard to guess that many linear algebraic results extend to modules.

Let R be a commutative ring and let $M = M_n(R)$ be the set of $n \times n$ matrices with coefficients in R . Then, M is a module over R .

We denote by $I_n \in M$ to be the **identity** matrix, and the **determinant** $\det A$ of A is defined in the same way, as an element of R . Particularly, there exists a matrix B such that

$$A \cdot B = B \cdot A = \det A \cdot I_n.$$

Here, B is the **adjoint** of A and it's defined as the matrix with (i, j) -th entry given by $(-1)^{i+j} \det A_{j,i}$, where $A_{j,i}$ minor attained from deleting the j -th row and i -th column.

Remark 2.3.1 — We note that A is invertible if and only if $\det A$ is a unit in R .

Definition 2.3.2. For any positive integers, we denote $\delta_{i,j}$ to be the (i, j) -**Kronecker symbol**; that is,

$$\delta_{i,j} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}.$$

Theorem 2.3.3 (Cayley–Hamilton's Theorem)

Let R be a commutative ring, let $I \subset R$ be an ideal and let M be a finitely generated R -module. Let $\varphi: M \rightarrow M$ be a homomorphism such that

$$\varphi(M) \subset I \cdot M.$$

Then there exist $a_0, \dots, a_{n-1} \in I$ such that

$$\varphi^n(m) + a_{n-1} \cdot \varphi^{n-1}(m) + \dots + a_1 \cdot \varphi(m) + a_0 \cdot m = 0 \quad \text{for all } m \in M.$$

In the theorem, φ^i denotes the i -fold composition of φ with itself.

Proof. We can consider M as a $R[X]$ -module, by setting

$$X \cdot m := \varphi(m) \quad \text{for } m \in M.$$

We need to show that there exist $a_0, \dots, a_{n-1} \in I$ such that if

$$f(X) := X^n + a_{n-1}X^{n-1} + \dots + a_0$$

then $f(X) \cdot m = 0$ for all $m \in M$.

Let $m_1, \dots, m_n$ be generators of M over R . By assumption, we have $\varphi(m_i) \in I \cdot M$ for all $i = 1, \dots, n$. Thus, we may write

$$\varphi(m_i) = \sum_{j=1}^n r_{i,j} m_j \quad \text{for some } r_{i,j} \in I.$$

It follows that

$$\sum_{j=1}^n (X\delta_{i,j} - r_{i,j}) \cdot m_j = 0 \quad \text{for } i = 1, \dots, n.$$

Thus, if we consider the matrix

$$A := (X\delta_{i,j} - r_{i,j}) \in M_n(R[X])$$

and its adjoint matrix B , then we obtain

$$0 = B \cdot A \cdot \begin{pmatrix} m_1 \\ \dots \\ m_n \end{pmatrix} \implies 0 = \det A \cdot m_j \quad \text{for } j = 1, \dots, n.$$

Thus, $\det A \in R[X]$ is such that $\det A \cdot m = 0$, for all $m \in M$. Thus, the claim follows. $\square$

Theorem 2.3.4 (Nakayama's Lemma)

Let R be a commutative ring, let $I \subset R$ be an ideal and let M be a finitely generated R -module such that $I \cdot M = M$. Then there exists $a \in I$ such that

$$a \cdot m = m \quad \text{for all } m \in M.$$

Proof. Let $\varphi := \text{Id} : M \rightarrow M$ be the identity homomorphism. Then, by Theorem 2.3.3, there exist $a_0, \dots, a_{n-1} \in I$ such that

$$(1 + a_{n-1} + \dots + a_0) \cdot m = 0 \quad \text{for all } m \in M.$$

Define $a := -a_{n-1} - \dots - a_0 \in I$. Then our claim follows. $\square$

Corollary 2.3.5

Let R be a commutative ring and let M be a finitely generated R -module. Let $\varphi : M \rightarrow M$ be a surjective morphism. Then φ is an isomorphism.

Proof. As in the proof of Theorem 2.3.3, we can consider M as a $R[X]$ -module, by setting

$$X \cdot m := \varphi(m) \quad \text{for } m \in M.$$

Since φ is surjective, we have that if $I := (X)$ then $I \cdot M = M$. By Nakayama's lemma (Theorem 2.3.4), there exists $f \in I$ such that $f \cdot m = m$, for all $m \in M$. Since $f \in (X)$, there exist $a_1, \dots, a_n \in R$ such that

$$f(x) = a_1x + a_2x^2 + \dots + a_nx^n.$$

Thus, we have

$$a_1\varphi(m) + a_2\varphi^2(m) + \dots + a_n\varphi^n(m) = m,$$

for all $m \in M$. Assume that $m \in \ker \varphi$. Then $\varphi(m) = 0$ which implies that $m = 0$. Thus, φ is injective, as claimed. $\square$

Corollary 2.3.6

Let R be a commutative ring and M be a finitely generated R -module. Let $I \subset R$ be an ideal such that $I \subset \mathcal{J}(R)$ and $I \cdot M = M$. Then, $M = 0$.

Proof. By Nakayama's lemma (2.3.4), there exists $a \in I$ such that $a \cdot m = m$ for all $m \in M$. Let $x := 1 - a$. Since $a \in I$ and since $\mathcal{J}(R)$ is the intersection of all the maximal ideals of R , it follows that $x \notin \mathfrak{m}$ for all maximal ideals $\mathfrak{m} \subset R$. Thus, Proposition 1.2.12 implies that $(x) = R$, and so it follows that x is a unit in R . Thus,

$$m = x^{-1} \cdot x \cdot m = x^{-1} \cdot (1 - a) \cdot m = 0 \quad \text{for all } m \in M,$$

and the claim follows. $\square$

Exercise 2.3.7. Let $(R, \mathfrak{m})$ be a local ring. Let M be a finitely generated R -module and let $N \subset M$ be a submodule such that $M = N + \mathfrak{m}M$. Show that $N = M$.

Solution: Write $\overline{M} = M/N$. The goal is to show that $\overline{M} = 0$. Since R is local, $\mathcal{J}(R) = \mathfrak{m}$. If we can show that $\mathfrak{m} \cdot \overline{M} = \overline{M}$, then we are done by Nakayama's lemma. This is indeed true because we can quotient $M = N + \mathfrak{m}M$ by N . $\square$

Remark 2.3.8 — Note that the containment $M \supset N + \mathfrak{m}M$ always holds, so $M \subset N + \mathfrak{m}M$ implies $N = M$.

§2.4 Artin Rings

Definition 2.4.1. Let R be a commutative ring and let

$$I_1 \supset I_2 \supset I_3 \supset \dots$$

be a decreasing chain of ideals of R . The chain satisfies the **descending chain condition** (in short **DCC**) if there exists a positive integer k such that $I_\ell = I_k$ for all $\ell \geq k$.

Definition 2.4.2. Let R be a commutative ring. Then R is called an **Artin ring** (or **Artinian**) if any decreasing chain of ideals of R satisfies the DCC.

Proposition 2.4.3

Let R be a commutative Artin ring. Then every prime ideal of R is maximal.

Proof. Let $\mathfrak{p} \subset R$ be a prime ideal. Then, we have that $S := R/\mathfrak{p}$ is an integral domain. It is easy to see that it is an Artin ring. Thus, it's enough to show that S is a field. Let $0 \neq x \in S$. Then there exists a positive integer n such that $(x^n) = (x^{n+1})$, and hence $x^n = x^{n+1}y$ for some $y \in S$. Since S is an integral domain and $x \neq 0$, it follows that $xy = 1$. Thus, x has an inverse in S , as claimed. $\square$

Recall that if R is a commutative ring, then the nilradical. $\mathcal{N}(R)$ of R is the radical of the ideal (0) , i.e.

$$\mathcal{N}(R) = \{x \in R \mid x^n = 0 \text{ for some } n \geq 1\},$$

the set of all nilpotent elements in R ; the Jacobson radical $\mathcal{J}(R)$ of R is the intersection of all maximal ideals of R . We have:

Corollary 2.4.4

Let R be a commutative Artin ring. Then

$$\mathcal{N}(R) = \mathcal{J}(R).$$

Proof. By Proposition 1.4.2, it follows that $\mathcal{N}(R)$ is the intersection of all prime ideals of R . Thus, Proposition 2.4.3 implies the claim. $\square$

Corollary 2.4.5

Every element of an Artin local ring is either nilpotent or a unit.

Proof. If $\mathfrak{m}$ is the unique maximal ideal, the $\mathfrak{m}$ tells us that it’s the only prime ideal. Then, the previous corollary tells us that every element of $\mathfrak{m}$ is nilpotent. Then, by Proposition 1.5.4, we have that the elements of R are either nilpotent or units. $\square$

Example 2.4.6

Every field is an Artin local ring (and Noetherian).

Example 2.4.7

$k[X]$ is Noetherian but not Artinian: $(X) \supset (X^2) \supset (X^3) \supset \dots$

Example 2.4.8

$\mathbb{Z}$ is Noetherian but not Artinian $(p) \supset (p^2) \supset \dots$

Example 2.4.9

If $p \in \mathbb{Z}$ is a prime number and n is a positive integer, then $\mathbb{Z}/(p^n)$ is a local Artin ring.

Example 2.4.10

If k is a field and n is a positive integer then $k[x]/(x^n)$ is a local Artin ring.

Remark 2.4.11 — Note that the Hopkins-Levitzki tells us that every Artinian ring is Noetherian ring. This is Exercise 2.6.22.

Exercise 2.4.12. Show that if R is an Artinian ring, it admits finitely many maximal ideals.

Solution: Consider the set $P = \{\text{finite intersections of maximal ideals of } R\}$. Then, $\mathcal{N}(R) \in P$, so P is nonempty. Now we order P based in inclusion. Then, R is Artinian, so any totally descending chain stabilizes. Thus, Zorn’s lemma implies that there is a minimal element in P . This is $\mathfrak{m}_1 \cap \dots \cap \mathfrak{m}_n$. Now, suppose $\mathfrak{m}$ is a maximal ideal of R not present as one of these intersections. Then, by minimality, $\mathfrak{m} \cap \mathfrak{m}_1 \cap \dots \cap \mathfrak{m}_n = \mathfrak{m}_1 \cap \dots \cap \mathfrak{m}_n$. Therefore, it follows that $\mathfrak{m}_1 \cap \dots \cap \mathfrak{m}_n \subseteq \mathfrak{m}$. The claim is that $\mathfrak{m} = \mathfrak{m}_i$ for some i then. If not, then there is some $x_i \in \mathfrak{m}_i \setminus \mathfrak{m}$, for all i . Then, we find that

$$x_1 \cdots x_n \in \mathfrak{m}_1 \cap \dots \cap \mathfrak{m}_n \subseteq \mathfrak{m},$$

but each $x_i \notin \mathfrak{m}$, contradicting that $\mathfrak{m}$ is prime. Therefore, $\mathfrak{m} = \mathfrak{m}_i$ for some i . $\square$

Exercise 2.4.13. Recall that, given two rings R and S , then the direct product $R \times S$ admits a natural structure of a ring. Let R be an Artin ring. Show that there exist $R_1, \dots, R_m$ Artin local rings such that R is isomorphic to the direct product of $R_1 \times \dots \times R_m$.

Solution: Note that if R is Artinian, there are finitely many prime ideals by the above exercise, $\mathfrak{p}_1, \dots, \mathfrak{p}_n$. Then,

$$\mathcal{N}(R) = \bigcap_{i=1}^n \mathfrak{p}_i,$$

but the nilradical is nilpotent in an Artinian ring, so suppose $\mathcal{N}(R)^q = 0$. Then, it follows that

$$\mathfrak{p}_1^q \cdots \mathfrak{p}_n^q \subseteq \bigcap_{i=1}^n \mathfrak{p}_i^q = \mathcal{N}(R)^q = 0.$$

Then, by coprimality of the primes (because they are maximal ideals), it follows that the natural map

$$R \cong R/(\mathfrak{p}_1^q \cdots \mathfrak{p}_n^q) \xrightarrow{\sim} \prod_{i=1}^n R/(\mathfrak{p}_i^q)$$

is an isomorphism. The claim is that each factor in the direct product is a Artin local ring. The fact that each component is Artinian is immediate. As for local-ness, a maximal ideal of $R/\mathfrak{p}_i^q$ corresponds to a maximal ideal of R containing $\mathfrak{p}_i^q$. Call this $\mathfrak{m}$, so $\mathfrak{p}_i^q \subseteq \mathfrak{m}$. Taking radicals, we can find the primes containing $\mathfrak{p}_i^q$. However, the radical of $\mathfrak{p}_i^q$ is $\mathfrak{p}_i$, so $\mathfrak{m} = \mathfrak{p}_i$ is the unique maximal ideal of $R/\mathfrak{p}_i^q$. $\square$

Exercise 2.4.14. Let R be a commutative ring. Show or disprove the following:

1. If $\mathcal{N}(R_{\mathfrak{p}}) = (0)$ for any prime ideal $\mathfrak{p} \subset R$ then $\mathcal{N}(R) = 0$ (True).
2. If $R_{\mathfrak{p}}$ is an integral domain for any prime ideal $\mathfrak{p} \subset R$ then R is an integral domain (False).
3. If $R_{\mathfrak{p}}$ is Noetherian for any prime ideal $\mathfrak{p} \subset R$ then R is Noetherian (False).

Solution: Recall that if R is a ring and $\mathfrak{p}$ is a prime ideal of R , then $R_{\mathfrak{p}} = (R \setminus \mathfrak{p})^{-1}R$. Note that $R_{\mathfrak{p}}$ is a local ring with maximal ideal $\mathfrak{p}R_{\mathfrak{p}}$.

1. The answer is yes. Suppose that there is a nonzero nilpotent element in R , say x , and $x^n = 0$ for some n . Define I to be the annihilator of x :

$$I = \text{ann}(x) = \{y \in R : xy = 0\}.$$

Then, $x^{n-1} \in I$. $1 \notin I$, since $x \neq 0$. Then, by Zorn's lemma, we know that there exists a maximal ideal $\mathfrak{m} = \mathfrak{p}$ such that $I \subset \mathfrak{p} \subsetneq R$. By assumption, $\mathcal{N}(R_{\mathfrak{p}}) = 0$. Then, $\frac{x}{1} \in R_{\mathfrak{p}}$, and taking the n -th power, this is $\frac{x^n}{1} = \frac{0}{1} \in R_{\mathfrak{p}}$. However, $\frac{x}{1}$ is nilpotent in $R_{\mathfrak{p}}$, implying that $\frac{x}{1} = 0$ in $R_{\mathfrak{p}}$. This means that there exists $t \in R \setminus \mathfrak{p}$ such that $t(x - 0) = tx = 0$ in R . This means that $t \in I \subset \mathfrak{p}$, so $t \in \mathfrak{p}$. However, $t \in R \setminus \mathfrak{p}$ and $\mathfrak{p}$, a contradiction.

2. Not true. Let k be a field and let $R = k \times k$. Then R is not an integral domain, as $(1, 0) \cdot (0, 1) = (0, 0)$. But it is easy to check that the prime ideals of R are $k \times (0)$ and $(0) \times k$. If $I \subset R$ is an ideal then $I \cap (k \times (0))$ and $I \cap ((0) \times k)$ are also ideals. But the localization of R at each of this ideal is isomorphic to k , which is an integral domain.

3. This is not true. Take $k = \mathbb{Z}/2\mathbb{Z}$ and let

$$R = k^N.$$

Then R is not Noetherian. Indeed, let $e_1, e_2, \dots$ be the standard basis of R and let $I_n \subset R$ be the ideal generated by $e_1, \dots, e_n$. Then

$$I_1 \subset I_2 \subset I_3 \subset \dots$$

is an increasing chain of ideals which does not satisfy ACC.

We now show that if $\mathfrak{p} \subset R$ is prime then $R_{\mathfrak{p}}$ is a field and, in particular, it is Noetherian. Note that $x(x-1) = 0$ for any $x \in R$. Thus, the same property holds for $R_{\mathfrak{p}}$. Let

$$\mathfrak{m}_{\mathfrak{p}} = \mathfrak{p}R_{\mathfrak{p}}$$

be the unique maximal ideal in $R_{\mathfrak{p}}$. If $\mathfrak{q} \subset R_{\mathfrak{p}}$ is a prime ideal and $x \in \mathfrak{m}_{\mathfrak{p}}$, then $x \cdot (x-1) \in \mathfrak{q}$ which implies that $x \in \mathfrak{q}$ or $x-1 \in \mathfrak{q}$. But since $x-1$ is a unit, we must have that $x \in \mathfrak{q}$. Thus,

$$\mathfrak{q} = \mathfrak{m}_{\mathfrak{p}},$$

which implies that $\mathfrak{m}_{\mathfrak{p}}$ is the only prime ideal of $R_{\mathfrak{p}}$. In particular,

$$\mathfrak{m}_{\mathfrak{p}} = N(R_{\mathfrak{p}}).$$

Thus, if $x \in \mathfrak{m}_{\mathfrak{p}}$, then $x^n = 0$ for some positive integer n , but since $x^2 = x$, it follows that $x = 0$ and therefore

$$\mathfrak{m}_{\mathfrak{p}} = (0).$$

Thus, any nonzero element in $R_{\mathfrak{p}}$ is invertible, i.e. $R_{\mathfrak{p}}$ is a field, as claimed. $\square$

Remark 2.4.15 — More generally, it is a natural question to ask if R satisfies a property P , does $R_{\mathfrak{p}}$ satisfy P , for all prime ideals $\mathfrak{p}$?

§2.5 Dimension of a Ring

We now extend the definitions of Noetherian rings and Artin rings to modules.

Definition 2.5.1. Let R be a commutative ring and let M be a module over R . Then

- M is **Noetherian** if any increasing sequence of submodules of M

$$M_1 \subset M_2 \subset \dots \subset M$$

satisfies the ACC.

- M is **Artinian** if any decreasing sequence of submodules of M

$$M \supset M_1 \supset M_2 \supset \dots$$

satisfies the DCC.

Remark 2.5.2 — Note that a ring R is an R -module over itself, and the submodules of R are ideals of R . Thus, this notion generalizes (and does not contradict) previous notions of Noetherian/ Artinian rings.

Example 2.5.3

Let k be a field. Recall that a k -module is just a vector space over k . Thus, the following are equivalent:

1. M is a finite-dimensional vector space.
2. M is Noetherian.
3. M is Artinian.

Remark 2.5.4 — By inspecting the proof of Proposition 2.1.3, we can see that the same proposition holds for Noetherian modules. That is, let M be an R -module. Then, the following are equivalent.

- (i) Every R -submodule of M is finitely generated.
- (ii) M is Noetherian.
- (iii) Every nonempty set of R -submodules of M has a maximal element.

Exercise 2.5.5. Let R be a commutative ring and let $I \subset R$ be an ideal. Show that

1. R is a Noetherian ring if and only if I and R/I are Noetherian R -modules;
2. R is an Artin ring if and only if I and R/I are Artinian R -modules.

Note that the same statement holds for modules; that is, replace R with an R -module M and ideals with R -submodule of M , N .

Solution: 1. ($\implies$): Let R be a Noetherian ring. Then, every submodule of I are just sub-ideals of I , which satisfies the ACC from R . An ascending chain of submodules of R/I correspond to submodules of R (regarded as an R -module) containing I . This chain then satisfies the ACC by R , so the original chain must stabilize as well.

($\impliedby$): Suppose we have an ascending chain of ideals:

$$J_1 \subset J_2 \subset \cdots \subset J_n \subset \cdots$$

Since I and R/I are Noetherian, it follows that $J_i \cap I = J_n \cap I$, for all $i \geq n$, and $J_i + I = J_n + I$ for all $i \geq n$, for some arbitrarily large n . It then follows that $J_i = J_n$, for all $i \geq n$.

2. Let R be Artinian. Then, every submodule of I is just a sub-ideal, so stabilizes by the DCC on R . Next, every submodule of R/I is a sub-ideal of R/I , so corresponds with a descending chain of sub-ideals of R which contain I . This stabilizes by the DCC as well, so we are done. Conversely, suppose we have a descending chain of ideals

$$J_1 \supset J_2 \supset \cdots$$

Then, I and R/I are Artinian modules. Thus, $J_i \cap I$ is a submodule of I , so stabilizes. Similar, $J_i + I = (J_i + I)/I$ is a descending chain of submodules of R/I , so stabilizes as well. Thus, $J_i = J_n$ for all $i \geq n$, for some sufficiently large n , so the original descending chain satisfies the DCC. $\square$

Exercise 2.5.6. Let R be Noetherian. Prove that an R -module M is Noetherian if and only if M is finitely generated.

Solution: If M is finitely generated, then it is isomorphic to a quotient of R^k , for some k . R^k is a Noetherian R -module, since R is Noetherian in its own right. Since quotients of Noetherian modules are Noetherian, M is Noetherian. Conversely, if M is Noetherian, then M must be finitely generated in its own right, say by a previous exercise. Thus, in particular, we require R to be Noetherian for the implication M is f.g. implies M is Noetherian. $\square$

Remark 2.5.7 — We can translate this to say that in the exact sequence below,

$$0 \longrightarrow I \longrightarrow R \longrightarrow R/I \longrightarrow 0$$

the middle is Noetherian/ Artinian iff the outer two are Noetherian/ Artinian.

Exercise 2.5.8. Let $p \in \mathbb{Z}$ be a prime number and consider the quotient $M = \mathbb{Z}[\frac{1}{p}]/\mathbb{Z}$.

1. Is M a Noetherian $\mathbb{Z}$ -module?
2. Is M a Artinian $\mathbb{Z}$ -module?

Solution: First note that $\mathbb{Z}[\frac{1}{p}] = \{\frac{a}{p^k} : a \in \mathbb{Z}\}$. We determine the submodules of M . We claim that every nonzero submodule of M is of the form

$$N_k := \left\{ \frac{a}{p^k} : a \in \{0, 1, \dots, p^k - 1\} \right\},$$

for some n . These are indeed submodules of M . Suppose that $\left[\frac{b}{p^k}\right] \in N$, for some integer b not divisible by p . Then, we want to show that $\left[\frac{1}{p^k}\right] \in N$. This is true, since there is a integer c such that $cb \equiv 1 \pmod{p^k}$. Thus, $c \cdot \left[\frac{b}{p^k}\right] \in N$, which is equal to $\left[\frac{cb}{p^k}\right]$. Then, $\frac{1}{p^k} \in N$, so this generates $N_k \subset N$. In particular, we achieve equality $N = N_\ell$, for some ℓ , where ℓ is the maximum of all the k such that $N_k \subset N$. Finally, note that

$$N_k \subset N_{k+1} \subset N_{k+2} \subset \dots$$

To answer the first part, M is not a Noetherian $\mathbb{Z}$ -module then by considering the chain

$$\left(\frac{1}{p} + \mathbb{Z}\right) \subset \left(\frac{1}{p^2} + \mathbb{Z}\right) \subset \dots \subset \left(\frac{1}{p^n} + \mathbb{Z}\right) \subset \dots$$

However, M is Artinian since any proper submodule of M has finitely many submodules. $\square$

Definition 2.5.9. Let X be a topological space. The **dimension** of X is

$$\dim X := \sup\{n \geq 0 : \exists X_0 \subsetneq X_1 \subsetneq X_2 \subsetneq \dots \subsetneq X_n \subset X \text{ with } X_i \text{ closed and irreducible}\}.$$

Definition 2.5.10. Let R be a commutative ring. The **dimension** of R is

$$\dim R := \sup\{n \geq 0 : \exists \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \dots \subsetneq \mathfrak{p}_n \subset R \text{ with } \mathfrak{p}_0, \dots, \mathfrak{p}_n \text{ prime ideals}\}.$$

Definition 2.5.11. Let R be a commutative ring and let $\mathfrak{p} \subset R$ be a prime ideal. The **height** of $\mathfrak{p}$ is

$$h(\mathfrak{p}) := \sup\{n \geq 0 : \exists \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \dots \subsetneq \mathfrak{p}_n = \mathfrak{p} \text{ with } \mathfrak{p}_0, \dots, \mathfrak{p}_n \text{ prime ideals}\}.$$

Example 2.5.12

$\dim \mathbb{Z} = 1$, since we can have $(0) \subset (p) \subset \mathbb{Z}$.

Example 2.5.13

We have seen that prime ideals of $\mathbb{Z}[X]$ are of the form (f) or (p, g) , where f is irreducible in $\mathbb{Z}[X]$ or g is irreducible in $\mathbb{Z}/p\mathbb{Z}[X]$. Therefore, $\dim \mathbb{Z}[X] = h((p, g)) = 2$, $h((f)) = 1$.

Example 2.5.14

For a field k , $\dim k = 0$, since we only have $(0) \subset k$.

Example 2.5.15

Let k be a field and let $R = k[x_1, \dots, x_n]$. Then we have a chain of prime ideals

$$(0) \subsetneq (x_1) \subsetneq (x_1, x_2) \subsetneq \cdots \subsetneq (x_1, \dots, x_n) \subsetneq R.$$

Thus, $\dim R \geq n$. We will show later that $\dim R = n$.

Example 2.5.16

Let $X = \mathbb{A}_k^n$. Then, we can find a sequence

$$\mathbb{A}_k^n \supset V(x_1) \supset V(x_1, x_2) \supset \cdots \supset V(x_1, \dots, x_n) \supset \emptyset.$$

This is a sequence of irreducible varieties. This shows that inside X , there is no sequence more than $n + 2$ irreducible subvarieties.

§2.5.i Jordan-Hölder Series

Definition 2.5.17. Let R be a commutative ring and let M be a module over R . Then a chain of submodules

$$\{0\} = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$$

is called a **Jordan-Hölder series** (or **composition series**) if each module M_i/M_{i-1} is a simple R -module for $i = 1, \dots, n$. The number n is called the **length** of the Jordan-Hölder series.

Example 2.5.18

Let k be a field and M be a k -module of finite dimension. Then, let $\{v_1, \dots, v_n\}$ be a basis of M over k . Then, we may take

$$0 = M_0 \subseteq \langle v_1 \rangle \subsetneq \langle v_1, v_2 \rangle \subsetneq \cdots \subsetneq \langle v_1, \dots, v_n \rangle = M.$$

Note that each $M_i/M_{i-1} \cong k$, which is a simple k -module.

Exercise 2.5.19. Let R be a ring and let M be a R -module. Let $P \subset N \subset M$ be submodules such that the natural map

$$N/P \rightarrow M/P$$

is an isomorphism. Show that $N = M$.

Solution: First, this natural map just maps $n + P$ to $n + P$. We want to show that $M \subset N$, so let $m \in M$. Then, consider $m + P \in M/P$. Then, since the map is an isomorphism, it is surjective, so there exists some element $n + P$ such that $n + P = m + P$. In other words, $m - n \in P$. However, $P \subset N$, so $(m - n) + n = m \in N$. Thus, we are done. $\square$

Proposition 2.5.20

Let R be a commutative ring and let M be a module over R . Let

$$\{0\} = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$$

be a Jordan-Hölder series. Then every Jordan-Hölder series has length n .

Proof. Given a module M which admits a Jordan-Hölder series, we denote by $\ell(M)$ the minimum length of a Jordan-Hölder series in M .

We first show that if $N \subset M$ is a submodule, then N admits a Jordan-Hölder series and $\ell(N) \leq \ell(M)$. Let

$$\{0\} = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$$

be a Jordan-Hölder series in M of minimum length and let $N_i := N \cap M_i$ for $i = 0, \dots, n$. Then

$$N_i/N_{i-1} \subset M_i/M_{i-1} \quad \text{for } i = 1, \dots, n.$$

Since M_i/M_{i-1} is simple, it follows that either $N_i = N_{i-1}$ or $N_i/N_{i-1} = M_i/M_{i-1}$ is simple. Hence, removing the repeated terms, we get a Jordan-Hölder series of length at most n . Note that if the equality holds, then Exercise 2.5.19 implies that $N_i = M_i$ for all $i = 0, \dots, n$ and, in particular, $N = M$.

Now let

$$\{0\} = M'_0 \subsetneq M'_1 \subsetneq \cdots \subsetneq M'_m = M$$

be a Jordan-Hölder series of M . Then, by what we have proven above, it follows that

$$0 = \ell(M'_0) \leq \ell(M'_1) \leq \cdots \leq \ell(M'_m) = \ell(M).$$

Thus, $m \leq \ell(M)$. Since $\ell(M)$ is the minimum length of a Jordan-Hölder series in M , the equality holds. Thus, our claim follows. $\square$

Definition 2.5.21. Let R be a commutative ring and let M be a module over R . If M admits a Jordan-Hölder series, we define the **length** $\ell(M)$ of M to be the length of any Jordan-Hölder series in M . If M does not admit any Jordan-Hölder series, we set $\ell(M) = \infty$.

Proposition 2.5.22

Let R be a commutative ring and let M be a module over R such that $\ell(M) < +\infty$. Then any chain of submodules

$$\{0\} = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$$

can be extended to a Jordan-Hölder series.

Proof. If $n \geq \ell(M)$, then as in the proof of Proposition 2.5.20 it follows that $n = \ell(M)$ and that

$$\{0\} = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$$

is a Jordan-Hölder series. If $n < \ell(M)$, then

$$\{0\} = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$$

is not a Jordan-Hölder series and therefore new submodules can be inserted until the length is $\ell(M)$. $\square$

Proposition 2.5.23

Let R be a commutative ring and let M be a module over R . Then $\ell(M) < +\infty$ if and only if M is Noetherian and Artinian.

Proof. Assume that $\ell(M) < +\infty$. Then any chain of submodules in M is bounded by $\ell(M)$ by the previous result, so both the ACC and DCC hold. Thus M is Noetherian and Artinian.

Assume now that M is Noetherian and Artinian. We aim to construct a Jordan-Hölder series. Since M is Noetherian, by Zorn's Lemma (Theorem 1.2.10), there exists a maximal submodule $M^1 \subsetneq M$. In particular, it follows that M/M^1 is simple. If $M^1 = \{0\}$ then we are done. Otherwise, using Zorn's Lemma again, there exists $M^2 \subsetneq M^1$ maximal submodule such that M^1/M^2 is simple. By continuing this way, we have a decreasing sequence

$$M \supsetneq M^1 \supsetneq M^2 \supsetneq \cdots$$

By the Artinian property, the sequence satisfies DCC. Thus, it is a Jordan-Hölder series and our claim follows. $\square$

§2.6 Primary Decomposition

Definition 2.6.1. Let R be a commutative ring and let $I \subsetneq R$ be a proper ideal. Then

- I is said to be **irreducible** if for any ideals $J_1, J_2 \subset R$ such that $I = J_1 \cap J_2$, we have that $I = J_1$ or $I = J_2$.
- I is said to be **primary** if, for any $x, y \in R$ such that $x \cdot y \in I$, we have that $x \in I$ or $y^n \in I$ for some $n > 0$.

Example 2.6.2

Let $p \in \mathbb{Z}$ be prime and $n \geq 2$. Then, $(p^n) \subset \mathbb{Z}$ is primary, but not prime.

Proposition 2.6.3

Let R be a commutative ring and let $I \subsetneq R$ be a proper ideal.

- (i) If I is prime then I is primary.
- (ii) I is primary if and only if every zero divisor in R/I is nilpotent.

Proof. (i) Just let $n = 1$ in the definition.

(ii) Let I be primary, and assume y is a zero divisor and $xy = 0$ in R/I , for some $x \notin I$. Then, $xy \in I$, so $x \in I$ or $y^n \in I$; thus, y is nilpotent. Conversely, suppose every zero divisor of R/I is nilpotent. That is suppose $xy \in I$ where $x \notin I$, then $x^n \in I$ or $y^m \in I$ for some n, m . Then, $y^m \in I$, which is exactly the definition of I being primary. $\square$

Proposition 2.6.4

Let R be a commutative ring and let $I \subset R$ be a primary ideal. Then $\sqrt{I}$ is the smallest prime ideal containing I .

Proof. By Proposition 1.4.2, it is enough to show that $\sqrt{I}$ is a prime ideal. Let $x, y \in R$ such that $x \cdot y \in \sqrt{I}$. Then $(x \cdot y)^m \in I$ for some positive integer m . Thus, $x^m \in I$ or $y^{mn} \in I$ for some positive integer n , which implies that $x \in \sqrt{I}$ or $y \in \sqrt{I}$. $\square$

Definition 2.6.5. Let R be a commutative ring and let $I \subset R$ be a primary ideal. Then, if $\mathfrak{p} := \sqrt{I}$, we say that I is **$\mathfrak{p}$ -primary**. (Here, $\mathfrak{p}$ is defined to be the prime ideal defined by $\sqrt{I}$ as in Proposition 2.6.4.)

Example 2.6.6 ($\sqrt{I}$ prime $\not\Rightarrow I$ is primary)

The converse of the result above is not true. Let k be a field, and suppose $R = k[X, Y]$. Then, consider $I = (X^2, XY) \subset R$. $V(I) = V(X)$. We also have that $\sqrt{I} = (X)$, which is prime. However, I is not primary, since if we consider $XY \in I$, $X \notin I$ and $Y^n \notin I$ for all $n \in \mathbb{N}$.

Exercise 2.6.7.

1. Show that if $R = \mathbb{Z}$ then the primary ideals are (0) and (p^n) where $p \in \mathbb{Z}$ is a prime number and n is a positive integer.
2. Let k be a field, let $R = k[x, y]$ and let $I = (x, y^2)$. Show that I is primary.

Solution: 1. Of course, 0 is a primary ideal. Now, it is also clear that (p^n) is a primary ideal of $\mathbb{Z}$. Let $m \in (p^n)$. Then, p^n must be in the prime factorization of m , so no matter how we *split* m , there will be one part which has p in it. Then, we just take the n -th power of that, so (p^n) is primary. We claim that these are actually all the primary

ideals. Let $I \subset \mathbb{Z}$ be a primary ideal, but since $\mathbb{Z}$ is a PID, every ideal is principal. Then, consider the ideal (pq) ; this is not primary, for $p \neq q$ prime. Note that $p \cdot q \in (pq)$, but $p, q \notin (pq)$, and no powers of p or q are in (pq) . Another way to do this is by setting $I = (m) = (p_1^{a_1} \dots p_\ell^{a_\ell})$. Then, $\sqrt{I}$ is a prime ideal, but $\sqrt{I} = (p_1 \dots p_\ell)$. This is prime if and only if $\ell = 1$.

2. We check that the zero divisors of R/I are nilpotent. We note that

$$R/I \cong k[y]/(y^2).$$

The zero divisors of this ring are therefore elements of the form $[y]$. These are nilpotent. Thus, I is primary. $\square$

Proposition 2.6.8

Let R be a commutative ring and let $I \subset R$ be an ideal such that $\sqrt{I}$ is maximal. Then I is primary.

Proof. Let $\mathfrak{m} = \sqrt{I}$. Then Proposition 1.4.2 implies that $\mathfrak{m}$ is the only prime ideal containing I . Thus, R/I has exactly one prime ideal and a previous exercise implies that every element of R/I is either a unit or nilpotent. Thus, every zero divisor of R/I is nilpotent and Proposition 2.6.3 implies that I is primary. $\square$

Definition 2.6.9. Let R be a commutative ring and let $I \subset R$ be an ideal. A **primary decomposition** of I is an expression of I as a finite intersection of primary ideals, i.e.

$$I = \bigcap_{i=1}^n J_i$$

with $J_1, \dots, J_n \subset R$ primary ideals.

Example 2.6.10

Let $R = k[X, Y]$ and let $I = (X^2, XY)$. The following is a primary decomposition of I :

$$I = (X) \cap (X^2, XY),$$

and (X) , (X^2, XY) are primary. However, we also note that

$$I = (X) \cap (X^2, XY, Y^n), \quad n \geq 2.$$

Therefore, primary decomposition is not unique! We also note that $\sqrt{I} = (x)$, so Proposition 2.6.4 tells us that we can't write I as a finite intersection of prime ideals of R , because if we could, then $I = (x)$, but $x \notin I$.

Example 2.6.11

$(XY) \subseteq k[X, Y]$ is not primary. However, it does have the primary decomposition $(X) \cap (Y) = (XY)$.

Lemma 2.6.12

Let R be Noetherian ring and let $I \subset R$ be an ideal. Then I is a finite intersection of irreducible ideals.

Proof. Consider the set

$$P := \{J \subset R \text{ ideal} : J \text{ is not a finite intersection of irreducible ideals}\}.$$

Assume by contradiction that P is not empty. Then since R is Noetherian, Zorn's Lemma implies that P admits a maximal element J_0 with respect to the inclusion. In particular, J_0 is not irreducible and there exist ideals J_1 and J_2 such that

$$J_0 = J_1 \cap J_2 \quad \text{and} \quad J_0 \neq J_i \quad i = 1, 2.$$

Thus, by maximality, $J_i \notin P$ and therefore it is a finite intersection of irreducible ideals, for $i = 1, 2$. But then also J_0 is a finite intersection of irreducible ideals, a contradiction. $\square$

Lemma 2.6.13

Let R be a Noetherian ring and let $I \subset R$ be an irreducible ideal. Then I is primary.

Proof. Let $S := R/I$. Since I is irreducible, it follows easily that $(0) \subset S$ is an irreducible ideal. By Proposition 2.6.3, it is enough to show that every zero divisor x of S is nilpotent. Let $y \neq 0$ such that $x \cdot y = 0$ and let $J_i := \text{ann}(x^i)$ for all $i \geq 1$. Then

$$J_1 \subset J_2 \subset \dots$$

is an ascending chain of ideals and by ACC, there exists a positive integer n such that $J_i = J_n$ for all $i \geq n$.

We want to show that

$$(x^n) \cap (y) = (0).$$

Indeed, let $r \in (x^n) \cap (y)$. Since $r \in (y)$ we have that $r = y \cdot s$ and since $r \in (x^n)$ we have that $r = x^n \cdot t$ for some $s, t \in S$ and hence $r \cdot x = y \cdot s \cdot x = 0$, since $x \cdot y = 0$. Next, $t \cdot x^{n+1} = rx = 0$. This implies $t \in J_{n+1} = J_n$, which implies that $r = t \cdot x^n = 0$ and so $(x^n) \cap (y) = (0)$ follows.

Finally, since $(0) \subset S$ is irreducible and $(y) \neq (0)$, it follows that $(x^n) = (0)$, i.e. $x^n = 0$. Therefore, x is nilpotent and our claim follows. $\square$

Theorem 2.6.14

Let R be a Noetherian ring and let $I \subset R$ be an ideal. Then I admits a primary decomposition.

Proof. The Theorem follows immediately from Lemma 2.6.12 and Lemma 2.6.13. $\square$

Example 2.6.15 (Primary $\not\Rightarrow$ Irreducible in Noetherian Ring)

Consider the ring $k[x, y]$ (where k is a field) and the ideal $I = (x, y)^2$. Then, I is primary because the radical is $\sqrt{I} = (x, y)$, so maximal. However, we can write $I = (x^2, y) \cap (y, x^2)$, which are both primary ideals. Therefore, I is not irreducible.

Lemma 2.6.16

Let R be a commutative ring and assume that

$$(0) = \mathfrak{m}_1 \dots \mathfrak{m}_n$$

where $\mathfrak{m}_1, \dots, \mathfrak{m}_n$ are maximal ideals (not necessarily distinct). Then R is Noetherian if and only if R is Artinian.

Proof. Let $k_i = R/\mathfrak{m}_i$, which is a field. Consider the chain

$$(0) = \mathfrak{m}_1 \dots \mathfrak{m}_n \subset \dots \subset \mathfrak{m}_1 \mathfrak{m}_2 \subset \mathfrak{m}_1 \subset R.$$

Then, each quotient

$$M_i := (\mathfrak{m}_1 \dots \mathfrak{m}_{i-1})/(\mathfrak{m}_1 \dots \mathfrak{m}_i)$$

is a module over k_i , and therefore a vector space, for each $i = 1, \dots, n$. Thus, as in Example 2.5.3, we have that M_i is Noetherian if and only if it is Artinian. The claim then follows by repeated applications of Exercise 2.5.5. To illustrate how it goes for Noetherian implies Artinian, we first use R Artinian to get that M_i are Artinian. Then, we start from (0) and get the $\mathfrak{m}_1 \dots \mathfrak{m}_i$ is Artinian from the fact that (0) is Artinian, so you can rise up. So it's like a descend then go back up argument. $\square$

Example 2.6.17

Let $R = k[X, Y]/(X, X - Y^2)$. Let $I = (X, X - Y^2)$, and $V(I) = \{(0, 0)\}$. Then, $\dim R = 0$. Consider $J = (X, Y)$. Then, $V(J) = \{(0, 0)\}$, so $I \neq J$, and also I carries more information than J . Now, note that $R \cong k[Y]/(Y^2)$, which is Artinian of dimension 0. Further, $R' = k[X, Y]/(X, Y) \cong k$.

Theorem 2.6.18

Let R be a Noetherian ring. Then $\dim R = 0$ if and only if R is an Artin ring.

Proof. Assume that R is Artinian. Then Proposition 2.4.3 implies that every prime ideal is maximal. Thus, $\dim R = 0$, since we can just take the one prime ideal to be equal $\sqrt{(0)}$, which always exists.

Assume now that $\dim R = 0$. Every ideal then admits a primary decomposition, so we may take

$$(0) = \bigcap_{i=1}^n J_i$$

with $J_1, \dots, J_n \subset R$ being primary ideals. Let $\mathfrak{p} \subset R$ be a prime ideal. We want to show that there exists $i = 1, \dots, n$ such that $J_i \subset \mathfrak{p}$. Suppose not. Then, for all $i = 1, \dots, n$, there exists $x_i \in J_i \setminus \mathfrak{p}$. We have

$$x_1 x_2 \dots x_n \in \bigcap_{i=1}^n J_i = (0) \subset \mathfrak{p}.$$

Thus, $x_1, \dots, x_n \in (0) \subset \mathfrak{p}$. Since $\mathfrak{p}$ is prime, we have that $x_i \in \mathfrak{p}$ for some $i = 1, \dots, n$, a contradiction.

$J_i \subset \mathfrak{p}$ for some i and $\mathfrak{p}$ is prime, so $\sqrt{J_i} \subset \mathfrak{p}$ because $\sqrt{J_i}$ is the minimal prime ideal which contains J_i . Now, since $\dim R = 0$, it must be that $\sqrt{J_i} = \mathfrak{p}$. Since the prime ideal was arbitrary, this implies that R admits finitely many prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_k$. Again, $\dim R = 0$, so they are also maximal ideals.

By Proposition 2.1.10, we have that the nilradical $\mathcal{N}(R)$ of R is nilpotent, i.e. there exists a positive integer q such that $\mathcal{N}(R)^q = 0$. On the other hand, Proposition 1.4.2 implies that

$$\mathcal{N}(R) = \bigcap_{i=1}^k \mathfrak{p}_i.$$

Thus, we have

$$\mathfrak{p}_1^q \dots \mathfrak{p}_k^q \subset \left(\bigcap_{i=1}^k \mathfrak{p}_i \right)^q = \mathcal{N}(R)^q = (0).$$

By Lemma 2.6.16, since R is Noetherian, it follows that R is Artinian. □

Example 2.6.19

A ring with only one prime ideal need not be Noetherian, and hence need not be an Artin ring. The key here is to find some ring with one prime ideal which is not finitely generated. The classic example to rely on for a non-Noetherian ring is $A = k[x_1, x_2, \dots]$, and we can find an example from this. Let $\mathfrak{a}$ be the ideal

$$\mathfrak{a} = (x_1, x_2^2, \dots, x_n^n, \dots).$$

The ring $B = A/\mathfrak{a}$ has only one prime ideal, namely the image of

$$(x_1, x_2, \dots, x_n, \dots),$$

hence B is a local ring of dimension 0. But B is not Noetherian, for it is not difficult to see that its prime ideal is not finitely generated.

Proposition 2.6.20

Let $(R, \mathfrak{m})$ be a Noetherian local ring. Then exactly one of the following is true:

- (i) $\mathfrak{m}^n \neq \mathfrak{m}^{n+1}$ for all $n \geq 0$; or
- (ii) $\mathfrak{m}^n = 0$ for some n and R is an Artin ring.

Proof. Assume that $\mathfrak{m}^n = \mathfrak{m}^{n+1}$ for some $n \geq 0$. Since $\mathfrak{m} = \mathcal{J}(R)$, Nakayama's Lemma (Corollary 2.3.6) implies that $\mathfrak{m}^n = 0$. Let $\mathfrak{p}$ be any prime ideal of R . Since $\mathfrak{m}^n = 0 \subset \mathfrak{p}$, we must have $\mathfrak{m} = \mathfrak{p}$, i.e. $\mathfrak{m}$ is the only prime ideal of R . Thus, $\dim R = 0$ and Theorem 2.6.18 implies that R is an Artin ring. $\square$

Similarly to Proposition 2.1.10 we have the following:

Proposition 2.6.21

Let R be an Artin ring. Then the nilradical $\mathcal{N}(R)$ of R is nilpotent. That is, there exists a positive integer m such that $\mathcal{N}(R)^m = (0)$.

Proof. Since R is an Artin ring, there exists a positive integer m and an ideal $I \subset R$ such that

$$\mathcal{N}(R)^k = I \quad \text{for all } k \geq m.$$

This is because the descending chain

$$\mathcal{N}(R) \supset \mathcal{N}(R)^2 \supset \dots$$

must stabilize. Suppose by contradiction that $I \neq (0)$ and let

$$P := \{J \subset R \text{ ideal} : J \cdot I \neq (0)\}.$$

Then, $\mathcal{N}(R) \in P$, so $P \neq \emptyset$ and by Zorn's lemma there exists a minimal element $J_0 \in P$ with respect to inclusion. Thus, there exists $x \in J_0$ such that $x \cdot I \neq (0)$. Since $(x) \subset J_0$ and J_0 is minimal, it follows that $(x) = J_0$. It follows that

$$(xI) \cdot I = xI^2 = xI \neq (0),$$

so $xI \in P$. Since $xI \subset (x)$, by minimality of J_0 we have that $xI = (x) = J_0$. Thus, $x \in xI$ and therefore, there exists $y \in I$ such that $x = xy$. Hence,

$$x = xy = \dots = xy^n \quad \text{for all } n \geq 1.$$

Since $y \in I = \mathcal{N}(R)^m \subset \mathcal{N}(R)$, it follows that y is nilpotent. Thus, if n is sufficiently large, we have $x = xy^n = 0$, a contradiction. Thus the claim follows. $\square$

Exercise 2.6.22 (Hopkins–Levitzki Theorem). Let R be a commutative ring. Show that R is Artinian if and only if R is Noetherian and $\dim R = 0$.

Solution. We already know that if R is Noetherian, then R is Artinian if and only if $\dim R = 0$. Thus, we have the if direction. For the forward implication, it suffices to show that every Artinian ring is Noetherian. Every prime ideal of R is maximal, so firstly, $\dim R = 0$ is clear.

Next, let $\mathfrak{m}_i$ be maximal ideals of R , and consider the chain of maximal ideals:

$$\mathfrak{m}_1 \supset \mathfrak{m}_1 \cap \mathfrak{m}_2 \supset \mathfrak{m}_1 \cap \mathfrak{m}_2 \cap \mathfrak{m}_3 \supset \dots$$

This is a descending sequence, so by R being Artinian, it stabilizes. What this tells us is that there are only finitely many maximal ideals, say n of them, since after some point,

$$\mathfrak{m}_1 \cap \dots \cap \mathfrak{m}_n = \mathfrak{m}_1 \cap \dots \cap \mathfrak{m}_{n+1},$$

so $\mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_n \subset \mathfrak{m}_{n+1}$. We now claim that $\mathfrak{m}_i \subset \mathfrak{m}_{n+1}$ for some i . If not, then there is some $x_i \in \mathfrak{m}_i \setminus \mathfrak{m}_{n+1}$, for all i . Then, taking the product, $x_1 \cdots x_n \in \mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_n \subset \mathfrak{m}_{n+1}$. $\mathfrak{m}_{n+1}$ is maximal, so it's prime, and so we get a contradiction. Thus, we achieve equality: $\mathfrak{m}_i = \mathfrak{m}_{n+1}$ for some i , by maximality. Now, note that in Artin rings, we also have that $\mathcal{J}(R) = \mathcal{N}(R) = \sqrt{(0)}$. Since the nilradical is nilpotent in Artin rings, it follows then that the Jacobson radical is nilpotent in Artin rings. Therefore,

$$\left(\bigcap_{i=1}^n \mathfrak{m}_i \right)^q = (0),$$

for some q . Now, since $\mathfrak{m}_1 \cdots \mathfrak{m}_n \subseteq \bigcap_{i=1}^n \mathfrak{m}_i$, it follows then that

$$\mathfrak{m}_1^q \cdots \mathfrak{m}_n^q = (0),$$

so R being Artinian implies that R is Noetherian. $\square$

Exercise 2.6.23. Let R be a commutative ring and let $I \subset R$ be an irreducible ideal. Show that the following are equivalent:

1. I is primary
2. For any $x \in R$, the chain defined by $I_n := (I : (x^n))$, with $n \geq 1$ is stationary, i.e. there exists a positive integer m such that $I_n = I_m$ for all $n \geq m$.

Solution: Recall that if $I = J_1 \cap J_2$, then $I = J_1$ or $I = J_2$. Also, we have the results that if R is Noetherian and $I \subset R$ is irreducible, then I is primary. First, note that if $yx^n \in I$, then $yx^{n+1} \in I$, so $I_n \subset I_{n+1}$. First assume that I is primary. We distinguish between two cases. First, $x \in \sqrt{I}$, i.e. $x^n \in I$ for some $n \geq 1$. Then, we have that $I_m = R$ for all $m \geq n$, since $I_m = \{y \in R : yx^m \in I\}$, but $yx^m \in I$ for $m \geq n$, so there is no condition on y . Thus, $I_m = R$. The next case is when $x \notin \sqrt{I}$. Then, we want to show that $I_n = I$ for all n . Indeed, if $y \in I_n$, then $y \cdot x^n \in I$. Then, $(x^n)^k = x^{nk} \notin I$, for any k , so since I is primary, $y \in I$. The reverse inclusion is clear; $I \subset I_n$ is clearly true. Thus, we have proved the forward implication.

Time for the reverse implication. Suppose that $I_n = I_m$, for all $m \geq n$. We want to show that I is primary. Take $xy \in I$. We want to show that $x^m \in I$ for some $m \geq 1$ or $y \in I$. We choose $m = n$, where n is the fixed integer at the beginning. Define $J_1 = I + (x^n)$ and $J_2 = I + (y)$. We now claim that $J_1 \cap J_2 = I$. The reason why we make this claim is because I is irreducible! This tells us that $I = J_1$ or $I = J_2$. In the former, we get that $x^n \in I$, and in the latter, we get $y \in I$, so this suffices. We now work to $J_1 \cap J_2 = I$. Of course, $I \subseteq J_1 \cap J_2$. Let $s \in J_1 \cap J_2$. $s \in J_2$, so $s = a + by$, where $a \in I$ and $b \in R$. Multiplying this by x , we get $sx = ax + byx$. Now, since we chose xy to be in I , we have that $sx \in I$. Next, $s \in J_1$ as well, so write $s = c + dx^n$, where $c \in I$ and $d \in R$. Multiplying by x , we get $sx = cx + dx^{n+1}$. However, $sx \in I$, so $dx^{n+1} \in I$. By definition, $d \in I_{n+1} = I_n$ by the hypothesis. Thus, $dx^n \in I$. Therefore, $s = c + dx^n \in I$, so we get $J_1 \cap J_2 \subset I$. $\square$

§2.7 Exercises from [2]

Exercise 2.7.1. If A is a ring in which every ideal has a primary decomposition, show that every ring of fractions $S^{-1}A$ has the same property.

Solution: For this, we note that ideals of the localization $S^{-1}A$ are of the form $S^{-1}I$, where $I \subseteq A$ is an ideal. We also wish to show that if I is primary and $S \cap I = \emptyset$, then $S^{-1}I$ is primary. Let $\frac{a}{s}, \frac{b}{t} \in S^{-1}R$ such that $\frac{a}{s}, \frac{b}{t} \in S^{-1}I$. Then, $ab \in I$, so $a \in I$ or $b^n \in I$ from I being primary. Further, $s, t \in S$, so $s \in S$ and $t^n \in S$. Therefore, $\frac{a}{s} \in S^{-1}I$ or $\left(\frac{b}{t}\right)^n \in S^{-1}I$. Now, suppose that $I = \cap_{i=1}^n J_i$, where each J_i is primary. Then, if $S \cap J_i \neq \emptyset$, then $S^{-1}J_i = S^{-1}A$. If $S \cap J_i = \emptyset$, then $S^{-1}J_i$ remains a proper ideal, and so the above says that it is primary. Therefore, $S^{-1}I = \cap_{i=1}^n S^{-1}J_i$. $\square$

Exercise 2.7.2. Let M be a Noetherian R -module. Then, show that $M[X]$ is a Noetherian $R[X]$ -module.

Solution: One way to prove this is by repeating the argument for the Hilbert basis theorem, but essentially replacing $R[X]$ with $M[X]$ and ideals of R with submodules of M . $\square$

3 The Nullstellensatz

§3.1 Algebras

Definition 3.1.1. Let R be a commutative ring. A R -algebra (or algebra over R) is a ring A which admits a morphism of rings $\varphi: R \rightarrow A$.

Remark 3.1.2 — Note that every R -algebra A is an R -module, where the R -module action is defined by $r \cdot a = \varphi(r)a$, for any $r \in R$ and $a \in A$. In fact, we often denote $r \cdot a := \varphi(r)a$

Example 3.1.3

If S is a commutative ring and $R \subset S$ is a subring then S is an R -algebra, which is called an **extension** of R .

Definition 3.1.4. Let R be a commutative ring and let $\varphi: R \rightarrow A$ be a R -algebra. Then we have the following definitions:

- A is said to be **finite** over R if it is finitely generated as an R -module.
- An element $a \in A$ is said to be **integral over** R if there exists a monic polynomial

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in R[x]$$

such that $f(a) = 0$.

- A is said to be **integral over** R if every $a \in A$ is integral.
- Let $S \subset A$ be a subset. Then we denote by $R[S] \subset A$ the **R -algebra generated by** S , i.e. the smallest subring of A containing $\varphi(R)$ and S .

Note that there is a big difference between being generated as an algebra or as a module, as shown in the following example.

Example 3.1.5

Let k be a field and let $A = k[x]$. Then A is a k -algebra and, therefore, it is also a k -module. The k -module generated by x is the k -module

$$\{a \cdot x : a \in k\} \subset k[x].$$

On the other hand, the k -algebra generated by x is the whole algebra A . Note that this is not finite, but finitely generated.

Lemma 3.1.6

Let R, S and T be commutative rings such that $R \subset S \subset T$. Assume that T is a finite S -algebra and S is a finite R -algebra. Then T is a finite R -algebra.

Proof. We have that $T = \langle t_1, \dots, t_s \rangle$ as an S -module and $S = \langle s_1, \dots, s_r \rangle$ as an R -module. Then, $T = \langle t_i s_j : 1 \leq i \leq s, 1 \leq j \leq r \rangle$. This is because every $t \in T$ can be written as a finite sum $t = \sum_{i=1}^s s_i t_i$, but each S_i can be written as a sum $\sum_{j=1}^r r_i s_j$. $\square$

Proposition 3.1.7

Let R be a commutative ring, let $\varphi: R \rightarrow A$ be an algebra over R and let $a \in A$. Then a is integral over R if and only if $R[a]$ is finite over R .

Proof. Assume that a is integral over R . Then $R[a]$ is the R -module generated by $1, a, \dots, a^{n-1}$. Thus, $R[a]$ is finite over R .

Assume now that $R[a]$ is finite over R . Then there exist $r_{ij} \in R$ with $i = 1, \dots, m$ and $j = 1, \dots, n_i$ such that if

$$f_i = \sum_{j=0}^{n_i} \varphi(r_{ij}) a^j \in R[a] \quad \text{for } i = 1, \dots, m,$$

then $f_1, \dots, f_m$ generate $R[a]$. Let $n = \max\{n_1, \dots, n_m\}$. Then, since $a^{n+1} \in R[a]$, it follows that

$$a^{n+1} = \sum_{i=0}^n \varphi(s_i) a^i$$

for some $s_0, \dots, s_n \in R$. Thus, a is integral over R . $\square$

Proposition 3.1.8

Let $R \subset S$ be commutative rings. Then S is finite over R if and only if $S = R[\alpha_1, \dots, \alpha_n]$ for some $\alpha_1, \dots, \alpha_n \in S$ which are integral over R . Moreover, in this case, S is integral over R .

Proof. Let $\alpha_1, \dots, \alpha_n$ be integral over R . Then by Proposition 3.1.7 and Lemma 3.1.6, it follows immediately that $S = R[\alpha_1, \dots, \alpha_n]$ is finite over R .

Assume now that S is finite over R and let $\alpha_1, \dots, \alpha_n$ be generators of S over R . We prove that every element of S is integral over R , so that S is integral over R and, in particular, $\alpha_1, \dots, \alpha_n$ are integral over R . Let $a \in S$ and consider the R -linear morphism

$$\varphi: S \longrightarrow S, \quad s \longmapsto a \cdot s.$$

Then by the Cayley–Hamilton theorem (Theorem 2.3.3) there exist $c_0, \dots, c_{k-1} \in R$ such that

$$\varphi(s)^k + c_{k-1} \varphi(s)^{k-1} + \dots + c_0 \cdot s = 0$$

for all $s \in S$. If $s = 1$, then $\varphi(1) = a$, so

$$a^k + c_{k-1} a^{k-1} + \dots + c_0 = 0.$$

Thus, a is integral over R , as claimed. $\square$

Remark 3.1.9 — Note that the converse doesn't hold; that is, S integral over R doesn't imply S being finite over R .

Example 3.1.10

Let k be a field and let

$$k[x^2] := \left\{ \sum_{i=0}^n a_i x^{2i} : a_i \in k \right\}.$$

Then $k[x]$ is an extension of $k[x^2]$, which is integral and finite.

Example 3.1.11 (Integral $\not\Rightarrow$ Finite)

Let k be a field, let $R = k[x]$ and consider the R -algebra

$$A = k[x, x^{1/2}, x^{1/4}, x^{1/8}, \dots]$$

which is the ring obtained by adjoining all 2^n -th roots of x to R . Then A is integral over R but it is not finite over R .

Example 3.1.12

Let R be a commutative ring of characteristic p . Then the **Frobenius morphism** is defined as

$$F: R \rightarrow R, \quad r \mapsto r^p.$$

Recall that R is said to be **perfect** if F is surjective. Note that the Frobenius morphism is not a ring morphism in rings of characteristic not p , due to the Freshman's dream. Assume now that k is a field of characteristic p and $R = k[x]$. Then, this is not perfect, since for example, $t^p = x$ has no solutions. However, R is an integral extension of the image $F(R)$.

Example 3.1.13

Recall that if p is a prime, we defined

$$\mathbb{Z}[1/p] := \left\{ \frac{a}{p^k} : a \in \mathbb{Z}, k \geq 0 \right\}.$$

Then $\mathbb{Z}[1/p]$ is a $\mathbb{Z}$ -algebra which is not integral.

Exercise 3.1.14. Let $A = \mathbb{Q}[X, Y, Z]/(XY - Z^2)$. Find a polynomial extension B of $\mathbb{Q}$ such that A is finite over B .

Solution: Note that in A , $Z^2 = XY$. Thus, let $B = \mathbb{Q}[x, y]$. Note that there is a natural inclusion $B \hookrightarrow A$, and A is a finitely generated B -module, with generators $1, z \in A$. Thus, A is finite over B . $\square$

Exercise 3.1.15. Let R be a ring and A be a finite R -algebra. Then, if R is Noetherian, A is Noetherian.

Solution: A is a finitely generated R -module, so is a Noetherian R -module, so A is a Noetherian A -module. Thus, A is a Noetherian ring. $\square$

Example 3.1.16

The ring of integers of any number field is Noetherian.

Exercise 3.1.17. Let B be a finitely generated A -algebra. If A is Noetherian, then so is B .

Solution: B is the image of some map $A[x_1, \dots, x_n] \rightarrow B$ which is surjective. The polynomial ring is Noetherian, by the Hilbert basis theorem. $\square$

§3.2 Normal Rings and Integral Closure

The following result is a variation of Lemma 3.1.6.

Lemma 3.2.1

Let R, S and T be commutative rings such that $R \subset S \subset T$. Assume that T is integral over S and S is integral over R . Then T is integral over R .

Proof. Let $t \in T$. Since T is integral over S , there exist $b_0, \dots, b_{n-1} \in S$ such that

$$t^n + b_{n-1}t^{n-1} + \dots + b_0 = 0.$$

Thus, t is integral over $R[b_0, \dots, b_{n-1}]$, so Proposition 3.1.7 implies that $R[b_0, \dots, b_{n-1}, t]$ is finite over $R[b_0, \dots, b_{n-1}]$.

Since S is integral over R , we have that $b_0, \dots, b_{n-1}$ are integral over R . Thus, Proposition 3.1.8 implies that $R[b_0, \dots, b_{n-1}]$ is finite over R . By Lemma 3.1.6, $R[b_0, \dots, b_{n-1}, t]$ is finite over R . By Proposition 3.1.8 again, we have that t is integral over R . $\square$

Proposition 3.2.2

Let R be a commutative ring and let A be an algebra over R . Then

$$\tilde{R} := \{a \in A : a \text{ is integral over } R\}$$

is a subring of A . Moreover, if $b \in A$ is integral over $\tilde{R}$ then $b \in \tilde{R}$.

Proof. Let $a, b \in \tilde{R}$ then $a \pm b, ab \in R[a, b]$. By Proposition 3.1.8, we have that $R[a, b]$ is finite over R , so integral over R . Thus, $R[a, b] \subset \tilde{R}$. Thus, $a \pm b, ab \in \tilde{R}$ and, therefore, $\tilde{R}$ is a subring of A .

The second part of the proposition follows immediately from Lemma 3.2.1. Simply construct a chain of inclusions

$$\tilde{R}[b] \supset \tilde{R} \supset R,$$

so b is integral of R , so $b \in \tilde{R}$. $\square$

Definition 3.2.3. Let R be a commutative ring and let A be an algebra over R . The ring $\tilde{R}$ in Proposition 3.2.2 is called the **integral closure** of R in A . If $\tilde{R} = R$ then R is called **integrally closed** in A .

Definition 3.2.4. Assume now that R is an integral domain and let $A = \text{Frac}(R)$. The integral closure of R in A is called the **normalization** of R . If R is integrally closed in A then R is called **normal**.

Exercise 3.2.5. Show that any UFD is normal.

Solution: Let R be a UFD, and let $A = \text{Frac}(R)$. We want to show that the roots in A of any monic polynomial in $R[X]$ are contained in R . Let $f(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0 \in R[X]$, and suppose $a \in A$ is a root of this. Then, we have that

$$a^n + a_{n-1}a^{n-1} + \cdots + a_0 = 0.$$

If $a = \frac{p}{q}$, with $p, q \in R$ and $q \neq 0$, and suppose that there is some irreducible element r dividing q but not p . If there weren't, then $a \in R$. We then multiply the equation by q^n . This then gives the following relation with all terms in R :

$$p^n + qa_{n-1}a^{n-1} + \cdots + a_0q^n = 0$$

Now, note that everything but the first term is divisible by r since R is a UFD, which is a contradiction. Thus, it must be that q is a unit, and so $a \in R$. $\square$

Example 3.2.6

Recall that a number field $K \supset \mathbb{Q}$ is a finite extension of the rational field $\mathbb{Q}$, i.e. an extension of $\mathbb{Q}$ which is a finite-dimensional vector space over $\mathbb{Q}$. In this case, the integral closure of $\mathbb{Z}$ in K is called the **ring of integers** $\mathcal{O}_K$ of K .

For example, it follows from elementary number theory that if n is a square-free positive integer (i.e. a product of distinct prime numbers) and $K = \mathbb{Q}(\sqrt{n})$ is the field extension generated by $\sqrt{n}$ then the ring of integers of K is

$$\mathcal{O}_K = \mathbb{Z}[\alpha]$$

where

$$\alpha = \begin{cases} \frac{1 + \sqrt{n}}{2}, & \text{if } n \equiv 1 \pmod{4}, \\ \sqrt{n}, & \text{if } n \equiv 2 \text{ or } 3 \pmod{4}. \end{cases}$$

Proof. Let $a + b\sqrt{-n} \in \mathcal{O}_K$. Then,

$$(a + b\sqrt{-n})^2 = a^2 - bn^2 + 2ab\sqrt{-n} = a^2 - bn^2 + 2a(a + b\sqrt{-n}) - 2a^2.$$

Thus, $a + b\sqrt{-n}$ satisfies the following:

$$(a + b\sqrt{-n})^2 - 2a(a + b\sqrt{-n}) + (a^2 + b^2n) = 0.$$

Here, $a, b \in \mathbb{Q}$. It is then clear that $a \in \frac{1}{2}\mathbb{Z}$. Thus, $b \in \frac{1}{2}\mathbb{Z}$, otherwise, we can't get the coefficients to be in $\mathbb{Z}$. Thus, we can just set $a, b = \frac{1}{2}\mathbb{Z}$. because if we add by an integer, it's still in $\mathcal{O}_K$. If $n \equiv 1 \pmod{4}$, then n is an odd square, so we may take $a \equiv b \pmod{2}$. If otherwise, then a, b must be even, but then we can just takeaway a , so $\sqrt{-n}$ remains. $\square$

Example 3.2.7

Let k be a field and let

$$R := k[x, y]/(x^2 - y^3).$$

We show that R is not normal. We want to find $t \in \text{Frac}(R)$ such that it's integral over R but not contained inside R . Then the field of fraction of R is

$$\text{Frac}(R) = k(t) \quad \text{where } t = \frac{x}{y}.$$

We have

$$t^2 - y = \frac{x^2 - y^3}{y^2} = 0 \quad \text{in } R.$$

On the other hand, t is not contained in R . Thus R is not normal. We now find the integral closure of R , $\tilde{R}$. Note that $R \subset \tilde{R} \subset \text{Frac}(R)$. We get that $\tilde{R} = R[t] \cong k[t]$.

Exercise 3.2.8. Let R be an integral domain. Show that the following are equivalent:

1. R is normal.
2. $R_{\mathfrak{p}}$ is normal for any prime ideal $\mathfrak{p} \subset R$.
3. $R_{\mathfrak{m}}$ is normal for any maximal ideal $\mathfrak{m} \subset R$.

Proof. Note that, for any prime ideal $\mathfrak{p} \subset R$, we have

$$K := \text{Frac}(R) = \text{Frac}(R_{\mathfrak{p}}).$$

(1) $\implies$ (2): Let $x \in K$ be an integral element over $R_{\mathfrak{p}}$. Then there exist

$$\frac{b_0}{c_0}, \dots, \frac{b_{n-1}}{c_{n-1}} \in R_{\mathfrak{p}}$$

such that $c_0, \dots, c_{n-1} \notin \mathfrak{p}$ and

$$x^n + \frac{b_{n-1}}{c_{n-1}}x^{n-1} + \dots + \frac{b_0}{c_0} = 0.$$

Then if

$$y = c_0 \cdots c_{n-1} x,$$

we have

$$y^n + a_{n-1}y^{n-1} + \dots + a_0 = 0$$

for some $a_0, \dots, a_{n-1} \in R$. Since R is normal, it follows that $y \in R$. Since $\mathfrak{p}$ is prime, we have

$$c := c_0 \cdots c_{n-1} \notin \mathfrak{p}.$$

Thus,

$$x = \frac{y}{c} \in R_{\mathfrak{p}}.$$

Hence $R_{\mathfrak{p}}$ is normal.

(2) $\implies$ (3): Since each maximal ideal is prime, the claim follows immediately.

(3) $\Rightarrow$ (1) Let $y \in K$ be integral over R . Then there exist $a_0, \dots, a_{n-1} \in R$ such that

$$y^n + a_{n-1}y^{n-1} + \dots + a_0 = 0.$$

Assume by contradiction that $y \notin R$ and define

$$I := \{b \in R \mid by \in R\}.$$

Then $I \subset R$ is a proper ideal. By Zorn's lemma, there exists a maximal ideal $\mathfrak{m} \subset R$ such that $I \subset \mathfrak{m}$. Since $R_{\mathfrak{m}}$ is normal, it follows that $y \in R_{\mathfrak{m}}$. Thus,

$$y = \frac{c}{b}$$

with $b, c \in R$ such that $b \notin \mathfrak{m}$. But then

$$by = c \in R,$$

which implies that $b \in I \subset \mathfrak{m}$, a contradiction. Therefore $y \in R$, so R is normal. $\square$

§3.3 Noether Normalization Lemma

Definition 3.3.1. If k is a field then a polynomial $f(x_1, \dots, x_n) \in k[x_1, \dots, x_n]$ is called **homogeneous** if it is a sum of monomials of the same degree.

Lemma 3.3.2

Let k be an infinite field and let $f(x_1, \dots, x_n) \in k[x_1, \dots, x_n]$ be a non-zero homogeneous polynomial. Then, there exist $c_1, \dots, c_{n-1} \in k$ such that

$$f(c_1, \dots, c_{n-1}, 1) \neq 0.$$

Proof. Indeed, we proceed by induction on n . If $n = 1$ then f is a non-zero multiple of a monomial and therefore $f(1) \neq 0$. Assume now that $n > 1$. We may write

$$f(x_1, \dots, x_n) = \sum_{i=0}^d x_1^i f_i(x_2, \dots, x_n)$$

for some homogeneous polynomials $f_i \in k[x_2, \dots, x_n]$ of degree $d - i$. Since f is non-zero, we may assume that f_i is non-zero for some $i = 0, \dots, d$. By induction, there exist $c_2, \dots, c_{n-1}$ such that

$$f_i(c_2, \dots, c_{n-1}, 1) \neq 0.$$

Let

$$g(x) := f(x, c_2, \dots, c_{n-1}, 1) \in k[x].$$

Then g is a non-zero polynomial in one variable and, in particular, it has only finitely many zeroes. Since k is infinite, there exists $c_1 \in k$ such that

$$f(c_1, c_2, \dots, c_{n-1}, 1) \neq 0,$$

as claimed. $\square$

Remark 3.3.3 — Note that the claim in the previous remark is not true over a finite field. For example, consider $k := \mathbb{Z}/p\mathbb{Z}$ for some prime p and let

$$f(x_1, x_2) := x_1^p - x_2^p \in k[x_1, x_2].$$

Then, by Fermat's little theorem, $f(c, 1) = 0$ for all $c \in k$.

Definition 3.3.4. Let k be a field and let A be an algebra over k . Then the elements $a_1, \dots, a_n \in A$ are said to be **algebraically independent** over k if there does not exist a non-zero polynomial $F \in k[x_1, \dots, x_n]$ such that

$$F(a_1, \dots, a_n) = 0.$$

Equivalently, the morphism

$$k[x_1, \dots, x_n] \longrightarrow k[a_1, \dots, a_n], \quad x_i \longmapsto a_i \quad \text{for } i = 1, \dots, n$$

is an isomorphism.

Remark 3.3.5 — In the case $n = 1$, x_1 is algebraically independent if and only if x_1 is not integral of k . If $F(x_1) = 0$, then we can say that F is monic because every element is a unit in k . Thus, x_1 is integral over k .

Remark 3.3.6 — $a_1, \dots, a_n \in A$ are algebraically independent if and only if the morphism

$$\begin{aligned} \varphi: k[x_1, \dots, x_n] &\rightarrow k[a_1, \dots, a_n] \\ x_i &\mapsto a_i \end{aligned}$$

is an isomorphism. Note that $k[x_1, \dots, x_n]$ is the polynomial ring, and $k[a_1, \dots, a_n]$ is the algebra generated by a_i ; in particular, this lives in A . We note that φ is always surjective. To be injective, the kernel must be zero; that is, if and only if there are no polynomials that vanish on the $(a_1, \dots, a_n)$.

The following theorem states that if A is a finitely generated algebra over a field k , then we can find a decomposition

$$k \subset k[a_1, \dots, a_n] \subset A$$

where $k[a_1, \dots, a_n]$ is a polynomial extension of k and A is a finite extension of it.

Theorem 3.3.7 (Noether Normalization Lemma)

Let k be a field and let A be a finitely generated k -algebra. Then there exist $a_1, \dots, a_n \in A$ such that

- (i) $a_1, \dots, a_n$ are algebraically independent over k ;
- (ii) A is finite over $k[a_1, \dots, a_n]$.

Proof. We first assume that k is an infinite field, as in this case the proof is easier. Since A is a finitely generated algebra over k , there exist $z_1, \dots, z_m$ be generators of A over k : $A = k[z_1, \dots, z_m]$. At this point, we know nothing regarding algebraic independence. We proceed by induction on m . Assume first that $m = 1$. If z_1 is integral (so not algebraically independent) over k , then Proposition 3.1.8 implies that $A = k[z_1]$ is finite over k and, thus, we are done, as $n = 0$. If z_1 is not integral over k then z_1 is algebraically independent and we can choose $a_1 = z_1$ so that $A = k[a_1]$.

Assume now that $m > 1$. If $z_1, \dots, z_m$ are algebraically independent over k then we may choose $m = n$ and $a_i = z_i$ for $i = 1, \dots, m$ and so $A = k[a_1, \dots, a_m] \supset k$, so we are done. Now, if $z_1, \dots, z_m$ are not algebraically independent over k , then there exists a non-zero polynomial $f \in k[x_1, \dots, x_m]$ such that

$$f(z_1, \dots, z_m) = 0.$$

So far, note that we haven't used the fact that k is infinite, and this is where we shall split off to use this.

Let $d = \deg f$ and let f_d be the homogeneous part of f of degree d , then we can write $f = f_d + \text{lot}$. Since k is infinite, by Lemma 3.3.2 we may find $c_1, \dots, c_{m-1} \in k$ such that

$$f_d(c_1, \dots, c_{m-1}, 1) \neq 0.$$

Define

$$y_i := z_i - c_i z_m \quad \text{for } i = 1, \dots, m-1.$$

Then

$$\begin{aligned} 0 &= f(z_1, \dots, z_m) = f(y_1 + c_1 z_m, \dots, y_{m-1} + c_{m-1} z_m, z_m) \\ &= \underbrace{f_d(c_1, \dots, c_{m-1}, 1)}_{\neq 0} z_m^d + (\text{lot in } z_m). \end{aligned}$$

Since we're working over a field, we may divide by $f_d(c_1, \dots, c_{m-1}, 1)$ to conclude that z_m is integral over $k[y_1, \dots, y_{m-1}]$. By Proposition 3.2.2, we have that $z_i = y_i + c_i z_m$ is also integral over $k[y_1, \dots, y_{m-1}]$ for $i = 1, \dots, m-1$, since the integral elements form a ring. Thus, Proposition 3.1.8 implies that A is finite over $k[y_1, \dots, y_{m-1}]$. By induction, $k[y_1, \dots, y_{m-1}] \supset k[a_1, \dots, a_n]$ a finite extension, where $a_1, \dots, a_n$ are algebraically independent over k . Thus, $A = k[z_1, \dots, z_m]$ is finite over $k[a_1, \dots, a_n]$.

We now consider the general case (with k not necessarily infinite). As above, we let $z_1, \dots, z_m$ be generators of A and we proceed by induction on m and, as above, may assume that $m > 1$ and that there exists a non-zero polynomial $f \in k[x_1, \dots, x_m]$ (polynomial ring) such that

$$f(z_1, \dots, z_m) = 0.$$

Pick some positive integers $r_1, \dots, r_m$ (we will impose some extra assumptions on the r_i 's later) and define

$$y_i := x_i - x_m^{r_i} \quad \text{for } i = 1, \dots, m-1.$$

Then, if $\alpha = (\alpha_1, \dots, \alpha_m) \in \mathbb{N}^m$, we consider the monomial $x^\alpha := x_1^{\alpha_1} \cdots x_m^{\alpha_m}$. We ave that

$$x^\alpha = (y_1 + x_m^{r_1})^{\alpha_1} \cdots (y_{m-1} + x_m^{r_{m-1}})^{\alpha_{m-1}} x_m^{\alpha_m} = x_m^{N(\alpha)} + (\text{lot in } x_m),$$

where

$$N(\alpha) = \sum_{i=1}^{m-1} r_i \alpha_i + \alpha_m.$$

Thus, if we write

$$\begin{aligned} f &= \sum_{\alpha \in \mathbb{N}^m} a_\alpha x^\alpha \\ &= \sum_{\alpha \in \mathbb{N}^m} (a_\alpha \cdot (x_m^{N(\alpha)} + (\text{lot in } x_m \text{ of degree } < N(\alpha))))). \end{aligned}$$

We need to be sure that the terms $\sum a_\alpha x_m^{N(\alpha)}$ does not cancel out. To do this, we choose the integers r_i to make the function

$$N: \{\alpha \in \mathbb{N}^m : a_\alpha \neq 0\} \rightarrow \mathbb{N}, \quad \alpha \mapsto N(\alpha)$$

injective, so that no cancellation among the top powers of x_m can take place. We show one way of doing this. Let r be a positive integer such that $r > \alpha_i$ and for all $\alpha \in \mathbb{N}^m$ such that $a_\alpha \neq 0$; define $r_i := r^{m-i}$. Then, it is easy to check that

$$N(\alpha) = r^{m-1} \alpha_1 + r^{m-2} \alpha_2 + \cdots + r \alpha_{m-1} + \alpha_m$$

satisfies the injectivity criterion. α_i are just integers, so this is essentially just a r -adic expansion. Thus, if $N = \max\{N(\alpha) \mid a_\alpha \neq 0\}$, we have

$$f = ax_m^N + \sum_{i=0}^{N-1} g_i x_m^i$$

where $a \in k$ is non-zero and $g_i \in k[y_1, \dots, y_{m-1}]$. Since $f(z_1, \dots, z_m) = 0$, it follows that z_m is integral over $k[y_1, \dots, y_{m-1}]$ and we can proceed as above. $\square$

Remark 3.3.8 — Note that we really use the fact that k is infinite in the first portion when we say that $f_d(c_1, \dots, c_{m-1}, 1) \neq 0$. This is due to the characteristic of k possibly dividing this number. Therefore, what we're really doing is case working over fields of characteristic zero or positive.

Remark 3.3.9 — What this tells us is that any finitely generated k -algebra splits into a finite extension and finitely generated part:

$$A \supset k[a_1, \dots, a_n] \supset k.$$

§3.4 Nullstellensatz Theorem

Lemma 3.4.1

Let A and B be rings such that $A \subset B$ and B is integral over A . Then A is a field if and only if B is a field.

Proof. Assume first that A is a field and let $b \in B$ be a non-zero element. Then b is integral over A and therefore there exist $a_0, a_1, \dots, a_{n-1} \in A$ such that

$$b^n + a_{n-1}b^{n-1} + \dots + a_0 = 0.$$

We may assume that $a_0 \neq 0$ as otherwise we can divide by b and we get an equation of lower degree. Define

$$c := -a_0^{-1}(b^{n-1} + a_{n-1}b^{n-2} + \dots + a_1) \in B.$$

It is easy to check that $b \cdot c = 1$, i.e. c is the inverse of b . Thus, B is a field.

Assume now that B is a field and let $a \in A$ be a non-zero element. Then $c := a^{-1} \in B$ and, by assumption, c is integral over A . Thus, there exist $a_0, \dots, a_{n-1} \in A$ such that

$$c^n + a_{n-1}c^{n-1} + \dots + a_0 = 0.$$

Thus, dividing by c^{n-1} , we get

$$c + a_{n-1} + \dots + a_0c^{-n+1} = 0.$$

It follows that

$$c = -a_{n-1} - a_{n-2}c^{-1} - \dots - a_0c^{-n+1} = -a_{n-1} - a_{n-2}a - \dots - a_0a^{n-1} \in A.$$

It follows that A is a field. □

Definition 3.4.2. Let k be a field and let $k \subset K$ be a field extension. Then we say that K is a **finite field extension** if K is a finite dimensional vector space over k , i.e. if K is a finitely generated module over k .

Remark 3.4.3 — Recall that if k is an algebraically closed field then any finite field extension of k coincides with k .

Proof. First, finite extensions are integral. Then, for any $x \in K$, then we can find a monic polynomial $f \in k[X]$ such that $f(x) = 0$. By algebraic closure, it follows that $x \in k$. □

Theorem 3.4.4 (Weak Nullstellensatz)

Let k be a field and let $k \subset K$ be a field extension such that K is a finitely generated algebra over k . Then $k \subset K$ is a finite field extension. Moreover, if k is algebraically closed, then $k = K$.

Proof. By the Noether normalization theorem (Theorem 3.3.7), there exist $x_1, \dots, x_m \in K$ algebraically independent and such that K is finite over $k[x_1, \dots, x_m]$. By Proposition 3.1.8, K is integral over $k[x_1, \dots, x_m]$, and since K is a field, Lemma 3.4.1 implies that also $k[x_1, \dots, x_m]$ is a field.

As the x_i 's are algebraically independent, we have that $k[x_1, \dots, x_m]$ is isomorphic to a polynomial ring of m variables and this is a field if and only if $m = 0$. This shows that $m = 0$. Therefore, since the intermediate ring $k[x_1, \dots, x_n]$ is not there, we get that $k \subset K$ is finite. In particular, if K is algebraically closed then $K = k$. □

Corollary 3.4.5

Let k be an algebraically closed field and let $\mathfrak{m} \subset k[x_1, \dots, x_n]$ be a maximal ideal. Then there exist $a_1, \dots, a_n \in k$ such that

$$\mathfrak{m} = (x_1 - a_1, \dots, x_n - a_n).$$

Proof. We have that $K := k[x_1, \dots, x_n]/\mathfrak{m}$ is a field. Moreover K is finitely generated over k , with generators $x_1 + \mathfrak{m}, \dots, x_n + \mathfrak{m}$. Since k is algebraically closed, Theorem 3.4.4 implies that the natural inclusion $i: k \hookrightarrow K$ is an isomorphism. Then, i is surjective, so there exist $a_1, \dots, a_n \in k$ such that $i(a_j) = x_j + \mathfrak{m}$, for $j = 1, \dots, n$. It then follows that

$$(x_1 - a_1, \dots, x_n - a_n) \subset \mathfrak{m},$$

and by maximality of the left ideal, equality holds. Therefore, $\mathfrak{m}$ is generated by the $(x_j - a_j) \in K$ for $j = 1, \dots, n$. $\square$

Remark 3.4.6 — Recall that the ideal $\mathfrak{m} = (x_1 - a_1, \dots, x_n - a_n) \subset k[x_1, \dots, x_n]$ is always maximal, even without assuming that k is algebraically closed. The opposite, in general (without algebraic closure) is not true.

Corollary 3.4.7

Let k be an algebraically closed field and let $I \subset k[x_1, \dots, x_n]$ be a proper ideal. Then

$$V(I) \neq \emptyset.$$

Proof. By Proposition 1.2.12, there exists a maximal ideal $\mathfrak{m} \subset k[x_1, \dots, x_n]$ such that $I \subset \mathfrak{m}$, since I is proper (Zorn’s lemma!). Thus, $V(\mathfrak{m}) \subset V(I)$. Then, applying the weak Nullstellensatz (especially Corollary 3.4.5), we get that there exist $a_1, \dots, a_n \in k$ such that

$$\mathfrak{m} = (x_1 - a_1, \dots, x_n - a_n).$$

Thus,

$$(a_1, \dots, a_n) \in V(\mathfrak{m}) \subset V(I).$$

and our claim follows. $\square$

Remark 3.4.8 — This is false if $k = \mathbb{R}$, as usual.

Remark 3.4.9 — Note that the converse, is always true, regardless of k being algebraically closed; that is, the statement that if $V(I) \neq \emptyset$, then $I \subsetneq R$ is proper. If it weren’t proper, then $1 \in I$, so then $V(I) = \emptyset$, since the 1-polynomial doesn’t vanish.

Theorem 3.4.10 (Strong Nullstellensatz)

Let k be an algebraically closed field and let $I \subset k[x_1, \dots, x_n]$ be an ideal. Then

$$I(V(I)) = \sqrt{I}.$$

Proof. Recall that by Lemma 1.3.15, we have that $\sqrt{I} \subset I(V(I))$. Let $f \in I(V(I))$. Then $f(x) = 0$ for all $x \in V(I)$. We consider the ring extension

$$k[x_1, \dots, x_n] \subset k[x_1, \dots, x_n, y]$$

and the ideal

$$J := (I, yf - 1) \subset k[x_1, \dots, x_n, y].$$

The claim is that $V(J) = \emptyset$. Assume by contradiction that there exists $(a_1, \dots, a_n, b) \in V(J) \subset \mathbb{A}_k^{n+1}$ if we denote $x := (a_1, \dots, a_n)$ we have that

$$x \in V(I) \quad \text{and} \quad b \cdot f(x) = 1.$$

This implies that $f(x) \neq 0$, a contradiction. Thus, $V(J) = \emptyset$ and Corollary 3.4.7 implies that $J = k[x_1, \dots, x_n, y]$, so $1 \in J$. Thus, there exist $g_0, \dots, g_k \in k[x_1, \dots, x_n, y]$, $h_1, \dots, h_k \in I \subseteq k[x_1, \dots, x_n]$ such that

$$1 = \sum_{i=1}^k g_i \cdot h_i + g_0 \cdot (yf - 1).$$

It is easy to check that there exists a sufficiently large integer m such that, after multiplying by f^m on both sides, we get

$$f^m = \sum_{i=1}^m G_i(x_1, \dots, x_n, fy) \cdot h_i + G_0(x_1, \dots, x_n, fy) \cdot (fy - 1)$$

for some $G_0, \dots, G_m \in k[x_1, \dots, x_n, y]$. The equality holds true for all $(x_1, \dots, x_n, y) \in \mathbb{A}_k^{n+1}$. Thus, we may restrict to the locus where $yf(x_1, \dots, x_n) = 1$ and we get that

$$f^m = \sum_{i=1}^m G_i(x_1, \dots, x_n, 1) \cdot h_i \in I.$$

Thus, $f \in \sqrt{I}$ and our claim follows. $\square$

Remark 3.4.11 — As noted in the proof, note that $\sqrt{I} \subset I(V(I))$ always holds. Note that we also have that if $V \subset \mathbb{A}_k^n$ is a variety, then $V(I(V)) = V$, for any field k .

Example 3.4.12

Suppose that $k = \mathbb{R}$ (not alg. closed). Suppose that $n = 1$, and consider the ideal $I = (x^2 + 1) \subset k[x]$. Then, $V(I) = \emptyset$, since there are no solutions to $x^2 + 1$ over $\mathbb{R}$. Thus,

$$I(V(I)) = I(\emptyset) = k[x].$$

This is different to the $\sqrt{I}$, which is actually equal to I in this case. Thus, algebraic closure is really crucial.

Recall by Proposition 1.4.2 that if R is a commutative ring and $I \subset R$ is an ideal then the radical $\sqrt{I}$ of I coincides with the intersection of all the prime ideals containing I . We now show that if R is a polynomial ring over an algebraically closed field, then it is enough to consider the intersection of all the maximal ideals containing I .

Proposition 3.4.13

Let k be an algebraically closed field, let $R = k[x_1, \dots, x_n]$ and let $I \subset R$ be a proper ideal. Then

$$\sqrt{I} = \bigcap_{I \subset \mathfrak{m}} \mathfrak{m}$$

where the intersection runs over all the maximal ideals $\mathfrak{m}$ containing I .

Proof. Since any maximal ideal is prime, Proposition 1.4.2 implies that $\sqrt{I} \subset \bigcap_{I \subset \mathfrak{m}} \mathfrak{m}$. Let $f \in \bigcap_{I \subset \mathfrak{m}} \mathfrak{m}$ and let $a := (a_1, \dots, a_n) \in V(I)$. Then, the ideal

$$\mathfrak{m}_a := I(\{a\}) = (x_1 - a_1, \dots, x_n - a_n) \subset k[x_1, \dots, x_n]$$

is maximal and Lemma 1.3.15 implies that

$$I \subset \sqrt{I} \subset I(V(I)) \subset \mathfrak{m}_a.$$

Thus, $f \in \mathfrak{m}_a$, since f is contained in all the maximal ideals of $k[x_1, \dots, x_n]$ containing I . It follows that $f(a) = 0$ for all $a \in V(I)$. Therefore, $f \in I(V(I))$. Theorem 3.4.10 implies that $f \in \sqrt{I}$ and our claim follows. $\square$

Remark 3.4.14 — The geometric point of view is that if $I(V(I)) = \sqrt{I} = \bigcap \mathfrak{m}$, then $V(I) = \bigcup V(\mathfrak{m})$.

Exercise 3.4.15. Let k be an algebraically closed field. A polynomial $f \in k[x_1, \dots, x_n]$ is said to be reduced if $f = f_1 \cdots f_k$ for some distinct irreducible polynomials $f_1, \dots, f_k \in k[x_1, \dots, x_n]$. Let $f \in k[x_1, \dots, x_n]$ be reduced and consider the hypersurface

$$V := \{x \in \mathbb{A}_k^n : f(x) = 0\} \subset \mathbb{A}_k^n.$$

Show that $I(V) = (f)$.

Solution: By the Nullstellensatz, we have that $I(V(f)) = \sqrt{(f)}$, so the real goal here is to show that f is radical. In particular, we must show that if $g \in \sqrt{(f)}$, then it's also in (f) . If $g \in \sqrt{(f)}$, then $g^n = f \cdot h$, where $h \in k[X_1, \dots, X_n]$. Now, note that $f = f_1 \cdots f_k$. Then, each f_i must divide g^n , by irreducibility of the f_i . In particular, by unique factorization, each of the f_i must divide g . Therefore, g is actually divisible by all the f_i , so $g \in (f)$. $\square$

Exercise 3.4.16. Let k be a field and A a finitely generated k -algebra. Prove that the following are equivalent:

1. A is Artinian;
2. A is a finite k -algebra.

Hint: To prove (i) $\Rightarrow$ (ii), reduce to the case where A is an Artin local ring. By the Nullstellensatz, the residue field of A is a finite extension of k . Now use the fact that A is of finite length as an A -module. To prove (ii) $\Rightarrow$ (i), observe that the ideals of A are k -vector subspaces and therefore satisfy DCC.

Solution: 2 $\Rightarrow$ 1: If A is a finite k -algebra, it is a finite dimensional k -vector space (since finite k -algebras are just finitely generated k -modules). Therefore, it is Artinian.

1 $\implies$ 2: For the forward, we wish to show that A is a finite dimensional k -vector space. For this, we use the structure theorem of Artinian rings. Note that any Artinian ring can be written as a product of (Artinian) local rings, so if we can show that each factor is a finite k -algebra, we are good. Thus, let $(A, \mathfrak{m})$ be our Artinian local ring. Note that A is also Noetherian and has $\dim A = 0$ by Hopkins-Levitzki. Then $\mathfrak{m}^n = 0$ for some n , so we have a sequence

$$0 = \mathfrak{m}^n \subseteq \mathfrak{m}^{n-1} \subseteq \dots \subseteq \mathfrak{m}^2 \subseteq \mathfrak{m} \subseteq A.$$

Since A is Noetherian, each quotient $\mathfrak{m}^i/\mathfrak{m}^{i+1}$ is a finite dimensional $A/\mathfrak{m}$ -vector space. However, $A/\mathfrak{m}$ is a finitely generated k -algebra, so the weak Nullstellensatz implies that $k \subset A/\mathfrak{m}$ is a finite field extension. That is, $A/\mathfrak{m}$ is a finite dimensional vector space over k . Since A is built on finitely many finite dimensional k -vector spaces, it follows that A is a finite dimensional k -vector space. $\square$

Remark 3.4.17 — Another way to prove 2 $\implies$ 1, which is similar in spirit is to use the Jordan Holder series. Artinian implies Noetherian, so a Jordan Holder series for A exists and is finite, and each quotient is a simple A -module. However, each simple A -module is isomorphic to $A/\mathfrak{m}_i$ for some maximal ideal $\mathfrak{m}_i$. Then, by weak Nullstellensatz again, $A/\mathfrak{m}_i$ is a finite dimensional k -vector space, so we use the same finite number of finite dimensional k -vector spaces argument again,

§3.5 Ax Theorem

Theorem 3.5.1

Let A be a finitely generated ring extension of $\mathbb{Z}$. Let $\mathfrak{m} \subset A$ be a maximal ideal. Then, $A/\mathfrak{m}$ is a finite field.

Proof. Let $\mathfrak{n} := \mathfrak{m} \cap \mathbb{Z}$. Then, $\mathfrak{n}$ is a prime ideal of $\mathbb{Z}$. Since $1 \notin \mathfrak{n}$, we have that $\mathfrak{n} \neq \mathbb{Z}$. Thus, $\mathfrak{n} = (p)$ for some prime $p \in \mathbb{Z}$ or $\mathfrak{n} = (0)$.

Assume first that $\mathfrak{n} = (p)$. Then, by the assumption it follows that $A/\mathfrak{m}$ is a field which is a finitely generated $\mathbb{Z}/(p)$ -algebra. Thus, by Theorem 3.4.4, we have that $A/\mathfrak{m}$ is a finite extension of $\mathbb{Z}/(p) = \mathbb{Z}/p\mathbb{Z}$ and, in particular, it is finite. Thus, $A/\mathfrak{m}$ is a finite field.

Assume now that $\mathfrak{n} = (0)$. We will see that this case leads to a contradiction. We have that $A/\mathfrak{m}$ is a field that contains $\mathbb{Z}/(0) = \mathbb{Z}$. Hence $\mathbb{Q} \subset A/\mathfrak{m}$. It follows that $A/\mathfrak{m}$ is a finitely generated algebra over $\mathbb{Q}$ and by Theorem 3.4.4, we have that $A/\mathfrak{m}$ is a finite extension of $\mathbb{Q}$, so integral. Let $x_1, \dots, x_n \in A/\mathfrak{m}$ be generators of $A/\mathfrak{m}$ over $\mathbb{Q}$. Then there exist monic polynomials $f_1, \dots, f_n \in \mathbb{Q}[x]$ such that $f_i(x_i) = 0$ for $i = 1, \dots, n$. We may assume that $a_i \in \mathbb{Z}$ is the common denominator of the coefficients of f_i for each $i = 1, \dots, n$. Therefore $A/\mathfrak{m}$ is an integral extension of the ring $\mathbb{Z}[a_1^{-1}, \dots, a_n^{-1}]$; the key here is that we don't need all of $\mathbb{Q}$. Since $A/\mathfrak{m}$ is a field, Lemma 3.4.1 implies that $\mathbb{Z}[a_1^{-1}, \dots, a_n^{-1}]$ is also a field. Thus, $\mathbb{Z}[a_1^{-1}, \dots, a_n^{-1}] = \mathbb{Q}$, which easily implies a contradiction. $\square$

Definition 3.5.2. Let k be a field. A **regular map** $F: \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n$ is given by polynomials $f_1, \dots, f_n \in k[X_1, \dots, X_n]$, so that $F = (f_1, \dots, f_n)$:

$$F: \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n, \quad (x_1, \dots, x_n) \mapsto (f_1(x_1, \dots, x_n), \dots, f_n(x_1, \dots, x_n)).$$

The following result is one of the many important applications of the Nullstellensatz.

Theorem 3.5.3 (Ax Theorem)

Let $p: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$ be an injective regular mapping. Then p is a bijection.

Before proceeding with the proof, we first consider the case of a general field k .

Remark 3.5.4 — If k is a finite field then $\mathbb{A}_k^n$ is finite and therefore any injective regular mapping $F: \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n$ is automatically surjective by basic set theory.

We now consider the case of a general algebraically closed field k and we assume that

$$F := (f_1, \dots, f_n): \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n$$

is a regular mapping, where $f_1, \dots, f_n \in k[x_1, \dots, x_n]$. We will need the following two remarks in the proof of Theorem 3.5.3:

Remark 3.5.5 (Injectivity...?) — Here, we ask what injectivity means for F . Let $I \subset k[x_1, \dots, x_n, y_1, \dots, y_n]$ be the ideal given by

$$I := (f_1(x_1, \dots, x_n) - f_1(y_1, \dots, y_n), \dots, f_n(x_1, \dots, x_n) - f_n(y_1, \dots, y_n)).$$

Assume that F is injective and let $(x_1, \dots, x_n, y_1, \dots, y_n) \in V(I) \subseteq \mathbb{A}_k^{2n}$. Then we have

$$f_i(x_1, \dots, x_n) = f_i(y_1, \dots, y_n) \quad \text{for all } i = 1, \dots, n.$$

Thus, since F is injective, it follows that $x_i = y_i$ for all i , which implies that

$$x_i - y_i \in I(V(I)).$$

Since k is algebraically closed, the Nullstellensatz (cf. Theorem 3.4.10) implies that

$$I(V(I)) = \sqrt{I}.$$

Thus, for each $i = 1, \dots, n$, there exists $\ell_i \in \mathbb{Z}_{>0}$ and $q_{i,j} \in k[x_1, \dots, x_n, y_1, \dots, y_n]$ for $j = 1, \dots, n$ such that

$$(x_i - y_i)^{\ell_i} = \sum_{j=1}^n (f_j(x_1, \dots, x_n) - f_j(y_1, \dots, y_n)) \cdot q_{i,j}(x_1, \dots, x_n, y_1, \dots, y_n). \quad (3.1)$$

Conversely, if Equation (3.1) is satisfied then F is injective, even without the assumption of k being algebraically closed: assume (3.1) hold. Let $x, y \in \mathbb{A}_k^n$ such that $F(x) = F(y)$. Then, $f_i(x) = f_i(y)$, so the hypothesis implies that $(x_i - y_i)^{\ell_i} = 0$, so $x_i - y_i = 0$, implying that $x = y$. Thus, F is injective. We emphasize again that this reverse implication is true for any field k . Thus, F injective $\iff$ (3.1) holds.

Remark 3.5.6 (Non-surjectivity?) — Let $(a_1, \dots, a_n) \in \mathbb{A}_k^n$ such that

$$F(x_1, \dots, x_n) \neq (a_1, \dots, a_n) \quad \text{for all } (x_1, \dots, x_n) \in \mathbb{A}_k^n.$$

Then if we consider the ideal

$$J := (f_1(x_1, \dots, x_n) - a_1, \dots, f_n(x_1, \dots, x_n) - a_n) \subset k[x_1, \dots, x_n],$$

we have that $V(J) = \emptyset$. Since k is algebraically closed, Corollary 3.4.7 implies that the ideal is not proper, so $J = k[x_1, \dots, x_n]$ and $1 \in J$. Therefore, there exist $h_1, \dots, h_n \in k[x_1, \dots, x_n]$ such that

$$\sum_{i=1}^n (f_i - a_i) \cdot h_i = 1. \quad (3.2)$$

Conversely, if Equation (3.2) holds then $(a_1, \dots, a_n) \notin \text{Im} F$ even without assuming that k is algebraically closed. Suppose that there exists some x such that $F(x) = a$. Then $f_i(x) - a_i = 0$, so $1 = 0$, which is a contradiction. Thus, F being non-surjective $\iff$ (3.2) holds.

Proof of Theorem 3.5.3. We assume by contradiction that $p: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$ is injective but not surjective. We want to find an extension A of $\mathbb{Z}$ which is finitely generated and such that we can replace $\mathbb{C}$ by A . Consider all the possible coefficients of all the p_i and call them a_i . We also know that (3.1) and (3.2) hold for $f_i = p_i$ because $\mathbb{C}$ is algebraically closed. This implies that there exist

- $q_{i,j} \in \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n]$ for $i, j = 1, \dots, n$
- $h_i \in \mathbb{C}[x_1, \dots, x_n]$ for $i = 1, \dots, n$

Now consider the set S to be the finite set consisting of the a_i and the coefficients of p_i , $q_{i,j}$, and h_i , for $i, j = 1, \dots, n$. Let $A \subset \mathbb{C}$ be an algebra over $\mathbb{Z}$ generated by S . Then, A is finitely generated. Now consider $p: A^n \rightarrow A^n$. Let $\mathfrak{m} \subset A$ be a maximal ideal. Then $k := A/\mathfrak{m}$ is a field. We can then apply Theorem 3.5.1 to say that k is finite. It follows that we have an induced regular mapping

$$F := \bar{p}: \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n$$

defined by

$$([x_1], \dots, [x_n]) \mapsto ([p_1(x_1, \dots, x_n)], \dots, [p_n(x_1, \dots, x_n)]).$$

We show that this is well defined. If $x - y \in \mathfrak{m}$, since p_i is a polynomial, $p_i(x) - p_i(y)$ is divisible by $x - y$; thus, $p_i(x) - p_i(y) = (x - y) \cdot t(x, y)$. Therefore, $p_i(x) - p_i(y) \in \mathfrak{m}$ as well. Since the p_i satisfy (3.1) and (3.2), then F also satisfies (3.1) and (3.2). We know that for any field, (3.1) implies F is injective and (3.2) implies that F is not surjective. However, this is an issue, since F is surjective, because Theorem 3.5.1 implies that k is a finite field. This gives a contradiction. $\square$

Remark 3.5.7 — We note that Ax's theorem holds for $\mathbb{C}$, $\mathbb{R}$ (for $n = 1$ by exercise and $n > 1$ much harder), and finite fields. In $\mathbb{Q}$, it definitely does not hold.

Exercise 3.5.8. Find a regular mapping $p: \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n$ which is injective but not surjective.

Solution: Note that when $k = \mathbb{F}_q$, a finite field, then injections are the same as surjections, so these don't work. Consider $k = \mathbb{Q}$, and $n = 1$. Then, we want a polynomial $\mathbb{A}_{\mathbb{Q}}^1 \hookrightarrow \mathbb{A}_{\mathbb{Q}}^1$ which is injective but not surjective. A natural choice is x^3 . This is not surjective, since there is no solution to $x^3 = 2$. $\square$

Exercise 3.5.9. Let $p: \mathbb{A}_{\mathbb{R}}^1 \rightarrow \mathbb{A}_{\mathbb{R}}^1$ be a regular mapping. Show that if it is injective then it is also surjective.

Solution: Firstly, the polynomial must be of odd degree. Further, this polynomial must be strictly monotonically increasing or decreasing. Then, $\lim_{x \rightarrow -\infty} f(x) = -\infty$ and $\lim_{x \rightarrow \infty} f(x) = \infty$. Then, surjectivity is clear by the intermediate value theorem. $\square$

Remark 3.5.10 — The same result holds more in general for any regular mapping $p: \mathbb{A}_{\mathbb{R}}^n \rightarrow \mathbb{A}_{\mathbb{R}}^n$, but the proof is much harder and $n \geq 1$.

4 Dimension of Local Rings

Recall that given a ring R , we defined the dimension of R to be

$$\dim R = \sup\{n \geq 0 : \exists \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n \subsetneq R\},$$

where each $\mathfrak{p}_i$ is prime. Now, we haven't really calculated $\dim \mathbb{A}_k^n$, i.e. $\dim k[X_1, \dots, X_n]$, where k is a field. For sure, this must be greater than or equal to n , since we can easily find a sequence that has length n :

$$0 \subsetneq (X_1) \subsetneq (X_1, X_2) \subsetneq \cdots \subsetneq (X_1, X_2, \dots, X_n) \subsetneq k[X_1, \dots, X_n].$$

However, can we get anything bigger? The answer is no...but we shall show this in the coming material. Particularly, we show that $\dim k[X_1, \dots, X_n] = n$, if k is an algebraically closed field.

Definition 4.0.1. Let $(R, \mathfrak{m})$ be a local ring. Let M be a finitely generated module over R . We define $\mu(M)$ to be the **minimum number** of generators of M over R .

Proposition 4.0.2

Let $(R, \mathfrak{m})$ be a local ring and let $k = R/\mathfrak{m}$ be its residue field. Let M be a finitely generated module over R . Then $M/\mathfrak{m}M$ is a vector space over k of dimension $\mu(M)$.

Proof. It is easy to check that $M/\mathfrak{m}M$ is a vector space over k . Further, it is clear that if $x_1, \dots, x_n$ are generators of M over R then $[x_1], \dots, [x_n]$ are generators of $M/\mathfrak{m}M$ over k . Thus,

$$\dim_k M/\mathfrak{m}M \leq \mu(M).$$

Conversely, assume that $x_1, \dots, x_n \in M$ are such that $[x_1], \dots, [x_n]$ are generators of $M/\mathfrak{m}M$ over k . The goal is to show that $x_1, \dots, x_n$ generate M . Let $N \subset M$ be the submodule generated by $x_1, \dots, x_n$. Consider the R -module

$$\overline{M} := M/N.$$

We want to show that $\overline{M} = 0$. We first show that $\mathfrak{m}\overline{M} = \overline{M}$. Let $x \in M$. Then there exist $r_1, \dots, r_n \in R$ such that

$$[x] = \sum_{i=1}^n [r_i][x_i] \quad \text{in } M/\mathfrak{m}M.$$

Thus,

$$x - \sum_{i=1}^n r_i x_i \in \mathfrak{m}M$$

and, in particular, the image of x in $\overline{M}$ is contained in $\mathfrak{m}\overline{M}$. Thus, $\mathfrak{m}\overline{M} = \overline{M}$.

Note that since $(R, \mathfrak{m})$ is a local ring, we have that $\mathcal{J}(R) = \mathfrak{m}$. Therefore, by Nakayama's lemma (Corollary 2.3.6), we have that $\overline{M} = 0$, which implies that $N = M$. It follows that

$$\dim_k M/\mathfrak{m}M \geq \mu(M).$$

Thus, the equality holds. □

As a special case, when $(R, \mathfrak{m})$ is a local Noetherian ring, then take $M = \mathfrak{m}$, which is an R -module. Since R is Noetherian, M is finitely generated, so applying the proposition above, we have

$$\mu(\mathfrak{m}) = \dim_k \mathfrak{m}/\mathfrak{m}^2.$$

Remark 4.0.3 — In algebraic geometry, $\mathfrak{m}/\mathfrak{m}^2$ is called the tangent space of $(R, \mathfrak{m})$. The idea comes from focus on a specific local Noetherian ring. Let $R = k[X_1, \dots, X_n]/I$, where $I \subset k[X_1, \dots, X_n]$ is an ideal. Let $\mathfrak{p} \subset R$ be a prime ideal corresponding to a point in $V(I)$, and we consider the localization at $\mathfrak{p}$ — $R_{\mathfrak{p}}$. This implies that $\mathfrak{m}/\mathfrak{m}^2$ is the usual tangent space.

Definition 4.0.4. Let $(R, \mathfrak{m})$ be a local Noetherian ring with residue field $k := R/\mathfrak{m}$. Then we define

$$\mu(R) := \dim_k \mathfrak{m}/\mathfrak{m}^2.$$

R is called **regular** if $\mu(R) = \dim R$.

Remark 4.0.5 — Note that the definition is compatible with Proposition 4.0.2. Indeed, since R is Noetherian, we have that $M := \mathfrak{m}$ is a finitely generated R -module and $\mathfrak{m}M = \mathfrak{m}^2$.

Definition 4.0.6. Let R be a commutative ring. The **spectrum** of R is the set

$$\mathrm{Spec}(R) := \{\mathfrak{p} \subset R: \mathfrak{p} \text{ is a prime ideal of } R\}.$$

Remark 4.0.7 — Note that this set can be endowed with a topology. Fix the prime ideal $\mathfrak{q} \subset R$.

Exercise 4.0.8. Let R be a commutative ring, let $\mathfrak{p} \subset R$ be a prime ideal and let $R_{\mathfrak{p}}$ be the localization of R at $\mathfrak{p}$. Then there exists a bijection between $\mathrm{Spec}(R_{\mathfrak{p}})$ and the set

$$\{\mathfrak{q} \in \mathrm{Spec} R: \mathfrak{q} \subset \mathfrak{p}\}.$$

In particular, we have

$$\dim R_{\mathfrak{p}} = h(\mathfrak{p}).$$

Solution: Let

$$\varphi: R \rightarrow R_{\mathfrak{p}}, \quad x \mapsto \frac{x}{1}$$

be the natural map and consider the function

$$\Psi: \mathrm{Spec}(R_{\mathfrak{p}}) \rightarrow \mathrm{Spec}(R), \quad \mathfrak{r} \mapsto \varphi^{-1}(\mathfrak{r}).$$

Note that Ψ is well-defined since if $\mathfrak{r} \subset R_{\mathfrak{p}}$ is prime then $\varphi^{-1}(\mathfrak{r}) \subset R$ is also prime.

Let $\mathfrak{r} \in \mathrm{Spec}(R_{\mathfrak{p}})$ and let $x \in \varphi^{-1}(\mathfrak{r})$. Then $x \in \mathfrak{p}$ as otherwise $1/x \in R_{\mathfrak{p}}$ and, therefore,

$$1 = \frac{1}{x} \cdot \frac{x}{1} \in \mathfrak{r},$$

and $\mathfrak{r}$ is not prime, a contradiction. Thus,

$$\Psi(\mathfrak{r}) \subset \mathfrak{p}.$$

Vice versa, for any $\mathfrak{q} \in \operatorname{Spec} R$ such that $\mathfrak{q} \subset \mathfrak{p}$, we define

$$\Phi(\mathfrak{q}) = \mathfrak{q}R_{\mathfrak{p}}.$$

It is then easy to check that

$$\Phi: \{\mathfrak{q} \in \operatorname{Spec} R \mid \mathfrak{q} \subset \mathfrak{p}\} \rightarrow \operatorname{Spec}(R_{\mathfrak{p}})$$

is the inverse of Ψ .

The $\dim R_{\mathfrak{p}}$ is the supremum of chains of prime ideals in $R_{\mathfrak{p}}$. However, by the bijection, such a supremum of chain of prime ideals correspond with a chain of prime ideals contained in $\mathfrak{p}$. Thus, $\dim(R_{\mathfrak{p}}) = h(\mathfrak{p})$. $\square$

Definition 4.0.9. The **height** of $\mathfrak{p}$ is

$$h(\mathfrak{p}) = \sup\{\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n = \mathfrak{p}\},$$

where each $\mathfrak{p}_i$ are prime.

Exercise 4.0.10. If $\mathfrak{m} \subset R$ is maximal, then $h(\mathfrak{m}) = \dim R$.

Solution: This is essentially by definition. We note that in a chain of prime ideals $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n \subsetneq R$ which is actually the supremum, the final ideal $\mathfrak{p}_n$ must be maximal. Therefore, $\dim R = n = h(\mathfrak{m})$. $\square$

Remark 4.0.11 — We consider the geometric point of view here. Let $R = k[X_1, \dots, X_n]/I$, and let $\mathfrak{p} \subset R$ be prime. Then, $h(\mathfrak{p})$ is equal to the codimension of $V(\mathfrak{p})$ inside $V(I)$; that is, $\dim V(I) - \dim V(\mathfrak{p})$. This is very soft at this moment, and we just state this idea without proof.

Exercise 4.0.12. Let R be an Artin local ring and let $k = R/\mathfrak{m}$ be the residue field. Then the following are equivalent:

1. every ideal in R is principal
2. the maximal ideal $\mathfrak{m}$ is principal
3. $\dim_k(\mathfrak{m}/\mathfrak{m}^2) \leq 1$

Solution: $1 \implies 2$ is clear, and $2 \implies 3$ is also immediate. Thus, we shall prove $3 \implies 1$ here. Note that Artinian rings are Noetherian. If $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 0$, then $\mathfrak{m} = \mathfrak{m}^2$, so $\mathfrak{m} = 0$ by Nakayama's lemma, so R is a field. Thus, suppose $\dim_k(\mathfrak{m}/\mathfrak{m}^2) > 0$. If $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 1$, then $\mathfrak{m} = (t)$, for some $t \in R$, because for local rings $(R, \mathfrak{m})$, $\dim_k(M/\mathfrak{m}M) = \mu(M)$. Now, let $I \subseteq R$ be proper nonzero ideal of R . By Artinian-ness, $\mathcal{N}(R) = \mathfrak{m}$, so $\mathfrak{m}$ is nilpotent. Therefore, there exists an integer r such that $I \subseteq \mathfrak{m}^r$ and $I \not\subseteq \mathfrak{m}^{r+1}$. Therefore, there exists $y \in I$ such that $y = xt^r$ but $y \neq x't^{r+1}$. Therefore, $x \notin (t)$, so it is a unit by local-ness. Thus, $t^r \in I$, so $\mathfrak{m}^r = (t^r) \subseteq I$. Therefore, we attain equality: $I = (t^r)$. $\square$

Example 4.0.13

The rings $\mathbb{Z}/(p^n)$, where p is prime, and $k[x]/(f^n)$, where f is irreducible, satisfy the conditions above.

Example 4.0.14

The Artin local ring $k[x^2, x^3]/(x^4)$ doesn’t satisfy the exercise above. $\mathfrak{m}$ is generated by x^2 and x^3 modulo x^4 , so that $\mathfrak{m}^2 = 0$ and $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 2$.

§4.1 Krull’s Principal Ideal Theorem**Theorem 4.1.1 (Krull’s Principal Ideal Theorem)**

Let R be a Noetherian ring and let $a \in R$ be a non-unit. Let $\mathfrak{p}$ be a prime ideal which is minimal among all the prime ideals such that $a \in \mathfrak{p}$. Then

$$h(\mathfrak{p}) \leq 1.$$

Proof. Let $R_{\mathfrak{p}}$ be the localization of R at $\mathfrak{p}$. Then $R_{\mathfrak{p}}$ is Noetherian and the maximal ideal of $R_{\mathfrak{p}}$ is $\mathfrak{m}_{\mathfrak{p}} := \mathfrak{p}R_{\mathfrak{p}}$. We have that $\mathfrak{m}_{\mathfrak{p}}$ is a minimal prime ideal containing $a \in R_{\mathfrak{p}}$. Since $\mathfrak{m}_{\mathfrak{p}}$ is maximal, we have that $\dim R_{\mathfrak{p}} = h(\mathfrak{m}_{\mathfrak{p}})$ and by Exercise 4.0.8, we have that $\dim R_{\mathfrak{p}} = h(\mathfrak{p})$. Therefore, it is enough to show that $h(\mathfrak{m}_{\mathfrak{p}}) \leq 1$. Thus, after replacing R by $R_{\mathfrak{p}}$ we may assume that R is a local ring with maximal ideal $\mathfrak{m} := \mathfrak{p}$.

Assume that $h(\mathfrak{p}) > 0$, as otherwise there is nothing to show. Then, there exists a prime ideal $\mathfrak{q} \subsetneq \mathfrak{p}$. In particular, $a \notin \mathfrak{q}$. We want to show that $h(\mathfrak{q}) = 0$. Let $R_{\mathfrak{q}}$ be the localization of R at $\mathfrak{q}$, with maximal ideal $\mathfrak{m}_{\mathfrak{q}} := \mathfrak{q}R_{\mathfrak{q}}$. We first show that

$$\mathfrak{m}_{\mathfrak{q}} \subset \sqrt{(0)} \quad \text{in } R_{\mathfrak{q}}.$$

To this end, we define the ideal

$$\mathfrak{q}^{(n)} := \{r \in R : rt \in \mathfrak{q}^n \text{ for some } t \in R \setminus \mathfrak{q}\} \subset R.$$

Note that if $\varphi: R \rightarrow R_{\mathfrak{q}}$ is the natural map, then

$$\mathfrak{q}^{(n)} = \varphi^{-1}(\mathfrak{q}^n R_{\mathfrak{q}}).$$

We have a descending chain of ideals in R

$$\mathfrak{q} = \mathfrak{q}^{(1)} \supset \mathfrak{q}^{(2)} \supset \mathfrak{q}^{(3)} \supset \dots$$

which induces the chain of ideals in R

$$\mathfrak{p} \supset (a) + \mathfrak{q}^{(1)} \supset (a) + \mathfrak{q}^{(2)} \supset \dots$$

which, in turn, induces the chain of ideals in $\overline{R} := R/(a)$

$$\overline{\mathfrak{p}} \supset \overline{\mathfrak{q}}^{(1)} \supset \overline{\mathfrak{q}}^{(2)} \supset \dots$$

Since $\bar{\mathfrak{p}}$ is the only maximal ideal of $\bar{R}$, and at the same time, it is a minimal prime ideal of $\bar{R}$ containing a , it follows easily that $\bar{\mathfrak{p}}$ is the only prime ideal of $\bar{R}$. Thus, $\dim \bar{R} = 0$ and, since $\bar{R}$ is Noetherian, by Theorem 2.6.20 we have that $\bar{R}$ is also Artinian. Thus, there exists a positive integer n such that

$$\bar{\mathfrak{q}}^{(i)} = \bar{\mathfrak{q}}^{(n)} \quad \text{for all } i \geq n.$$

and, therefore,

$$(a) + \mathfrak{q}^{(i)} = (a) + \mathfrak{q}^{(n)} \quad \text{for all } i \geq n.$$

Thus, if $x \in \mathfrak{q}^{(n)}$, there exists $r \in R$ and $y \in \mathfrak{q}^{(n+1)}$ such that $x = ra + y$ and, therefore, $ra = x - y \in \mathfrak{q}^{(n)}$. Since $\mathfrak{q}^{(n)} = \varphi^{-1}(\mathfrak{q}^n R_{\mathfrak{q}})$, it follows that

$$\frac{ra}{1} \in \mathfrak{q}^n R_{\mathfrak{q}}.$$

Since $a \notin \mathfrak{q}$, we have that $\frac{1}{a} \in R_{\mathfrak{q}}$ and, therefore, $\frac{r}{1} \in \mathfrak{q}^n R_{\mathfrak{q}}$. Since $\mathfrak{q}^{(n)} = \varphi^{-1}(\mathfrak{q}^n R_{\mathfrak{q}})$, it follows that $r \in \mathfrak{q}^{(n)}$. Thus, since $x = ra + y$ and $a \in \mathfrak{p}$, we have proven that

$$\mathfrak{q}^{(n)} = \mathfrak{p}\mathfrak{q}^{(n)} + \mathfrak{q}^{(n+1)}.$$

Since $(R, \mathfrak{p})$ is a local ring, by Nakayama's Lemma (Corollary 2.3.6) it follows that $\mathfrak{q}^{(n)} = \mathfrak{q}^{(n+1)}$ and, in particular,

$$\mathfrak{q}^n R_{\mathfrak{q}} = \mathfrak{q}^{n+1} R_{\mathfrak{q}} = (\mathfrak{q} R_{\mathfrak{q}}) \cdot (\mathfrak{q}^n R_{\mathfrak{q}}).$$

By applying Corollary 2.3.6 again, it follows that $\mathfrak{q}^n R_{\mathfrak{q}} = 0$, which implies that

$$\mathfrak{m}_{\mathfrak{q}} = \mathfrak{q} R_{\mathfrak{q}} \subset \sqrt{(0)},$$

i.e. $\mathfrak{m}_{\mathfrak{q}}$ is nilpotent, as claimed.

By Proposition 1.4.2, it follows that $\sqrt{(0)}$ is the intersection of all the prime ideals of $R_{\mathfrak{q}}$. Since $\mathfrak{m}_{\mathfrak{q}}$ is maximal in $R_{\mathfrak{q}}$, it follows that $\mathfrak{m}_{\mathfrak{q}}$ is the only prime ideal of $R_{\mathfrak{q}}$. Thus, $h(\mathfrak{q}) = \dim R_{\mathfrak{q}} = 0$ and, therefore, $h(\mathfrak{p}) \leq 1$. $\square$

Example 4.1.2

Consider $R = k[X_1, \dots, X_n]$, and take $f \in R$ be nonzero and not a unit. Then, $\deg f \geq 1$. Then, $a \in \mathfrak{p}$ if and only if $V(\mathfrak{p}) \subset V(f) = \{f = 0\}$. Then, $\mathfrak{p}$ is minimal if and only if $V(\mathfrak{p})$ is maximal with this property, and $\mathfrak{p}$ is prime if and only if $V(\mathfrak{p})$ is irreducible. This is saying that $\text{codim } V(\mathfrak{p}) \leq 1$.

Exercise 4.1.3. Let R be a Noetherian ring. Let

$$\mathfrak{p} \subsetneq \mathfrak{r}_0 \subsetneq \mathfrak{q} \subset R$$

be prime ideals. Show that there are infinitely many prime ideals $\mathfrak{r} \subset R$ such that

$$\mathfrak{p} \subsetneq \mathfrak{r} \subsetneq \mathfrak{q} \subset R.$$

Solution: Suppose not. Let $S := R/\mathfrak{p}$. Suppose S admits only finitely many prime ideals

$$\mathfrak{p}_1, \dots, \mathfrak{p}_n \neq (0)$$

properly contained in $\bar{\mathfrak{q}}$. We want to show by induction on n that there exists $a \in \bar{\mathfrak{q}}$ such that a is not contained in

$$\mathfrak{p}_1 \cup \dots \cup \mathfrak{p}_n.$$

If $n = 1$ then it is trivial. Thus, by induction, we may assume that, for any $i = 1, \dots, n$, there exists

$$x_i \in \bar{\mathfrak{q}} \setminus \left(\bigcup_{j \neq i} \mathfrak{p}_j \right).$$

Let

$$a := \sum_{i=1}^n x_1 x_2 \cdots x_{i-1} x_{i+1} \cdots x_n.$$

Then,

$$a \in \bar{\mathfrak{q}} \setminus \left(\bigcup_{i=1}^n \mathfrak{p}_i \right).$$

Thus, $\bar{\mathfrak{q}}$ is minimal among the prime ideals containing (a) and by Krull’s principal ideal theorem, we have that

$$h(\bar{\mathfrak{q}}) \leq 1.$$

However, this contradicts the fact that $\bar{\mathfrak{r}}$ exists between (0) and $\bar{\mathfrak{q}}$. □

Remark 4.1.4 — Note that when R is a PID, this says that $\dim R \leq 1$, since every nonzero prime ideal is maximal in a PID. Now, we present a counter example to this observation; that is, we present a ring which has dimension 1 but is not a PID. We want something that is Noetherian but not a PID, so our next natural guess might be $\mathbb{Z}[X]$, but this has dimension 2. $k[X]$ also doesn’t work because it is a PID. We next consider Dedekind domains, and the simplest type of Dedekind domains we know are quadratic extensions of $\mathbb{Z}$. In particular, the ring $\mathbb{Z}[\sqrt{-5}]$ works. This is not a PID because it’s not even a UFD:

$$2 \cdot 3 = (1 - \sqrt{-5})(1 + \sqrt{-5}).$$

These can be checked to be irreducible using the norm $N(a + b\sqrt{-5}) = a^2 - 5b^2$. By the Cohen-Seidenberg Theorem, $\dim \mathbb{Z}[\sqrt{-5}] = \dim \mathbb{Z} = 1$.

Exercise 4.1.5. Calculate the dimension of $k[x, y, z]/(xy - z^2)$, where k is a field.

Solution: First, note that $(xy - z^2)$ is a prime ideal of $k[x, y, z]$. This is because by Gauss’s lemma, $k[x, y, z]$ is a UFD, and $xy - z^2$ is an irreducible polynomial, and principal ideals generated by irreducibles in UFDs are prime. By Krull’s height theorem, $h(xy - z^2) = 1$, because (0) is a prime ideal, since $k[x, y, z]$ is an integral domain. Therefore, the dimension of this quotient ring is 2. □

Example 4.1.6

Let $I = (x^2, xy) \subseteq k[x, y]$. Then, $\sqrt{I} = (x)$. This is a minimal prime containing just one nonunit, so $h(\sqrt{I}) = 1$.

Example 4.1.7

Consider $k[x, y, z]/(xy, xz)$. Then, to calculate the dimension of this, we want the maximal prime chain in the quotient, which corresponds to calculating the minimum in $k[x, y, z]$ contained in (xy, xz) . Since $(xy, xz) = (x) \cap (y, z)$, $h(x) = 1$ and $h(y, z) = 2$, so $\dim k[x, y, z]/(xy, xz) = 2$.

§4.2 Krull's Height Theorem**Theorem 4.2.1 (Krull's Height Theorem)**

Let R be a Noetherian ring, let $a_1, \dots, a_r \in R$ be such that $I := (a_1, \dots, a_r) \subset R$ is a proper ideal. Let $\mathfrak{p}$ be a prime ideal which is minimal among all the prime ideals such that $a_1, \dots, a_r \in \mathfrak{p}$. Then

$$h(\mathfrak{p}) \leq r.$$

Proof. We proceed by induction on r . Note that the case $r = 1$ follows from Theorem 4.1.1. Thus, we can assume that $r > 1$. As in the proof of Theorem 4.1.1, we may assume that $(R, \mathfrak{p})$ is a local ring with maximal ideal $\mathfrak{p}$.

Assume, by contradiction, that $h(\mathfrak{p}) > r$. Then there exists a chain of prime ideals

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}'_1 \subsetneq \dots \subsetneq \mathfrak{p}'_r \subsetneq \mathfrak{p}.$$

Let $\mathfrak{p}_1 \subsetneq \mathfrak{p}$ be a prime ideal which is maximal with respect to all the prime ideals properly contained in $\mathfrak{p}$ and containing $\mathfrak{p}'_r$. Then $h(\mathfrak{p}_1) \geq r$. Since $\mathfrak{p}$ is a minimal prime ideal containing I , we have that I is not contained in $\mathfrak{p}_1$ and, after reordering the generators, we may assume that $a_1 \notin \mathfrak{p}_1$. Let

$$J := (a_1) + \mathfrak{p}_1 \subset \mathfrak{p}.$$

Since $h(\mathfrak{p}_1) = h(\mathfrak{p}) - 1$, we have that $\mathfrak{p}$ is the minimal prime ideal containing J .

We want to show that $\sqrt{J} = \mathfrak{p}$. By primary decomposition (Theorem 2.6.14), there exist $\mathfrak{q}_1, \dots, \mathfrak{q}_m \subset R$ prime ideals such that

$$\sqrt{J} = \bigcap_{i=1}^m \mathfrak{q}_i.$$

Since $\mathfrak{p}$ is the only maximal ideal in R , we have that $J \subset \sqrt{J} \subset \mathfrak{q}_i \subset \mathfrak{p}$ for all $i = 1, \dots, m$, and since $\mathfrak{p}$ is minimal among the prime ideals containing J , we have that $\mathfrak{p} = \mathfrak{q}_i$ for all $i = 1, \dots, m$. Thus, $\sqrt{J} = \mathfrak{p}$ and, in particular, there exists a positive integer n such that

$$a_i^n \in J \quad \text{for all } i = 1, \dots, r.$$

Hence, for all $i = 1, \dots, r$, there exist $x_i \in R$ and $y_i \in \mathfrak{p}_1$ such that

$$a_i^n = a_1 x_i + y_i.$$

Let

$$I_1 := (y_2, \dots, y_r) \subset \mathfrak{p}_1 \subset R.$$

We want to show that $\mathfrak{p}_1$ is a minimal prime ideal containing I_1 . Let $S = R/I_1$ and let $\bar{\mathfrak{p}}$ and $\bar{\mathfrak{p}}_1$ be the image of $\mathfrak{p}$ and $\mathfrak{p}_1$ in S respectively and let $\bar{a}_1 := [a_1] \in S$. We shall note that $\bar{a}_1$ is not zero. If it were equal to zero, then $a_1 \in I_1 \subset \mathfrak{p}_1$; however, we took a_1 to be an element not in $\mathfrak{p}_1$. Then

$$(0) \subsetneq (\bar{a}_1) \subsetneq \bar{\mathfrak{p}} \subset S.$$

We first show that $\bar{\mathfrak{p}} = \sqrt{(\bar{a}_1)}$. We have

$$(a_1, y_2, \dots, y_r) = (a_1, a_2^n - a_1 x_2, a_3^n - a_1 x_3, \dots, a_r^n - a_1 x_r) = (a_1, a_2^n, a_3^n, \dots, a_r^n).$$

Since $\mathfrak{p}$ is the minimal ideal containing $I = (a_1, \dots, a_r)$, we can mimic the proof above for showing $\sqrt{J} = \mathfrak{p}$, to attain that $\mathfrak{p} = \sqrt{I}$. Thus, Thus,

$$\sqrt{(a_1, y_2, \dots, y_r)} = \mathfrak{p}$$

which implies that $\bar{\mathfrak{p}} = \sqrt{(\bar{a}_1)}$, as claimed. This implies that $\bar{\mathfrak{p}}$ is the minimal prime ideal in S containing $\bar{a}_1$. Thus, by Theorem 4.1.1, it follows that $h(\bar{\mathfrak{p}}) \leq 1$. Since $\bar{\mathfrak{p}}_1 \subsetneq \bar{\mathfrak{p}}$, it follows that $h(\bar{\mathfrak{p}}_1) = 0$. Thus, $\mathfrak{p}_1$ is the minimal prime ideal in R containing I_1 , so $h(\mathfrak{p}_1) \leq r - 1$. However, this contradicts the earlier inequality that $h(\mathfrak{p}_1) \geq r$. Thus, $h(\mathfrak{p}) \leq r$ as claimed. $\square$

Corollary 4.2.2

Let k be an algebraically closed field. Then, $\dim k[X_1, \dots, X_n] = n$.

Proof. Let $\dim k[X_1, \dots, X_n] = N$. Recall that we've shown that $\dim k[X_1, \dots, X_n] = N \geq n$. Let

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \mathfrak{p}_2 \subsetneq \dots \subsetneq \mathfrak{p}_N \subsetneq R = k[X_1, \dots, X_n].$$

We want to show that $N \leq n$. We may ensure that $\mathfrak{p}_N$ is maximal. Since k is algebraically closed, $\mathfrak{p}_N = (x_1 - a_1, \dots, x_n - a_n)$, where $a_1, \dots, a_n \in k$. $\mathfrak{p}_n$ is the minimal prime ideal containing $x_1 - a_1, \dots, x_n - a_n$, so we're in the set up of Krull's height theorem. Thus, $\dim k[X_1, \dots, X_n] = h(\mathfrak{p}) \leq n$, so we achieve equality. $\square$

Let us unveil the geometric point of view. Let k be a field and $R = k[X_1, \dots, X_n]$. Let $(f_1, \dots, f_r) \subsetneq R$. If k is algebraically closed, this condition is equivalent to the fact that $V(I) \neq \emptyset$. Let $\mathfrak{p}$ now be the minimal prime ideal with $f_1, \dots, f_n \in \mathfrak{p}$. Then, $V(\mathfrak{p}) \subseteq V(I)$ is the maximal irreducible variety contained in $V(I)$. Then, the theorem says that $\text{codim } V(\mathfrak{p}) \leq n$. That is, $\text{codim } V(\mathfrak{p}) = \dim \mathbb{A}_k^N - \dim V(\mathfrak{p})$. Thus,

$$\dim V(\mathfrak{p}) \geq N - n.$$

Corollary 4.2.3

Let $(R, \mathfrak{m})$ be a local Noetherian ring. Then

$$\dim R \leq \mu(R) = \dim_k \mathfrak{m}/\mathfrak{m}^2,$$

where $k = R/\mathfrak{m}$. In particular, R has finite dimension.

Proof. Recall that we have that

$$\mu(R) := \dim_k \mathfrak{m}/\mathfrak{m}^2$$

and by Proposition 4.0.2, we have that $\mu(R) = \mu(\mathfrak{m})$, that is the minimum number of generators of $\mathfrak{m}$ as a R -module. Since R is Noetherian, we have that $\mu(R) < \infty$. If we let $r = \mu(R)$ and let $x_1, \dots, x_r \in \mathfrak{m}$ be generators of $\mathfrak{m}$. Then $\mathfrak{m}$ is the minimal prime ideal containing $x_1, \dots, x_r$, and, by Theorem 4.2.1, we have that

$$\dim(R) = h(\mathfrak{m}) \leq r = \mu(R)$$

as claimed. Note that the equality on the LHS follows from the fact that $\mathfrak{m}$ is maximal. $\square$

Remark 4.2.4 — Recall that in the geometric point of view, this is saying that $\dim R$ is less than or equal to the dimension of the tangent space of a variety V at x , where $x \in V$ is a point. This is because $\mathfrak{m}/\mathfrak{m}^2$ is the tangent space of V at x .

Exercise 4.2.5. Let k be an algebraically closed field and let $\mathfrak{m} := (x, y) \subset k[x, y]$. Show that

1. the localization

$$(k[x, y]/(y - x^2))_{\mathfrak{m}}$$

of $k[x, y]/(y - x^2)$ at $\mathfrak{m}$ is regular.

2. the localization

$$(k[x, y]/(x^2 - y^3))_{(x, y)}$$

of $k[x, y]/(x^2 - y^3)$ at $\mathfrak{m}$ is not regular.

Solution: 1. Let $R := (k[x, y]/(y - x^2))_{\mathfrak{m}}$. This is a local Noetherian ring. Thus, we want to show that $\dim R = \dim_k \mathfrak{m}/\mathfrak{m}^2$, where $k = R/\mathfrak{m}$. We first calculate the dimension, which is done by calculating the height of $\mathfrak{m}$ as a maximal ideal of $k[x, y]/(y - x^2)$. For this, we use the isomorphism $k[x, y]/(y - x^2) \cong k[t]$ defined by

$$x \mapsto t, \quad y \mapsto t^2,$$

and see where $\mathfrak{m}$ maps to. Then, $\mathfrak{m} = (x, y) \mapsto (t, t^2) = (t)$. $(0) \subsetneq (t)$, so by Krull's height theorem, this (t) has height 1. Now, as for $\dim_k \mathfrak{m}/\mathfrak{m}^2$, we first of all have that $R/\mathfrak{m} \cong k[t]/(t) \cong k$. Thus, $\dim_k(t)/(t)^2 = 1$, since $(t)/(t)^2$ is just the k -vector space with $k \cdot t$. Thus, regular.

2. Let $R := (k[x, y]/(x^2 - y^3))_{(x, y)}$. We try to do something similar. First, $k[x, y]/(x^2 - y^3) \cong k[t^3, t^2]$ via $x \mapsto t^3, y \mapsto t^2$. Then, (x, y) maps to $(t^3, t^2) \subseteq k[t^3, t^2]$. Then, $R/\mathfrak{m} \cong k[t^3, t^2]/(t^2, t^3) \cong k$. Since $(t^2, t^3)^2 = (t^4, t^5, t^6)$, we have that $(t^2, t^3)/(t^2, t^3)^2 = \text{Span}\{t^2, t^3\}$, so

$$\dim_k(t^2, t^3)/(t^2, t^3)^2 = 2.$$

We also want to calculate the height of $\mathfrak{m} \subseteq k[x, y]$. For this, we note that $\dim k[x, y] = 2$, and that $(0) \subsetneq (x^2 - y^3) \subsetneq (x, y) \subsetneq k[x, y]$. This provides a maximal chain. Therefore, we can't squeeze in another prime in between $(x^2 - y^3)$ and (x, y) . Since $(x^2 - y^3)$ is prime, it follows that $\dim R = h(\mathfrak{m}) = 1$, since if we take the quotient, the ideal (0) -disappears. $\square$

Example 4.2.6

Consider the ring $(k[x, y]/(xy))_{(x, y)}$. Let $R = k[x, y]/(xy)$ and $\mathfrak{m} = (x, y)$. We claim this is not regular. First, $\dim k[x, y]/(xy) = 1$ because the minimal primes which contain (xy) are (x) and (y) , which have height 1 by Krull’s height theorem. Since $\dim R = \dim R_{\mathfrak{m}}$, the dimension of the ring is 1. Now, as for the tangent space, we have that $\mathfrak{m}^2 = (x^2, xy, y^2)$. However, in R , we have that $xy = 0$, so $\dim_2 \mathfrak{m}/\mathfrak{m}^2 = 2$.

Exercise 4.2.7. Is the ring $R = (k[x, y]/(x^2, xy))_{(x, y)}$ regular?

Solution: We first calculate the dimension. To calculate the dimension, we first figure out the height of the minimal prime containing (x^2, xy) . Well, $\sqrt{I} = (x)$, which is prime. $h(x) = 1$, so $\dim k[x, y]/(x^2, xy) = 1$. Thus, the dimension of the local ring R is 1. Now, $\mathfrak{m}^2 = (x^2, xy, y^2)$, but in the quotient, x^2 and xy vanish. Therefore, $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 2$ because $\mathfrak{m}/\mathfrak{m}^2$ is generated by the classes of x, y in the quotient. $\square$

Remark 4.2.8 — Note that in the solution above, we used the radical to calculate the dimension. This fact can be used more generally: $\dim R/I = \dim R/\sqrt{I}$. Why is this true? Well, this is because the prime ideals of R/I correspond to prime ideals containing I in R . As for $\sqrt{I}$, it is equal to the intersection of prime ideals containing $\mathfrak{p}$, so prime ideals of $R/\sqrt{I}$ correspond to prime ideals containing $\sqrt{I}$. However, this is just prime ideals containing I . Thus, the dimension of R/I and $R/\sqrt{I}$ are the same!

Exercise 4.2.9. Is $R = (k[x, y, z]/(xy - z^2))_{(x, y, z)}$ regular?

Solution: We repeat what we’ve been doing above. The ideal $(xy - z^2)$ is prime because the generating polynomial is irreducible. Thus, $\dim R = 2$. However, $\mathfrak{m}^2 = (x^2, y^2, z^2, xy, yz, zx)$. In the quotient, $z^2 = xy$. Therefore, $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 3$. Thus, not regular. $\square$

Exercise 4.2.10. Let $f \in k[x_1, \dots, x_n]$ be an irreducible polynomial over an algebraically closed field k . A point P on the variety $f(x) = 0$ is *non-singular* if and only if not all the partial derivatives of f vanish at P . Let

$$A = k[x_1, \dots, x_n]/(f),$$

and let $\mathfrak{m}$ be the maximal ideal of A corresponding to the point P . Prove that P is non-singular if and only if $A_{\mathfrak{m}}$ is a regular local ring.

Hint: Show that $\dim A_{\mathfrak{m}} = n - 1$. Then,

$$\mathfrak{m}/\mathfrak{m}^2 \cong (x_1, \dots, x_n)/(x_1, \dots, x_n)^2 + (f)$$

and has dimension $n - 1$ if and only if $f \notin (x_1, \dots, x_n)^2$.

Solution: WLOG, assume P corresponds to the maximal ideal $(x_1, \dots, x_n)$ (by algebraic closure). Since f vanishes at P , we note that $f \in \mathfrak{m}$. Now, let $\mathfrak{m} = (x_1, \dots, x_n)/(f)$. Before proceeding, we shall calculate $\dim A_{\mathfrak{m}}$. We have that $\dim A_{\mathfrak{m}} = h(\mathfrak{m}A_{\mathfrak{m}}) = n - 1$, because quotienting by (f) killed the prime zero. Now, we must show the claim. Suppose

P is nonsingular to begin with. Then, $f \notin \mathfrak{m}^2$, because f cannot contain powers of the x_i . Then,

$$\mathfrak{m}/\mathfrak{m}^2 \cong (x_1, \dots, x_n)/((x_1, \dots, x_n)^2 + (f)).$$

Before modding by (f) , $(x_1, \dots, x_n)/(x_1, \dots, x_n)^2$ had the basis $\bar{x}_1, \dots, \bar{x}_n$. Therefore, if we have a relation

$$\sum_i \frac{\partial f}{\partial X_i}(P) \bar{y}_i = 0$$

which comes from writing f using its Taylor expansion (and modding out by f). This has a nontrivial relation, since not all the coefficients are zero. Therefore, $\dim_k \mathfrak{m}/\mathfrak{m}^2 = n - 1$ via some linear algebra.

Conversely, suppose that $\dim_k \mathfrak{m}/\mathfrak{m}^2 = n - 1$. This implies that as a k -vector space, $\mathfrak{m}/\mathfrak{m}^2$ is generated by $n - 1$ basis elements. This implies that the linear relation above has a nontrivial solution, implying that not all the partials vanish. Consequently, $f \notin \mathfrak{m}^2$ as well. $\square$

Example 4.2.11

Consider $R = k[x, y]_{(x)}$. Then, $\dim R = 1$ because the spectrum of R bijectively corresponds to primes contained by (x) . We note that $R/(x)R$ is isomorphic to $k(y)$. Therefore, $\dim_{k(y)} \mathfrak{m}/\mathfrak{m}^2 = 1$.

Exercise 4.2.12. Let R be a Noetherian ring. Then

$$\dim R[x] = 1 + \dim R,$$

and hence, by induction on n ,

$$\dim R[x_1, \dots, x_n] = n + \dim R.$$

Solution: We create bounds on both sides. First, if we have a maximal chain of prime ideals in A , say

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \dots \subsetneq \mathfrak{p}_n,$$

then this can be extended to a maximal chain of prime ideals in $R[X]$ by attaching $(\mathfrak{p}_n, X)$ at the end:

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \dots \subsetneq \mathfrak{p}_n \subsetneq (\mathfrak{p}_n, X),$$

where now the primes above are primes in $R[X]$ (that is, polynomials in $R[X]$ with coefficients in $\mathfrak{p}_i$). Thus, $\dim R[X] \geq \dim R + 1$. For the reverse inequality, we use Krull's principal ideal theorem. Let $\mathfrak{q} \in \text{Spec } R[X]$. Then, set $\mathfrak{p} := \mathfrak{q} \cap R$. Then, we have the isomorphism

$$R[X]/\mathfrak{p}R[X] \cong (R/\mathfrak{p})[X].$$

Inside this ring, $\mathfrak{q}/(\mathfrak{p}R[X])$ is a prime ideal. Furthermore, $\bar{\mathfrak{q}} \cap (R/\mathfrak{p}) = (0)$, because $\mathfrak{p} = \mathfrak{q} \cap R$. Since $R/\mathfrak{p}$ is an integral domain, it follows that $h(\mathfrak{q}) \leq 1$. To justify this last step, we prove that claim that if $\mathfrak{q} \in \text{Spec } A[X]$, where A is a domain such that $\mathfrak{q} \cap A = (0)$, then $h(\mathfrak{q}) \leq 1$. Let $S := A \setminus 0$. Then, $\mathfrak{q} \cap S = \emptyset$, so $\mathfrak{q}$ remains a proper prime ideal under the localization of $A[X]$ with respect to S . Now, let $K = S^{-1}A = \text{Frac}(A)$, so that $S^{-1}A[X] \cong K[X]$. By the localization, we have a bijection between primes of $A[X]$ disjoint with S and primes of $K[X]$. However, since K is a field, $K[X]$ is a Euclidean domain, so a PID, so $\dim K[X] = 1$. Therefore, $h(\mathfrak{q}) \leq 1$, and height is invariant under localization. $\square$

5 Discrete Valuation Rings

Definition 5.0.1. Let K be a field. A **discrete valuation** of K is a surjective map

$$\nu: K \setminus \{0\} \rightarrow \mathbb{Z}$$

such that, for all $x, y \in K \setminus \{0\}$, we have

1. $\nu(xy) = \nu(x) + \nu(y)$;
2. $\nu(x + y) \geq \min\{\nu(x), \nu(y)\}$.

We usually denote $\nu(0) = \infty$ so that the above two conditions continue to hold.

Remark 5.0.2 — It is easy to deduce that $\nu(1) = 0$.

Proposition 5.0.3

Let K be a field and let $\nu: K \setminus \{0\} \rightarrow \mathbb{Z}$ be a discrete valuation of K . Then

- (i) $\nu(x^{-1}) = -\nu(x)$ for all $x \in K \setminus \{0\}$.
- (ii) The set

$$R_\nu := \{x \in K : \nu(x) \geq 0\}$$

is a local ring, with maximal ideal

$$\mathfrak{m}_\nu := \{x \in K : \nu(x) > 0\} \subset R_\nu.$$

- (iii) If $t \in K$ is such that $\nu(t) = 1$ then $\mathfrak{m}_\nu = (t)$.
- (iv) If $I \subset R_\nu$ is an ideal then there exists an integer $n \geq 0$ such that $I = (t^n)$. In particular, R_ν is Noetherian and $\dim R_\nu = 1$.
- (v) Let $k_\nu := R_\nu / \mathfrak{m}_\nu$. Then $\dim_{k_\nu} \mathfrak{m}_\nu / \mathfrak{m}_\nu^2 = 1$, and, in particular, R_ν is regular.

The local ring $(R_\nu, \mathfrak{m}_\nu)$ is called a **discrete valuation ring (DVR)**. A generator t of $\mathfrak{m}_\nu$ is called a **parameter** or **uniformizer** of R .

Proof. (i) is easy.

(ii). The fact that R_ν is a ring is easy to verify. To show that $\mathfrak{m}_\nu$ is the unique maximal ideal, we show that $R_\nu \setminus \mathfrak{m}_\nu$ are invertible (in R_ν), so then Proposition 1.5.5 implies the result. If $x \in R_\nu \setminus \mathfrak{m}_\nu$, then $\nu(x) = 0$. This implies that $\nu(x^{-1}) = -\nu(x) = 0$, so $x^{-1} \in R_\nu$. The fact that $\mathfrak{m}_\nu$ is an ideal is clear as well.

We now prove (iii). By surjectivity of ν , it follows that such a t exists in the first place. If $x \in \mathfrak{m}_\nu$ then $\nu(x) = n \in \mathbb{Z}_{>0}$. Let

$$u := \frac{x}{t^n} \in K.$$

Then $\nu(u) = \nu(ut^n) - \nu(t^n) = 0$ and, therefore, $u \in R_\nu \setminus \mathfrak{m}_\nu$. Thus, u is invertible in R_ν , so $x = ut^n$ is in (t) and (iii) follows.

To prove (iv), if $I \subsetneq R_\nu$ is an ideal, then there exists $n \geq 0$ such that $I = (t^n)$, by taking

$$n = \min\{\nu(x) : x \in I\}.$$

Note that $n > 0$, since $I \subset R_\nu$. Then, (iv) follows that if $x \in I$, then $x \in (t^n)$. R_ν is therefore Noetherian. The dimension is 1 because there are no intermediate prime ideals between (0) and $\mathfrak{m}_\nu$.

For (v), since R_ν is a local ring and $\mathfrak{m}_\nu = (t)$, Proposition 4.0.2 implies that $\dim_{k_\nu} \mathfrak{m}_\nu / \mathfrak{m}_\nu^2 = \mu(\mathfrak{m}_\nu) = 1$. Thus, we have that $\dim R_\nu = \dim_{k_\nu} = \dim_{k_\nu} \mathfrak{m}_\nu / \mathfrak{m}_\nu^2$; thus, R_ν is regular. $\square$

Remark 5.0.4 — In particular, we have shown in the above that R_ν is a PID and regular.

Example 5.0.5 (*p*-adic valuation)

Let $k = \mathbb{Q}$ and fix a prime number p . Then any number $x \in k \setminus \{0\}$ can be written as

$$x = p^m \cdot \frac{a}{b} \quad \text{where } m \in \mathbb{Z} \text{ and } \gcd(a, p) = \gcd(b, p) = 1.$$

Note that $m \in \mathbb{Z}$ is uniquely defined by x . Thus, we can define the ***p*-adic valuation** as

$$\nu_p(x) := m.$$

The DVR of $\mathbb{Q}$ with respect to ν_p is $\mathbb{Z}_{(p)}$, which we've seen to be a local ring.

Solution: I will now just verify that this is indeed a valuation. Let $x = p^n \frac{a}{b}$, $y = p^m \frac{c}{d}$, where $p \nmid a, b, c, d$ and $a, b, c, d \in \mathbb{Z}$ and nonzero (WLOG). Assume WLOG that $n \geq m$. Then,

$$v(x + y) = v\left(p^m \left(p^{n-m} \frac{a}{b} + \frac{c}{d}\right)\right) = m + \dots \geq \min\{v(x), v(y)\}.$$

The multiplicativity is clear. $\square$

Lemma 5.0.6

Let R be a Noetherian integral domain and let $t \in R$ be a non-unit. Then

$$\bigcap_{n \geq 1} (t^n) = (0).$$

Proof. We may assume that $t \neq 0$, as otherwise, there is nothing to prove. Assume, by contradiction that there exists

$$0 \neq x \in \bigcap_{n \geq 1} (t^n).$$

Then, for each $n \geq 1$, there exists $x_n \in R$ such that

$$x = x_n t^n.$$

Since we're working in an integral domain, it follows that $x_{n-1} = x_n \cdot t$ and, in particular,

$$(x) \subset (x_1) \subset (x_2) \subset \dots$$

Since R is Noetherian, there exists a positive integer m such that $(x_m) = (x_{m+1})$, which implies that

$$x_{m+1} = ax_m \quad \text{for some } a \in R.$$

Thus,

$$x = x_{m+1}t^{m+1} = ax_mt^{m+1} = ax_{m-1}t^m = \dots = atx.$$

Since R is an integral domain, it follows that $at = 1$, i.e. t is a unit, a contradiction. $\square$

Lemma 5.0.7

Let $(R, \mathfrak{m})$ be a local ring, such that R is an integral domain and $\mathfrak{m} = (t)$ for some nonzero $t \in R$. Assume that

$$\bigcap_{n \geq 1} (t^n) = 0$$

and let $k := \text{Frac}R$. Then

- (i) If $x \in R$ is non-zero then $x = t^n u$ for some integer $n \geq 0$ and unit $u \in R$.
- (ii) Define $\nu(x) := n$ and if $y \in k$, define

$$\nu\left(\frac{x}{y}\right) := \nu(x) - \nu(y).$$

Then ν is a discrete valuation of k and $(R, \mathfrak{m})$ is its discrete valuation ring.

Proof. We first prove (i). If $x \in R$ is a unit, then we may write $x = t^0 \cdot u$ where $u = x$. Assume now that x is not a unit. Since $\mathfrak{m}$ is the unique maximal ideal of R , by Proposition 1.2.12, it follows that $x \in \mathfrak{m} = (t)$. Since $\bigcap_{n \geq 1} (t^n) = 0$, there exists a positive integer n such that

$$x \in (t^n) \setminus (t^{n+1}).$$

Thus, $x = t^n \cdot u$ and $u \notin \mathfrak{m}$. Thus, u is a unit and (i) follows.

We now prove (ii). Let $z \in k$. Then there exists $n \in \mathbb{Z}$ such that

$$z = t^n u \quad \text{where } u \in R \text{ is a unit.}$$

In particular $\nu(z) = n$. It is then easy to show that ν is a discrete valuation of k and $(R, \mathfrak{m})$ is its DVR. $\square$

Exercise 5.0.8. Show that R is a DVR if and only if R is a local PID that is not a field.

Solution: We've shown that if R is a DVR, then it is a local PID. Particularly, it is not a field because there is a nonzero proper ideal (the maximal ideal). Conversely, if R is a local PID that is not a field, the assumptions of the proposition above are met. $\square$

Remark 5.0.9 — Note that (ii) of the lemma above and Proposition 5.0.3 tells us that DVRs and discrete valuations correspond with each other (note that it's not bijective because the valuation we get from a DVR need not to be normalized).

Lemma 5.0.10

Let R be a Noetherian ring and let M be a non-zero R -module. Then there exists $0 \neq x \in M$ such that $\text{ann}(Rx)$ is a prime ideal of R .

Proof. Note that $\text{ann}(Rx) = \{y \in R: rxy = 0, \forall r \in R\}$. However, we can set $r = 1$, so this is just equal to the set $\{y \in R: xy = 0\} \subset R$. Consider the set

$$P := \{\text{ann}(Rx): 0 \neq x \in M\} \neq \emptyset.$$

Since R is Noetherian, by Zorn's Lemma, there exists $x \in M$ such that $\text{ann}(Rx)$ is maximal in P . We want to show that $\mathfrak{p} := \text{ann}(Rx)$ is a prime ideal. Assume by contradiction that there exist $a, b \in R$ such that $a, b \notin \mathfrak{p}$ and $ab \in \mathfrak{p}$. That is, we have

$$ax \neq 0 \quad bx \neq 0 \quad \text{and} \quad ab \cdot x = 0.$$

In particular, $\text{ann}(Rx) \subset \text{ann}(R(a \cdot x))$, since if $zx = 0$, then $zax = 0$ (by commutativity). Further, $b \in \text{ann}(R(a \cdot x)) \setminus \text{ann}(Rx)$, because $bax = 0$ but $bx \neq 0$. This means that $\text{ann}(Rx) \subsetneq \text{ann}(R(ax))$, which is a contradiction to the maximality of $\mathfrak{p}$ in P . $\square$

Theorem 5.0.11

Let $(R, \mathfrak{m})$ be a local ring. Then $(R, \mathfrak{m})$ is a DVR if and only if R is Noetherian, normal and

$$\text{Spec } R = \{(0), \mathfrak{m}\}.$$

Proof. Assume first that $(R, \mathfrak{m})$ is a DVR. The Proposition 5.0.3 immediately implies that R is Noetherian and $\text{Spec } R = \{(0), \mathfrak{m}\}$. Note that Proposition 5.0.3 tells us that every ideal of R is of the form (t^n) , so implies that R is a PID and, in particular, a UFD. Thus, R is normal.

Assume now that R is Noetherian, normal and $\text{Spec } R = \{(0), \mathfrak{m}\}$. Let $k = \text{Frac}(R)$. Since $\mathfrak{m} \neq 0$, if $\mathfrak{m} = \mathfrak{m}^2$, then Nakayama's lemma tells us that $\mathfrak{m}\mathfrak{m} = \mathfrak{m}$, so $\mathfrak{m} = 0$, which is not true. Thus, $\mathfrak{m} \neq \mathfrak{m}^2$. Let $x \in \mathfrak{m} \setminus \mathfrak{m}^2$. We want to show that $\mathfrak{m} = (x)$. Consider the R -module $M := \mathfrak{m}/(x)$ and assume by contradiction that $M \neq 0$. Then by Lemma 5.0.10 there exists $y \in \mathfrak{m} \setminus (x)$ such that if $\bar{y} \in M$ is the projection of y in M then $\text{ann}(R\bar{y})$ is a prime ideal of R . Since $\mathfrak{m}$ is the only non zero prime ideal of R , it follows that $\text{ann}(R\bar{y}) = \mathfrak{m}$, so $\bar{y}\mathfrak{m} = (0)$, implying that $y\mathfrak{m} = (x)$ in R . Define

$$z := \frac{y}{x} \in k \setminus R.$$

Since $y \cdot \mathfrak{m} = (x)$, we have that $z \cdot \mathfrak{m} \subset \frac{y}{x}\mathfrak{m} \subset R$. Assume that $z \cdot \mathfrak{m} = R$. Then there exists $w \in \mathfrak{m}$ such that $wz = 1$ and, therefore, we have $x = wy \in \mathfrak{m}^2$, a contradiction to $x \notin \mathfrak{m}^2$. Thus, $z \cdot \mathfrak{m} \neq R$ and since $\mathfrak{m}$ is the only maximal ideal in R , we have that $z \cdot \mathfrak{m} \subset \mathfrak{m}$.

We now want to show that z is integral over R . Consider the R -linear map

$$\varphi: \mathfrak{m} \rightarrow \mathfrak{m} \quad t \mapsto z \cdot t.$$

By Cayley–Hamilton’s Theorem, there exist $a_0, \dots, a_{n-1} \in R$ such that

$$\varphi^n(t) + a_{n-1} \cdot \varphi^{n-1}(t) + \dots + a_1 \cdot \varphi(t) + a_0 \cdot t = 0$$

for all $t \in \mathfrak{m}$. Thus

$$(z^n + a_{n-1}z^{n-1} + \dots + a_1z + a_0) \cdot t = 0 \quad \forall t \in \mathfrak{m}.$$

Since (0) is a prime ideal, note that R is an integral domain. Thus, $t \in \mathfrak{m}$, $z^n + \dots + a_0 = 0$ which implies that z is integral over R .

Since R is normal, we have that $\frac{y}{x} = z \in R$. Thus, $y \in (x)$, which contradicts our assumption. Hence, $M = 0$ which implies that $\mathfrak{m} = (x)$.

Finally, Lemma 5.0.6 and Lemma 5.0.7 imply that $(R, \mathfrak{m})$ is a DVR, as claimed. $\square$

Remark 5.0.12 — In algebraic geometry, regular rings are always normal. However, if the dimension of the ring is 1, then we have the reverse implication.

Example 5.0.13

Let k be a field and $R = k[x, y]/(x^2 - y^3)$. We want to localize R at the point $p = (0, 0) \in V(I)$, where $I = (x^2 - y^3)$. Note that $\dim R = 1$ by Krull’s height theorem. We want to show that $R_{\mathfrak{m}}$ is not regular, where $\mathfrak{m}$ is the maximal ideal at the point $(0, 0)$: $\mathfrak{m} = (x, y)$. By the previous theorem, it is enough to show that $R_{\mathfrak{m}}$ is not normal. Then, it can’t be a DVR, so can’t be regular. Let $t = \frac{x}{y}$. Then, $t^3 = \frac{x^3}{y^3} = x$ in R . Thus, $t^3 - x = 0$, which is monic. At the same time $t \notin R$, so R is not normal (that is, we’ve found an integral element which is not in R).

Exercise 5.0.14. Show that $\mathbb{Z}[i]_{(1+i)}$ is a DVR.

Solution: We do this by using the theorem above. First, we note that $(1 + i)$ is a prime ideal of $\mathbb{Z}[i]$ because $\mathbb{Z}[i]/(1 + i) \cong \mathbb{Z}/2\mathbb{Z}$. Set $R := \mathbb{Z}[i]_{(1+i)}$ and $\mathfrak{m} := (1 + i)\mathbb{Z}[i]_{(1+i)}$. Then, $\mathbb{Z}[i]$ is a Euclidean domain, so it is Noetherian and normal (from PID and UFD). Therefore, these properties carry over to the localization by previous results. $\text{Spec}(R)$ bijectively corresponds to ideals in $\text{Spec}(\mathbb{Z}[i])$ contained in $(1 + i)$; however, by Krull’s principal ideal theorem, $h(1 + i) = 1$. Thus, the only ideal contained in $(1 + i)$ is the (0) ideal, so $\text{Spec}(R) = \{0, \mathfrak{m}\}$. Thus, we have shown all conditions and so $(R, \mathfrak{m})$ is a DVR. $\square$

6 Completions

Let K be a field and $\nu: K \rightarrow \mathbb{Z} \cup \{\infty\}$ be a discrete valuation. Let $r \in \mathbb{R}$ with $r > 1$. We set

$$|x|_\nu := r^{-\nu(x)}, \quad \text{for } x \in K.$$

We can verify that $|\cdot|_\nu: K \rightarrow \mathbb{R}_{\geq 0}$ is a norm on K . Namely, for all $x, y \in K$, the following properties are satisfied:

1. $|xy|_\nu = |x|_\nu |y|_\nu$;
2. $|x + y|_\nu \leq \max\{|x|_\nu, |y|_\nu\}$; and
3. $|x|_\nu = 0 \implies x = 0$.

Recall that the second property is called the **non-archimedean property**. In particular, it implies the usual triangular inequality $|x + y|_\nu \leq |x|_\nu + |y|_\nu$. Hence, we can define a distance function $d(x, y) = |x - y|_\nu$ and to equip K with a topology with basic open sets

$$B_{x_0, \epsilon} := \{x \in K : |x - x_0|_\nu < \epsilon\}.$$

Note that the topology does not depend on the choice of $r > 1$. We next show that in non-archimedean geometry every triangle is isosceles:

Lemma 6.0.1

If $|x|_\nu < |y|_\nu$ then $|x - y|_\nu = |y|_\nu$.

Proof. We have $|x - y|_\nu \leq \max\{|x|_\nu, |y|_\nu\} = |y|_\nu$ since $|x|_\nu < |y|_\nu$. On the other hand, $y = (y - x) + x$, hence $|y|_\nu \leq \max\{|x - y|_\nu, |x|_\nu\} = |x - y|_\nu$ since $|x|_\nu < |y|_\nu$. $\square$ $\square$

Definition 6.0.2. Let K be a field and let $\nu: K \rightarrow \mathbb{Z} \cup \{\infty\}$ be a discrete valuation. A sequence $x_n \in K$, with $n \geq 1$, is a **Cauchy sequence** if for any $\epsilon > 0$ there exists $n_0 \geq 1$ such that $|x_n - x_m|_\nu < \epsilon$ for all $n, m \geq n_0$.

Remark 6.0.3 — This is equivalent to the condition that for all $N > 0$ there exists $n_0 \geq 1$ such that $\nu(x_n - x_m) > N$ for all $n, m \geq n_0$.

Definition 6.0.4. We say that the sequence $(x_n)_{n \geq 1}$ **converges** to a limit $x \in K$ if for all $\epsilon > 0$ there exists $n_0 \geq 0$ such that $|x_n - x|_\nu < \epsilon$ for all $n \geq n_0$. Or equivalently, for all $N > 0$ there exists $n_0 \geq 0$ such that $\nu(x_n - x) > N$ for all $n \geq n_0$.

Definition 6.0.5. The field K is called **complete** with respect to the valuation ν if every Cauchy sequence has a limit.

Definition 6.0.6. Let K be a field and $\nu: K \rightarrow \mathbb{Z} \cup \{\infty\}$ be a discrete valuation. We say that a field $\widehat{K}_\nu \supset K$ is a **completion** of K if the valuation ν extends to $\widehat{K}_\nu$, it is complete and it is the closure of K with respect to the topology induced by ν .

Given a field K with a valuation ν , there is a canonical construction of a completion of K_ν . We outline the construction. First, we set

$$\mathcal{C}_\nu := \{(x_n)_{n \geq 1} : x_n \in K, (x_n) \text{ is a Cauchy sequence}\},$$

and

$$\mathcal{I}_\nu := \{(x_n)_{n \geq 1} \in \mathcal{C}_\nu : \lim_{n \rightarrow \infty} x_n = 0\}.$$

Note that $\mathcal{C}_\nu$ is a commutative ring with

- $(x_n) + (y_n) = (x_n + y_n)$
- $(x_n) \cdot (y_n) = (x_n y_n)$;
- $0 = (0)_{n \geq 1}$; and
- $1 = (1)_{n \geq 1}$.

Moreover, $\mathcal{I}_\nu$ is a maximal ideal in $\mathcal{C}_\nu$, hence $\widehat{K}_\nu := \mathcal{C}_\nu / \mathcal{I}_\nu$ is a field. We have that the map $K \rightarrow \widehat{K}_\nu$ given by $x \mapsto (x)_{n \geq 1}$ is an embedding, hence $K \subset \widehat{K}_\nu$ is a field extension.

Lemma 6.0.7

Let K be a field and $\nu: K \rightarrow \mathbb{Z} \cup \{\infty\}$ be a discrete valuation. Let $0 \neq \alpha \in \widehat{K}_\nu$ be represented by the class of a sequence $(x_n)_{n \geq 1} \in \mathcal{C}_\nu / \mathcal{I}_\nu$. Then there exists $n_0 \geq 1$ such that $\nu(x_n) = \nu(x_{n_0})$ for all $n \geq n_0$. We define $\widehat{\nu}: \widehat{K}_\nu \rightarrow \mathbb{Z} \cup \{\infty\}$ by letting

$$\widehat{\nu}(\alpha) = \lim_{n \rightarrow \infty} \nu(x_n) = \nu(x_{n_0}).$$

Then $\widehat{\nu}$ is a valuation extending ν .

Proof. Since $\alpha \neq 0$, $(x_n) \notin \mathcal{I}_\nu$. Hence, there exists $c > 0$ such that for all $n \geq 1$, there exists $i_n > n$ with $\nu(x_{i_n}) \leq c$. On the other hand, (x_n) is a Cauchy sequence and, therefore, there exists $N \geq 1$ such that $\nu(x_n - x_m) > c$ for all $n, m \geq N$. Let $n_0 = i_N > N$. Then, for all $n \geq n_0$, we have

$$\nu(x_n) = \nu(x_n - x_{n_0} + x_{n_0}) \geq \min\{\nu(x_n - x_{n_0}), \nu(x_{n_0})\} = \nu(x_{n_0})$$

and

$$\nu(x_{n_0}) = \nu(x_{n_0} - x_n + x_n) \geq \min\{\nu(x_{n_0} - x_n), \nu(x_n)\} = \nu(x_n).$$

Hence, $\nu(x_n) = \nu(x_{n_0})$ for all $n \geq n_0$. We define $\widehat{\nu}(\alpha) := \nu(x_{n_0})$.

It is an easy exercise to check that $\widehat{\nu}$ is well-defined and a valuation. □

Remark 6.0.8 — With this lemma, we can show that $\mathcal{I}_\nu$ is maximal. Since $x_n \neq 0$ for all $n > n_0$, we can define a new sequence (y_n) by

$$y_n = \begin{cases} 1/x_n & \text{if } n \geq n_0, \\ 0 & \text{otherwise.} \end{cases}$$

Then $x_n y_n = 1$ for $n \geq n_0$, hence $(x_n)(y_n) - 1 \in \mathcal{I}_\nu$. Thus, $\mathcal{I}_\nu$ is maximal.

Thus, giving a field K with a discrete valuation $\nu: K \rightarrow \mathbb{Z} \cup \{\infty\}$, we have constructed a field extension $K \subset \widehat{K}_\nu$ and a valuation $\widehat{\nu}: \widehat{K}_\nu \rightarrow \mathbb{Z} \cup \{\infty\}$ such that all the Cauchy sequences in $\widehat{K}_\nu$ converge. In particular, if $\alpha \in \widehat{K}_\nu$ is represented by $(x_n)_{n \geq 1} \in \mathcal{C}_\nu/\mathcal{I}_\nu$ then $\lim_{n \rightarrow \infty} x_n = \alpha$.

As in Proposition 5.0.3, let

$$\widehat{R}_\nu := \{x \in \widehat{K}_\nu : \widehat{\nu}(x) \geq 0\}$$

be the corresponding discrete valuation ring in $\widehat{K}_\nu$. Let $t \in R_\nu$ such that $\nu(t) = 1$. Proposition 5.0.3 implies that $\mathfrak{m}_\nu = (t)$ is the maximal ideal of R_ν . Since $\widehat{\nu}|_{R_\nu} = \nu$, we have $\widehat{\nu}(t) = 1$ and $\widehat{\mathfrak{m}}_\nu = (t)$ is the unique maximal ideal of $\widehat{R}_\nu$. Thus, we have a diagram of inclusions as follows:

$$\begin{array}{ccccc} K_\nu & \supset & R_\nu & \supset & \mathfrak{m}_\nu \\ \cap & & \cap & & \cap \\ \widehat{K}_\nu & \supset & \widehat{R}_\nu & \supset & \widehat{\mathfrak{m}}_\nu \end{array}$$

Exercise 6.0.9. Let k be a field. Show that the completion of the discrete valuation ring $k[x]_{(x)}$ is isomorphic to the formal power series ring $k[[x]]$.

Solution: We note that if we take the field of rational function $K = k(x)$, $R := k[x]_{(x)}$ is the valuation ring with respect to the x -adic valuation

$$\nu\left(x^m \frac{f(x)}{g(x)}\right) = m,$$

where $x \nmid f(x), g(x)$ and $g(0) \neq 0$. Now, the idea is that we want to map $\widehat{R}$ to $k[[x]]$ using polynomial division, and the inverse being just taking the equivalence class of a formal power series. Thus, let $\phi_n: R \rightarrow k[x]/(x^n)$ be the map which takes $\frac{f}{g}$ and maps it to $fg^{-1} \pmod{x^n}$. Now if $\alpha \in \widehat{R}$, then write α as a sequence $\left(\frac{f_n}{g_n}\right)_n$. Then, we let Φ be the map which takes α to be the unique formal power series $\Phi(\alpha)$ which is equal to $\phi_n\left(\frac{f_n}{g_n}\right)$, for all n . Formally,

$$\Phi\left(\frac{f_n}{g_n}\right) = \sum_{i=0}^{\infty} a_i x^i,$$

where

$$\sum_{i=0}^{m-1} a_i x^i = f_n g_n^{-1} \pmod{x^m},$$

for large n . Note that such a_i are uniquely determined by completeness of $\widehat{R}$. As for the inverse map, we take some formal power series and map it to its equivalence class in $\widehat{R}$, truncated at each index. That is, $\sum_{i=0}^{\infty} a_i x^i$ maps to

$$\left[\left(\sum_{i=0}^{m-1} a_i x^i \right)_m \right].$$

We note that this is well-defined because such a sequence is Cauchy. Because the difference between any two partial sums is contained in some ideal (x^m) , so tends to 0. These maps are clearly ring homomorphisms. Seeing that $\Phi\Psi = \text{id}$ is clear, so we check the other composition. For this, we just need to examine how the power series α and $\Psi\Phi(\alpha)$ differ x -adically. Fix $N \geq 0$. Then, for all $n \geq N$, we have

$r_n := \frac{f_n}{g_n} \equiv a_0 + \cdots + a_{N-1}x^{N-1} \pmod{x^N}$. If $m \geq N$, then let t_m be what we get after applying Ψ to $\Phi(\alpha)$ and taking modulo x^N . We get that the coefficients appearing should match up, so $r_n \equiv t_m \pmod{x^N}$, so taking $n = m$ and taking n to be big enough, we find that $r_n - t_n \in (x^N)$. N was arbitrary, so this holds for all N . Therefore, $r_n - t_n \rightarrow 0$, so they represent the same element in the completion. Thus, $\hat{R} \cong k[[x]]$. $\square$

Remark 6.0.10 — A less constructive way to attack the above is to define a map from $k[[x]] \rightarrow \hat{R}$. For this, we notice that any formal power series is Cauchy in K . Therefore, we can just map any formal power series to its truncation modulo x^n . This gives us a ring homomorphism

$$\Phi: k[[x]] \rightarrow \widehat{K}_\nu, \quad f(x) \mapsto (f_n(x))_{n \geq 1}.$$

Because $(f_n(x))_{n \geq 1}$ is a Cauchy sequence in R , the image of Φ is contained in $\hat{R}_\nu$. Let

$$\mathfrak{m} := (x)k[x]_{(x)}$$

be the unique maximal ideal in R . Let $\bar{x}$ be the image of x in R . Then $\bar{x}$ is a uniformizer and we have an isomorphism $R/\mathfrak{m} \cong k$. Thus, every element of the completion of $(R, \mathfrak{m})$ can be written uniquely as

$$\sum_{i=0}^{\infty} a_i \bar{x}^i,$$

where $a_i \in k \subset R$.

Example 6.0.11

$\mathbb{F}_p[[x]]$ is a complete DVR with fraction field $\mathbb{F}_p((x))$, the field of formal Laurent series. This is an example of a *local field*. $\mathbb{Q}_p$ is also an example of a *local field*.

Let K be a field and let $\nu: K \rightarrow \mathbb{Z} \cup \{\infty\}$ be a discrete valuation. Let $(x_n)_{n \geq 1}$ be a sequence in K . If the sequence of partial sums $s_N := \sum_{n \geq 0}^N x_n$ converges, then we will call the **limit** to be the infinite sum $\sum_{n \geq 0} x_n$. In a complete discrete valuation ring $\hat{R}_\nu$, the sum converges if and only if $\lim_{n \rightarrow \infty} x_n = 0$. Indeed, if $x_n \rightarrow 0$, then the partial sums form a Cauchy sequence since

$$|s_N - s_M|_\nu = |x_{N+1} + \cdots + x_M|_\nu \leq \max_{N < n \leq M} |x_n|_\nu$$

and conversely if s_N converge, then $s_N - s_{N-1} = x_N \rightarrow 0$.

Proposition 6.0.12

Let K be a field and let $\nu: K \rightarrow \mathbb{Z} \cup \{\infty\}$ be a discrete valuation. Let $A \subset R_\nu$ be a set of representatives of $k_\nu = R_\nu/\mathfrak{m}_\nu$. Then every element of $\hat{R}_\nu$ can be written uniquely as $\sum_{m=0}^{\infty} a_m t^m$ with $a_m \in A$ where a representative Cauchy sequence $(x_n) \geq 0$ is given by the partial sums $x_n = \sum_{m=0}^n a_m t^m$.

Proof. Let $x \in \hat{R}_\nu$ be given. There is exactly one $a_0 \in A$ such that $|x - a_0|_\nu < 1$. Then, we let $x = a_0 + tb_1$ for some $b_1 \in \hat{R}_\nu$. Then, there is exactly one $a_1 \in A$ such that $|b_1 - a_1|_\nu < 1$, and then $b_1 = a_1 + tb_2$ for some $b_2 \in \hat{R}_\nu$. In this way, we get for every

n , $x = a_0 + a_1t + a_2t^2 + \cdots + a_nt^n + b_{n+1}t^{n+1}$ with $a_i \in A$ and $b_{n+1}t^{n+1} \rightarrow 0$, hence $x = \sum_{m=0}^{\infty} a_mt^m$. $\square$

Thus, given two elements $x = \sum_{m=0}^{\infty} a_mt^m$, $y = \sum_{m=0}^{\infty} b_mt^m$, using the addition and multiplication in R_ν , we can write

$$\sum_{m=0}^{\infty} a_mt^m + \sum_{m=0}^{\infty} b_mt^m = \sum_{m=0}^{\infty} c_mt^m$$

and

$$\left(\sum_{m=0}^{\infty} a_mt^m \right) \cdot \left(\sum_{m=0}^{\infty} b_mt^m \right) = \sum_{m=0}^{\infty} d_mt^m.$$

Remark 6.0.13 — In general, determining $(c_m)_{m \geq 0}$ and $(d_m)_{m \geq 0}$ from $(a_m)_{m \geq 0}$ and $(b_m)_{m \geq 0}$ can be quite complicated. In particular, the above proposition does not mean that every $\hat{R}_\nu$ is isomorphic to $k_\nu[[t]]$. On the other hand, if we want to compute c_m and d_m for all $m \leq n$ for some fixed n , we can do this in R_ν by working modulo t^n , thanks to the following isomorphism of rings:

$$R_\nu/t^n R_\nu \cong \hat{R}_\nu/t^n \hat{R}_\nu \quad \text{for all } n \geq 1.$$

Proof. Let us prove this ring isomorphism. We shall construct a ring isomorphism $\varphi_n: \hat{R}_\nu \rightarrow R_\nu/t^n R_\nu$. Let α represent a class in $\hat{R}_\nu$. Then, $\alpha = (f_n)$, where $f_n = \sum_{i=0}^n a_i t^i$, where t is the uniformizer and $a_i \in A$, a set of representatives of the residue field. We let φ_n be the natural map

$$\alpha \mapsto \alpha \pmod{t^n} = f_{n-1}.$$

This is clearly surjective, due to the inclusion map $R_\nu \hookrightarrow \hat{R}_\nu$. If $\alpha \mapsto 0$, then actually $\nu(\alpha) \geq n$, so $\alpha \in t^n \hat{R}_\nu$. Thus, the kernel is $t^n \hat{R}_\nu$. Therefore, they lie in the same class. This is also clearly a morphism of rings, with the product/sum defined above the remark. $\square$

Example 6.0.14

As in Example 5.0.5, let p be prime number and consider the discrete valuation ring $\mathbb{Z}_{(p)}$ with the valuation $\nu_p: \mathbb{Q} \rightarrow \mathbb{Z} \cup \{\infty\}$. The field of p -adic numbers is

$$\mathbb{Q}_p := \hat{\mathbb{Q}}_\nu$$

and we define p -adic integers as

$$\mathbb{Z}_p := \{a \in \mathbb{Q}_p : \hat{\nu}_p(a) \geq 0\},$$

which is a local ring with maximal ideal (p) , and $\mathbb{Z}_p/(p) \cong \mathbb{Z}/p\mathbb{Z}$.

Exercise 6.0.15. Describe what the elements of $\mathbb{Q}_p$ and $\mathbb{Z}_p$ look like, using Proposition 6.0.12.

Solution: We note that $\mathbb{Q}_p$ is the completion of $\mathbb{Q}$ with respect to $|\cdot|_p$ with complete discrete valuation ring $\mathbb{Z}_p$. The uniformizer of ν_p is the prime p . Therefore, every element

of $\mathbb{Z}_p$ can be described as

$$\sum_{i=0}^{\infty} a_i p^i, \quad a_i \in \{0, 1, \dots, p-1\}.$$

As for $\mathbb{Q}_p$, this just extends to negative powers:

$$\sum_{i=-m}^{\infty} a_i p^i, \quad a_i \in \{0, 1, \dots, p-1\}.$$

□

Exercise 6.0.16. Show that the completion of $k[x, y]_{(x)}$ is isomorphic to $k(y)[[x]]$.

Solution: Let $K = \text{Frac}(k[x, y])$, and $\nu: K^\times \rightarrow \mathbb{Z}$ measure the power of x dividing $x^n \frac{f(x, y)}{g(x, y)} \in K$, where $x \nmid f, g$. Then, $k[x, y]_{(x)}$ is the discrete valuation with respect to this setup. We now define a map from $k(y)[[x]]$ to $\widehat{R}_\nu$. Given a formal power series $S = \sum_{i=0}^{\infty} f_i(y)x^i$, let us denote the partial sums $S_n = \sum_{i=0}^n f_i(y)x^i$. Then S_n forms a Cauchy sequence in $k(x, y)$, since the difference of any two partial sums is contained in some ideal, so the induced norm tends to 0. Then, we send S to its class in the completion

$$[(f_n)_n].$$

This is well-defined, because any such sequence is Cauchy. The two partial sums is contained in some ideal (x^m) , so as m increases, the associated norm tends to zero. This is also clearly a morphism of rings. Now, we note that the image is contained inside $\widehat{R}_\nu$. Let $\mathfrak{m} = (x)k[x, y]_{(x)}$ be the unique maximal ideal in $\widehat{R}_\nu$, and $\bar{x}$ be the image of x in $R = k[x, y]_{(x)}$. Then, $\bar{x}$ is a uniformizer and $R/\mathfrak{m} \cong k(y)$. Therefore, every element of the completion can be written as $\sum_{i=0}^{\infty} h_i(y)\bar{x}^i$, where $h_i(y) \in k(y) \subseteq k[x, y]_{(x)}$. □

Exercise 6.0.17. Let A be an intermediate ring

$$k[x] \subset A \subset k[[x]]$$

where $k[[x]]$ is the ring of formal power series. Suppose that A is local with maximal ideal (x) . Prove that A is a DVR, and in particular A is Noetherian.

Solution: We shall show that A is an integral domain and that $\cap_{n \geq 1} (x^n) = 0$. A is clearly an integral domain because it is a subring of $k[[x]]$, which is an integral domain. As for the intersection condition, suppose that there was some element $f \in \cap_{n \geq 0} (x^n)$. Since f is in A , which is, in particular in $k[[x]]$, there is some lowest starting term, say of degree m . On the other hand, f is in the intersection, so $f = xg_1 = x^2g_2 = \dots$, for $g_i \in A$, but then we have that $g_1 = xg_2 = x^2g_3 = \dots$, because A is an integral domain. In particular, we would find that $f = x^{m+1}g_{m+1}$, for some $g_{m+1} \in A$, but g_{m+1} is not in A due to lowest degree reasons, since it is not even in $k[[x]]$. Therefore, $\cap_{n \geq 1} (x^n) = 0$. Therefore, A is a DVR. Therefore, A is Noetherian, because a local DVR is Noetherian, normal, and has $\text{Spec } A = \{(0), \mathfrak{m} = (x)\}$. □

§6.1 Hensel’s Lemma

Exercise 6.1.1. Show that $\mathbb{Z}[[x]]/(x-p)$ is isomorphic to $\mathbb{Z}_p$.

Solution. Consider the map $\varphi: \mathbb{Z}[[x]] \rightarrow \mathbb{Z}_p$ which maps $f \mapsto f(p)$ (formally). We just need to show that $f \in \ker \varphi$ if and only if $f = g \cdot (x-p)$. First, if f is of this form, it is clear that $f(p) = 0$. Thus, we prove other implication. If $f(p) = 0$, then this means that

$$\sum_{i=0}^{\infty} a_i p^i = 0.$$

In particular, this means that the finite sum $\sum_{i=0}^n a_i p^i$ is divisible by p^{n+1} , which is crucial. We want to show that $f = g \cdot (x-p)$, so we want to show that we can write

$$f(x) = \sum_{i=1}^{\infty} (b_{i-1} - p b_i) x^i - p b_0.$$

We find these b_i s by iterating. First, taking modulo p of the above, we find that $a_0 \equiv 0 \pmod{p}$, so we let $b_0 := -\frac{a_0}{p}$. Next, we define $b_1 := -\frac{a_0 + a_1 p}{p}$. Then, it is clear that

$$a_1 = b_0 - p b_1.$$

Now, suppose we've defined up to b_{k-1} . Continuing, we set $b_k = -\frac{\sum_{i=0}^k a_i p^i}{p^{k+1}} = -\frac{a_k}{p} + \frac{b_{k-1}}{p}$, so we get that

$$a_k = b_{k-1} - p b_k.$$

Therefore, we have shown what is desired. This is also unique by the theorem above. $\square$

Theorem 6.1.2 (Hensel's lemma)

Let K be a field and let $\nu: K \rightarrow \mathbb{Z} \cup \{\infty\}$ be a discrete valuation and $|\cdot|$ be the induced norm from ν . Let $f(X) = \sum_{i=0}^n a_i X^i \in R_\nu[X]$ with $a_n \in R_\nu \setminus \mathfrak{m}_\nu$. Suppose that there exists $x_0 \in R_\nu$ such that

$$|f(x_0)| < |f'(x_0)|^2, \quad (6.1)$$

where $f'(X) := \sum_{i=1}^n i a_i X^{i-1}$ is the (formal) derivative of $f(X)$. Then, there exists a unique root x of f in $\widehat{R}_\nu$, satisfying $|x - x_0| < |f'(x_0)|$. Furthermore, we have

$$|x - x_0| = \frac{|f(x_0)|}{|f'(x_0)|} < |f'(x_0)| \quad \text{and} \quad |f'(x)| = |f'(x_0)|.$$

Proof. Since $\left| \frac{f(x_0)}{f'(x_0)} \right| < |f'(x_0)| \leq 1$, there exists $y_0 \in \mathfrak{m}_\nu$ such that $f(x_0) + y_0 f'(x_0) = 0$. Let us define polynomials $f_1, f_2, \dots \in R_\nu[X]$ via the identity

$$f(X + Y) = f(X) + f_1(X)Y + f_2(X)Y^2 + \dots$$

Then $f_1(X) = f'(X)$. We have

$$|f(x_0 + y_0)| \leq \max_{j \geq 2} |f_j(x_0) y_0^j|.$$

Since $f_j(X) \in R_\nu[X]$ and $x_0 \in R_\nu$, we have $|f_j(x_0)| \leq 1$. Hence,

$$|f(x_0 + y_0)| \leq |y_0|^2 = \frac{|f(x_0)|^2}{|f'(x_0)|^2} < |f(x_0)|.$$

Similarly,

$$|f'(x_0 + y_0) - f'(x_0)| \leq |y_0| < |f'(x_0)|$$

and so $|f'(x_0 + y_0)| = |f'(x_0)|$ by the non-archimedean property of the norm. We now put $x_1 = x_0 + y_0$ and note that, similarly as in (6.1), we have

$$|f(x_1)| < |f'(x_1)|^2.$$

Thus, we can repeat this process. In this way, we obtain a sequence $(x_n)_{n \geq 0}$ using Newton's formula $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ with the properties:

1. $|x_n| \leq 1$ hence $x_n \in R_\nu$.
2. $|f'(x_n)| = |f'(x_0)|$.
3. $|f(x_{n+1})| \leq \frac{|f(x_n)|^2}{|f'(x_n)|} = \frac{|f(x_n)|^2}{|f'(x_0)|} < |f(x_n)|$ and so $f(x_n) \rightarrow 0$ as $n \rightarrow \infty$.
4. $|x_{n+1} - x_n| = |y_n| = \frac{|f(x_n)|}{|f'(x_n)|} = \frac{|f(x_n)|}{|f'(x_0)|} \rightarrow 0$ as $n \rightarrow \infty$.

It follows that $(x_n)_{n \geq 0}$ is a Cauchy sequence with respect to ν . Then, the element $x := \lim_{n \rightarrow \infty} x_n \in \widehat{K}_\nu$ exists and $f(x) = 0$. To see that $|x - x_0| = \frac{|f(x_0)|}{|f'(x_0)|}$, let us observe that

$$|x_{n+1} - x_n| = \left| \frac{f(x_n)}{f'(x_n)} \right| < \frac{|f(x_0)|}{|f'(x_0)|}$$

for all $n \geq 1$. Now, note that $|x_1 - x_0| = \frac{|f(x_0)|}{|f'(x_0)|}$ and if $|x_n - x_0| = \frac{|f(x_0)|}{|f'(x_0)|}$ for some n , then it follows that $|x_{n+1} - x_0| = \frac{|f(x_0)|}{|f'(x_0)|}$ since we have $|x_{n+1} - x_0| = |x_n - x_0|$ by Lemma 6.0.1.

Finally to see the uniqueness, suppose $\tilde{x}$ is another solution with $|\tilde{x} - x_0| < |f'(x_0)|$ and $x \neq \tilde{x}$. Then, let $\tilde{x} = x + \tilde{y}$. We have

$$0 = f(x + \tilde{y}) - f(x) = f'(x)\tilde{y} + f_2(x)\tilde{y}^2 + \dots$$

But,

$$|\tilde{y}| = |x - x_0 + x_0 - \tilde{x}| \leq \max\{|x - x_0|, |\tilde{x} - x_0|\} < |f'(x_0)| = |f'(x)|$$

and $|f_i(x)| \leq 1$, the first term on the right side of the above equation has strictly greater norm than the others, hence the sum cannot be 0, which is a contradiction. $\square$

Remark 6.1.3 — An easier condition to check than (6.1) is

$$\overline{f}(x_0) = 0 \quad \text{and} \quad \overline{f'}(x_0) \neq 0 \tag{6.2}$$

Since $x_0 \in R_\nu$, it is clear that these conditions imply that $f(x_0) \in \mathfrak{m}_\nu$ and $f'(x_0) \notin \mathfrak{m}_\nu$. Therefore, $|f(x_0)| < 1$ and $|f'(x_0)| = 1$, so our assumption is met. Here,

$$\bar{\cdot}: R_\nu \rightarrow R_\nu/\mathfrak{m}_\nu = k_\nu$$

is the reduction homomorphism to the residue field, and $\overline{f}(X) = \sum_{i=0}^n \overline{a_i} X^i \in k_\nu[X]$.

Corollary 6.1.4

Let

$$f(X) = X^{p-1} - 1 \in \mathbb{Z}[X] \subset \mathbb{Z}_p[X].$$

Then, for every $0 \neq \bar{\xi} \in \mathbb{Z}/p\mathbb{Z}$ there exists a unique $\xi \in \mathbb{Z}_p$ such that $f(\xi) = 0$ and $[\xi] = \bar{\xi} \in \mathbb{Z}_p/(p) = \mathbb{Z}/p\mathbb{Z}$.

Proof. Let $0 \neq \bar{\xi} \in \mathbb{Z}/p\mathbb{Z}$ and let ξ_0 be a lift of $\bar{\xi}$ in $\mathbb{Z}_p$. Since

$$\bar{f}(X) = X^{p-1} - 1 = \prod_{0 \neq x \in \mathbb{Z}/p\mathbb{Z}} (X - x),$$

we have that

$$\bar{f}(\bar{\xi}) = 0 \quad \text{and} \quad \bar{f}'(\bar{\xi}) = (p-1)\bar{\xi}^{p-2} \neq 0 \quad \text{in } \mathbb{Z}/p\mathbb{Z}.$$

Hence, by Theorem 6.1.2, f has a root ξ in $\mathbb{Z}_p$ such that $|\xi - \xi_0| < 1$ and therefore, ξ is also a lift of $\bar{\xi}$. $\square$

Exercise 6.1.5. Show that 2 is a square in the ring of 7-adic integers.

Solution: Let $f(X) = X^2 - 2 \in \mathbb{Z}_7[X]$. In $\mathbb{F}_7$, we $3^2 - 2 \equiv 0 \pmod{7}$. Furthermore, we have that

$$0 = |f(3)|_7 < |f'(3)|_7 = |6|_7 = 1.$$

Therefore, we can apply Hensel's lemma to say that there exists a unique lift of 3 in $\mathbb{Z}_7$ such that it is a root of $f(X)$. Thus, 2 is a square in $\mathbb{Z}_7$. $\square$

Bibliography

- [1] Dennis Gaitsgory. *Course Notes: Week 3, Localization*. <https://math.uchicago.edu/~may/PEOPLE/DENNIS/week3.pdf>. Course notes, accessed 2026-05-08. n.d.
- [2] W. Jonsson. “Introduction to Commutative Algebra. by M. F. Atiyah and I. G. Macdonald. Addison-Wesley, Reading, Mass. (1969). xx IX+128 pp.” In: *Canadian Mathematical Bulletin* 13.1 (1970), pp. 162–162. DOI: [10.1017/S0008439500031039](https://doi.org/10.1017/S0008439500031039).